Dimensional Asymptotics of Euler-Based Simulated Likelihood for Multidimensional Diffusions

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Abstract

A widely used simulation method for likelihood inference in discretely observed diffusions approximates each transition density by simulating an Euler scheme to one step before the observed endpoint and averaging the final Gaussian density. We show that this endpoint estimator is a shrinking-bandwidth kernel estimator whose effective simulation size is $S/M^{K/2}$, not S, where K is the state dimension, M the number of Euler substeps and S the number of simulations. We derive exact Brownian moments, the general local moment law, the corrected density-level normalisation $\sqrt{S/M^{K/2}}$, and the pointwise mean squared error, minimised at $M \asymp S^{2/(K+4)}$. We then give a direct counterexample to the joint transition-density statements of Brandt and Santa-Clara (2002): along $M = S \to \infty$, a sequence satisfying their rate condition and their own regularity assumptions, the simulated density converges in probability to zero for every $K > 2$ although the true density is strictly positive. This refutes the consistency asserted in their Lemma 2 and, since $\sqrt{S}$ times the centred density then diverges, the central-limit statement of their Lemma 3. The proofs of their parameter-estimator theorems consequently fail, and in four dimensions their rate condition is incompatible with mean-square consistency of the simulated density. **We do not show that the estimator itself is inconsistent**: the results refute two published statements about the transition density and remove inputs on which the published parameter-estimator proofs rely, but the behaviour of the maximiser remains open, and we set out precisely what a proof either way would require. Issues specific to the exchange-rate application are analysed separately in appendices.

Keywords: simulated maximum likelihood; diffusion processes; Euler scheme; transition density; Monte Carlo variance; curse of dimensionality; diffusion bridges; importance sampling.

JEL classifications: C13; C15; C32; G12; G15.

1 Introduction

Continuous-time diffusion models are often observed only at discrete dates. Their exact transition densities are rarely available, which makes exact maximum likelihood difficult even when simulation of approximate sample paths is straightforward. One influential response is the simulation method introduced by Pedersen (1995a) and Pedersen (1995b) and developed and applied by Brandt and Santa-Clara (2002). The method divides each observation interval into M Euler substeps, simulates the process through the first $M - 1$ steps, evaluates the Gaussian density of the final Euler step at the observed endpoint, and averages this density over S simulated paths.

For fixed M, the construction is elementary: the sample average converges to the transition density of the M-step Euler chain as $S \to \infty$. For fixed S it is also clear that making the final time step smaller turns the Gaussian endpoint density into an increasingly narrow and increasingly high function. The mathematical question is therefore not whether the two separate approximations are individually meaningful. It is whether they remain compatible when M and S increase jointly.

The starting point is not new and we do not claim it. The final Gaussian density is a kernel with bandwidth $h^{1/2}$, where $h = 1/M$, so the whole construction is a kernel density estimate built from simulated draws. That reading is due to Milstein et al. (2004) and is stated for this estimator by Detemple et al. (2006). In K dimensions the kernel has effective volume of order $h^{K/2}$ and height of order $h^{-K/2}$, so only simulations falling in a shrinking neighbourhood of the endpoint contribute materially, and the effective number of such simulations is of order

$$Sh^{K/2} = \frac{S}{M^{K/2}}.$$

This quantity, rather than S alone, controls the Monte Carlo approximation. That a dimension-dependent rate follows, and that the rate condition of Brandt and Santa-Clara (2002) is inadequate, are also prior results; Section 1.1 sets out exactly what was already known, what this paper adds, and what remains open.

The paper makes six main contributions.

First, we derive exact finite-M moments for the estimator under K-dimensional Brownian motion. The Euler approximation is exact in this benchmark, so any failure is caused by the endpoint Monte Carlo construction and not by discretisation bias, boundary behaviour, nonlinear coefficients, or parameter identification.

Second, we construct a direct counterexample to the joint transition-density convergence asserted in Lemma 2 of Brandt and Santa-Clara (2002). For every $K > 2$, choosing $M = S = n$ makes the simulated density at the Brownian origin converge in probability to zero. Standard Brownian motion satisfies the regularity assumptions of that paper exactly, and the sequence satisfies its condition $\sqrt{S}/M \to 0$, so the counterexample cannot be excluded by strengthening the smoothness, boundedness or ellipticity assumptions alone, since Brownian motion already satisfies all of them in their strongest form. It is excluded only by imposing a joint simulation–discretisation rate condition strong enough to make $S/M^{K/2}$ diverge, which is a restriction on the design rather than on the model. The same sequence also refutes the joint central-limit statement of their Lemma 3, since $\sqrt{S_n}(\widehat{q}_{n,n} - p) \to -\infty$ in probability, so that no finite limit distribution exists. Consequently the published proofs of Theorems 1 and 2 fail, since both rest on those lemmas. In the four-dimensional application, their condition requires $S/M^2 \to 0$, while the variance calculation requires $S/M^2 \to \infty$ for mean-square consistency of the simulated transition density.

We are careful about what this does and does not establish. The counterexample concerns the simulated transition-density estimator at a point. It does not by itself exhibit a sequence along which the simulated-likelihood argmax estimator fails to converge, and we do not claim one. Section 1.2 states the logical status of each result, and Section 7.7 sets out what an argmax counterexample would require.

Third, we establish the generic local moment law for a smooth uniformly elliptic diffusion. If G_h denotes one endpoint-density summand, then for $r > 1$,

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow (2\pi)^{-(r-1)K/2} r^{-K/2} p(y \mid x)|V(y)|^{-(r-1)/2}.$$

In particular, $\mathrm{Var}(G_h)$ is of order $h^{-K/2}$. This gives the corrected triangular-array central-limit normalisation

$$\sqrt{Sh^{K/2}}\,(\widehat{q}_{M,S} - q_M),$$

not $\sqrt{S}(\widehat{q}_{M,S} - q_M)$.

Fourth, we combine the $O(M^{-1})$ Euler density bias used by Brandt and Santa-Clara (2002) with the corrected simulation variance. The resulting pointwise mean squared error is

$$O(M^{-2}) + O\left(\frac{M^{K/2}}{S}\right),$$

which is minimised at $M \asymp S^{2/(K+4)}$ and yields the rate $S^{-4/(K+4)}$. The estimator therefore exhibits an explicit curse of dimensionality.

Fifth, we trace the density-level result into the log likelihood. The nonlinearity of the logarithm operates on the effective Monte Carlo size $S/M^{K/2}$. Whenever a standard second-order delta expansion is valid, the leading log-density bias is of order $M^{K/2}/S$, rather than uniformly of order $1/S$. This distinction matters because simulated likelihood sums a log transformation over all observations.

Sixth, we examine the mathematical status of the exchange-rate application. We first give finite-sample scaling diagnostics showing that the ratios corresponding to the paper's own rate conditions are not small at the implemented design, so the implementation does not resemble a point along a sequence in which those conditions operate. The empirical model then contains square-root diffusions that violate the smoothness and nondegeneracy assumptions of the generic theorem. For the interest rates the Feller ratios exceed one, so the exact origin is unattainable, and at the benchmark states we display the one-step Euler negativity probabilities are numerically negligible; we do not reconstruct the observed rate path, so that is not a sample-wide bound. The market-incompleteness state is different. The square-root specification exposes a boundary and well-posedness issue: its Feller ratio equals one half for every parameter value, so direct Euler simulation of that state is not positivity preserving, while the implemented transformed chain instead requires a reconstructed radicand to remain nonnegative at every intermediate state. The original paper does not specify how intermediate violations are handled. We derive the exact worst-case one-step violation probability for the directly discretised state; that is a statement about the direct scheme, and we do not claim a crossing frequency for the implemented chain, which would require the simulated state distribution we do not have. Under the time-varying risk-price specification the variance identity becomes an implicit quadratic in volatility, and at the reported U.S.–U.K. estimates its leading coefficient is negative, so the inverse mapping from the incompleteness state to positive volatility is not globally single-valued without a branch-selection condition. This concerns the inverse map, not the implemented estimator, which takes volatility as primitive and reconstructs incompleteness uniquely. The four-dimensional Brownian correlation matrix is positive definite at the reported estimates and at every algebraically feasible reference point we examined, and robust to perturbation by the reported standard errors; we find no algebraically feasible point at which it is invalid. The residual point is that the specification supplies no parameterisation guaranteeing validity globally. The model has five primitive correlation parameters, four of them entries of the reduced 4×4 matrix and the fifth, ρ_{zz^*}, entering through the currency-risk variance term instead; the matrix also carries two state-dependent implied entries. The variance identity bounds those two implied entries automatically, but the primitive parameters need their own restrictions, and in the degenerate configuration positive semidefiniteness requires both a compatibility relation that fixes ρ_{xy} and a separate magnitude condition on ρ_{ww^*}, ρ_{wy} and ρ_{w^*y}, neither of which is imposed. Finally, the minimum-incompleteness rule is affine in the selected parameter, so its solution lies on the boundary of the feasible set; we distinguish the constraints the published estimator imposes from those a valid model requires, and together with the binding constraint this yields a hybrid and potentially nonregular estimator not covered by the unconstrained SML theorem.

The aim is not to relitigate the substantive economics of a paper published more than two decades ago. Nor do we infer from a false theorem that every numerical estimate obtained with finite M and S must be poor. The aim is narrower and mathematical: to identify the correct dimensional scale of the endpoint estimator, separate sequential from joint convergence, and determine what the published asymptotic arguments do and do not establish.

1.1 Relation to existing asymptotic analyses

The estimator analysed here originates with Pedersen (1995a) and Pedersen (1995b) and is the one implemented by Brandt and Santa-Clara (2002). A substantial literature has already studied it, and it is important to be exact about what that literature establishes, because several statements that might look like the contribution of this paper are older than it.

A note on notation. The cited papers use M, N and δ for quantities that are not the M, N and h of this paper, and silently carrying their symbols across would invert the meaning of several statements. Detemple et al. in particular write M for the number of simulations, which is this paper's S. Throughout this subsection we therefore use neutral symbols: n_{data} for the number of observations, n_{MC} for the number of simulated trajectories, m_{Euler} for the number of Euler substeps, and b_{ker} for a kernel bandwidth, meaning always a standard-deviation scale rather than a variance. In this paper's own notation, $n_{\text{data}} = N$, $n_{\text{MC}} = S$, $m_{\text{Euler}} = M$ and $b_{\text{ker}} = \sqrt{h}$. Footnotes record the original symbol wherever the translation could matter.

What is already known. Milstein et al. (2004, §1, eqs. (1.5)–(1.7)) observe that a simulated transition density of the form $n_{\text{MC}}^{-1} \sum_n \varphi_{b_{\text{ker}}}(\bar{X}_n - y)$, where $\bar{X}_n$ are Euler realisations at the terminal time and $\varphi_{b_{\text{ker}}}$ is a Gaussian kernel, is a special case of a Parzen–Rosenblatt estimator with the bandwidth set equal to the integration step; they then construct forward–reverse representations that are root-n_{MC} consistent and so avoid the resulting curse of dimensionality.[1] Two features of their setting differ from the one studied here, and the difference is where the exponent below comes from.

Their kernel is an *imposed* smoother. In their equation (1.5) it is

$$\varphi_{b_{\text{ker}}}(z) = \left(2\pi b_{\text{ker}}^2\right)^{-K/2} \exp(-\|z\|^2/2b_{\text{ker}}^2),$$

so its covariance matrix is $b_{\text{ker}}^2 I$ and its bandwidth, in the standard-deviation sense used throughout, is b_{ker}; with b_{ker} set to the integration step this is a covariance of $h^2 I$. It is applied to paths simulated all the way to the terminal time. The endpoint estimator of Pedersen (1995a) and Pedersen (1995b) instead stops one step short and uses the model's own one-step Gaussian density, whose covariance is hV and whose bandwidth is therefore $\sqrt{h}$. Writing both as covariances, theirs is $h^2 I$ and the endpoint kernel's is hV; writing both as bandwidths, theirs is h and the endpoint kernel's is $\sqrt{h}$. Because a K-dimensional kernel of bandwidth b_{ker} has effective volume b_{ker}^K, the effective sample sizes are $n_{\text{MC}} b_{\text{ker}}^K = n_{\text{MC}} h^K$ in their case and $n_{\text{MC}} h^{K/2}$ here. That factor-of-two difference in the exponent is the whole source of the $K/2$ below, and it is a property of which Gaussian is being averaged, not of any choice made here.

The dimension-dependent rate they quote, an optimal bandwidth of order $n_{\text{MC}}^{-1/(4+K)}$ and accuracy $n_{\text{MC}}^{-2/(4+K)}$, is the classical Parzen–Rosenblatt rate for a *free* bandwidth, which they attribute to the nonparametric-statistics literature rather than claiming as their own.

Detemple et al. (2006, p. 32) restate the kernel interpretation for the estimator of Pedersen (1995a) and Pedersen (1995b) and Brandt and Santa-Clara (2002) specifically, and draw three consequences from it. First, the convergence rate is dimension dependent: they give a score rate of order $n_{\text{MC}}^{-2/(K+4)}$, which is the same exponent family as the pointwise rate derived

[1] They write N for what we call n_{MC} and δ for what we call b_{ker}; their N is a simulation count, not a sample size.

below.[2] Second, optimality requires a joint design linking sample size, simulation size and discretisation level, and they state the conditions explicitly on the same page. Third, and most directly, they observe that Brandt and Santa-Clara (2002) impose one of those two conditions and not the other, and write that this "assumption, unfortunately, is not sufficient" (Detemple et al., 2006, p. 32). They also note, on the same page, that with a fixed discretisation level and a fixed number of simulations the second-order bias explodes as the sample grows, so that asymptotic confidence intervals cover the true value with probability zero.

Separately, Durham and Gallant (2002) observe that unconditioned Euler paths rarely arrive near the observed endpoint and introduce bridge-based proposals for that reason, and Elerian et al. (2001) augment the path with a Metropolis–Hastings block on the same reasoning. Lindström (2012) regularises the Pedersen proposal towards a bridge. Stramer and Yan (2007b) compare the modified-Brownian-bridge simulation approach with closed-form expansions for univariate diffusions, and prove that under constant volatility that sampler is exactly a Brownian bridge; the endpoint estimator appears there only in passing, at their equation (9), as a sampler that "ignores the end point information" (Stramer and Yan, 2007b, p. 677).

Stramer and Yan (2007a) require particular care, because although their theorem concerns a different sampler their remarks about this one are pointed. Their Theorem 1 gives an efficient allocation for the *modified Brownian bridge* sampler, showing that its importance-weight variance is bounded uniformly in m_{Euler} and that the optimal choice is $n_{\text{MC}} \asymp m_{\text{Euler}}^2$ with total error $O(1/m_{\text{Euler}}) + O(1/\sqrt{n_{\text{MC}}})$; they note that this accuracy holds "regardless of the dimension d" (Stramer and Yan, 2007a, Theorem 1, p. 489).[3] About the endpoint sampler they observe three things. It "may result a random variable R_M with unbounded $\text{Var}_q(R_M)$ which leads to inefficiency" (Stramer and Yan, 2007a, p. 486); in its case "a limiting quantity for G_M does not exist and, therefore, even a huge number N can give poor approximations" (Stramer and Yan, 2007a, p. 490); and numerically "for any fixed N, the approximation error of Pedersen's sampler is dominated by the Monte Carlo error and increases as M increases" (Stramer and Yan, 2007a, Fig. 2, p. 494). In the last two quotations their N is again a simulation count. They also record the kernel connection for this estimator specifically, identifying it with a kernel method in which the evaluation points are the preterminal states at time $\Delta - h$ (Stramer and Yan, 2007a, p. 486).

We therefore claim *none* of the following as new: the kernel interpretation of the endpoint density; the dimension dependence of the resulting rate, including the exponent $2/(K+4)$; the existence of a bias–variance trade-off between discretisation and simulation and the associated optimal allocation; the observation that the rate condition of Brandt and Santa-Clara (2002) is inadequate; the need for joint rather than sequential limits; the case for bridge proposals; the possibility that the endpoint estimator's importance-weight variance is unbounded in M; and the observation that increasing S alone need not rescue it.

The same pathology in a different literature. The mechanism identified here is not peculiar to Pedersen's estimator, and saying so places it correctly. In sequential Monte Carlo, Bengtsson et al. (2008) and Snyder et al. (2008) analyse the collapse of importance weights in high dimension for a particle filter whose proposal does not use the observation: the largest weight comes to dominate the sum, the effective sample size falls to one, and avoiding this requires the particle count to grow exponentially in an effective dimension. The endpoint estimator is that construction. Its proposal is the unconditioned Euler path, which ignores the endpoint exactly

[2] Their notation is $M^{-2/(d+4)}$, where their M is the number of simulations and their d the state dimension; in this paper's symbols that is $S^{-2/(K+4)}$. Their N denotes Euler substeps and their L the number of observations, so all four symbols differ from ours in meaning.

[3] They write M for Euler substeps, as we do, but N for the number of simulated trajectories, which is our S; their optimal allocation therefore reads $S \asymp M^2$ in this paper's symbols.

as an untuned particle filter ignores the observation; its weights are the one-step Gaussian densities; and the shrinking bandwidth $\sqrt{h}$ plays the role of an observation with vanishing error variance, which is the regime in which the weight collapse is sharpest. What Theorem 3.3 adds to that literature is a setting in which the degeneracy is exactly computable, so that a specific admissible sequence can be exhibited rather than a scaling law asserted; what that literature adds here is the assurance that the phenomenon is structural rather than an idiosyncrasy of one estimator.

To restate the point of detail on which the dimensional exponent turns, now in this paper's own symbols: the kernel in the endpoint construction is the model's own one-step Euler density, of covariance hV and therefore of bandwidth $\sqrt{h}$, not h. That is why the effective simulation size is $Sh^{K/2}$ rather than Sh^{K}, and it is the reason a dimension-dependent exponent appears here where Stramer and Yan (2007a) find none for the bridge, whose importance weights carry no shrinking kernel at all.

What this paper adds. Against that background the contributions are the following, each stated for the unconditioned endpoint estimator.

(a) Exact finite-M moments of every order for the endpoint summand under K-dimensional Brownian motion (Proposition 3.1). The prior literature is asymptotic; these are closed forms valid at every M and S, and they are what make the counterexample below computable rather than merely plausible.

(b) The exact local moment constant $c_{r,K} = (2\pi)^{-(r-1)K/2} r^{-K/2}$ for general uniformly elliptic diffusions (Theorem 4.2), together with the identification of $R_{M,S} = S/M^{K/2}$ as the governing quantity. Prior work gives the exponent; this gives the constant.

(c) An explicit collapse-in-probability sequence (Theorem 3.3). The prior negative results are of a different kind: inefficiency, exploding second-order bias in the estimating function, possibly unbounded importance-weight variance, and the warning that a huge N may not help. Each says the estimator may be bad. Theorem 3.3 exhibits a specific sequence, admissible under the published rate condition and the published regularity assumptions, along which the density estimator converges in probability to a value it should not, with the true density strictly positive. That the variance may be unbounded is prior work; that there is an admissible path to the wrong limit is not.

(d) A direct refutation of the joint transition-density statements in Lemmas 2 and 3 of Brandt and Santa-Clara (2002) (Theorem 3.3 and Corollary 3.5), under those authors' own assumptions.

(e) The explicit four-dimensional incompatibility between the published condition, equivalent to $S/M^2 \to 0$, and mean-square consistency, which requires $S/M^2 \to \infty$. Detemple et al. (2006) state that the published condition is insufficient; the contribution here is to show that in the dimension actually used no sequence can satisfy both the published rate condition and mean-square consistency of the simulated transition density. The published condition is not vacuous: many sequences satisfy $\sqrt{S}/M \to 0$, including $S = M$, which is exactly the sequence Theorem 3.3 uses. What fails is their intersection with consistency.

(f) The mathematical reassessment of the exchange-rate application in Section 8 and Section B to Section G.

The right description of this paper's relation to Detemple et al. (2006) is therefore sharpening rather than discovery: it derives exact finite-discretisation moments, identifies the local moment constants, and constructs an explicit admissible sequence along which the simulated transition density converges to the wrong value. What distinguishes that last item from an efficiency observation is its logical force. An inefficiency says a design is wasteful and can be repaired by spending more. A collapse along an admissible sequence says the density estimator converges to a value it should not; divergence of S alone does not repair it, and only a fast enough rate of S relative to $M^{K/2}$ does.

Table 1 summarises the comparison.

1.2 Scope of the results

Because this paper both proves theorems and reassesses a published application, it matters which claims are which. Every result below falls into one of three categories, and the text signals the category wherever a result is stated.

Proved. The exact Brownian moments of Proposition 3.1, the collapse theorem Theorem 3.3, the general local moment law Theorem 4.2, the density variance scale of Corollary 4.6, the density-level triangular-array central limit theorems Theorem 3.10 and Theorem 4.9, and the pointwise mean-squared-error balance of Proposition 5.1. These are theorems about the simulated *transition density*, proved under stated assumptions.

Conditional. The log-density expansion Proposition 6.1, which requires uniform integrability and lower-tail control that we assume rather than verify; the resulting statements about scores and Hessians; and the abstract local perturbation result Proposition 6.2, whose hypotheses include consistency of the simulated maximiser and uniform score and Hessian agreement, none of which we verify for the endpoint estimator. It is a conditional abstract result about any pair of maximisers meeting those hypotheses, not an asymptotic theorem for the simulated estimator, and Remark 6.3 states what would be needed to make it one.

Diagnostic. Everything in Section 8 concerning the exchange-rate model: the finite-sample scaling ratios, the square-root boundary behaviour, correlation-matrix feasibility, the volatility state transformation, minimum-incompleteness geometry, and constrained inference. These are calculations about a specific published model, not asymptotic theorems, and they are reported as such.

One boundary deserves emphasis because it is easy to blur. All the proved results concern the simulated transition density. None of them concerns the maximiser of the simulated log likelihood. We do not prove that the endpoint simulated-likelihood estimator is inconsistent, and Section 7.7 and Section H set out precisely what such a proof would require.

The remainder is organised as follows. Section 2 defines the estimator. Section 3 gives exact Brownian results and the counterexample. Section 4 derives the general diffusion moment law and corrected CLT. Section 5 studies the bias–variance trade-off. Section 6 discusses log likelihood and parameter estimation. Section 7 analyses the published proof. Section 8 studies the exchange-rate diffusion. Section 9 presents reproducible numerical evidence, and Section 10 discusses variance-reduction alternatives.

Table 1: Relation to existing asymptotic analyses of Pedersen-type simulated likelihood. **All entries are translated into this paper's notation**: K state dimension, M Euler substeps, S simulated trajectories, N observations. Several cited papers use these symbols for different quantities, Detemple et al. writing M for the simulation count in particular, so the rates below do not read as in the originals. "Error quantity" names what is converging, since a score rate and a density rate are not comparable. "Joint" asks whether the limits in M, S and N are taken jointly rather than one after another, a sequential limit being weaker. "Counterexample" asks whether a sequence is exhibited along which the estimator converges to a wrong value.

Reference	Estimator	Dim.	Error quantity	Rate or variance result	Joint	Counter-example
Pedersen (1995a) and Pedersen (1995b)	Endpoint	general	estimator	consistency and asymptotic normality	seq.	no
Brandt and Santa-Clara (2002)	Endpoint	applied at $K = 4$	density and estimator	$\sqrt{S}/M \to 0$, $N/S^{1/4} \to 0$	asserted, proved seq.	no
Milstein et al. (2004)	Terminal-time kernel; forward–reverse	explicit in K	density	kernel reading at bandwidth h, covariance $h^2 I$; classical Parzen–Rosenblatt rate; root-S forward–reverse alternative	yes	no
Detemple et al. (2006)	Euler and Doss-transformed, incl. Pedersen and Brandt–Santa-Clara	explicit in K	score	score rate $S^{-2/(K+4)}$; optimal design; published condition "not sufficient"; second-order bias correction	yes	no
Durham and Gallant (2002)	Bridge proposals	general	density, numerically	numerical efficiency comparisons	—	no
Stramer and Yan (2007a)	Modified Brownian bridge; remarks on the endpoint sampler	dimension-free for the bridge	importance weights	bridge weight variance bounded in M, optimal $S \asymp M^2$; endpoint variance may be unbounded and large S may not rescue it	yes	no
Stramer and Yan (2007b)	Modified Brownian bridge; closed-form expansion	univariate	log likelihood, numerically	bridge is exact under constant volatility; comparison run at $S \asymp M^2$	conjectured	no
Lindström (2012)	Regularised bridge	general	density, numerically	sparse-sampling performance	—	no
This paper	Endpoint	explicit in K	density	exact finite-M moments; exact constant $c_{r,K}$; $\mathrm{Var} \asymp M^{K/2}/S$	yes	yes, Theorem 3.3

2 The endpoint simulated-likelihood estimator

Let $Y_t \in \mathbb{R}^K$ satisfy the stochastic differential equation

$$dY_t = \mu(Y_t, t; \theta)\, dt + \Sigma(Y_t, t; \theta)\, dW_t, \qquad V = \Sigma\Sigma^\top, \tag{1}$$

where W_t is a Brownian motion of suitable dimension and $\theta \in \mathbf{\Theta} \subset \mathbb{R}^L$. Coefficients are assumed regular enough for a unique strong solution in the sense of Karatzas and Shreve (1991). We suppress t and θ where no ambiguity arises. Consider one observation interval of length one, from $Y_0 = x$ to $Y_1 = y$. The normalisation to unit length is only notational.

For $M \geq 2$, set $h = 1/M$ and define the Euler chain (Kloeden and Platen, 1992)

$$Y^M_{m+1} = Y^M_m + \mu(Y^M_m, mh)h + \Sigma(Y^M_m, mh)\sqrt{h}\,\varepsilon_{m+1}, \qquad m = 0, \ldots, M-1, \tag{2}$$

with $Y^M_0 = x$ and independent $\varepsilon_m \sim \mathcal{N}(0, I)$. Conditional on $Y^M_m = z$, the one-step density is

$$g_h(y \mid z) = \varphi_K(y; z + \mu(z, 1-h)h, hV(z, 1-h)). \tag{3}$$

Let $Z_h = Y^M_{M-1}$ denote the preterminal Euler state and let $r_h(z \mid x)$ be its density. The M-step Euler transition density is

$$q_M(y \mid x) = \mathbb{E}[g_h(y \mid Z_h)] = \int_{\mathbb{R}^K} g_h(y \mid z) r_h(z \mid x)\, dz. \tag{4}$$

The endpoint simulation estimator is

$$\widehat{q}_{M,S}(y \mid x) = \frac{1}{S}\sum_{s=1}^{S} g_h(y \mid Z_{h,s}), \tag{5}$$

where $Z_{h,1}, \ldots, Z_{h,S}$ are independent draws from the preterminal Euler law. This is equation (8) of Brandt and Santa-Clara (2002), with notation adapted to the present analysis.

For each fixed M, the strong law yields

$$\widehat{q}_{M,S}(y \mid x) \longrightarrow q_M(y \mid x) \quad \text{almost surely as } S \to \infty,$$

provided the summand is integrable. Under regularity conditions such as those of Bally and Talay (1996a) and Bally and Talay (1996b),

$$q_M(y \mid x) - p(y \mid x) = O(M^{-1}),$$

where p is the diffusion transition density. These two facts establish a sequential limit: first $S \to \infty$ for fixed M, then $M \to \infty$. They do not establish convergence along arbitrary joint sequences $M, S \to \infty$.

The geometry of (3) explains why. Its covariance is $hV(z)$. Under nondegeneracy, it is concentrated in a ball of radius $O(\sqrt{h})$ around the endpoint and has maximum height $O(h^{-K/2})$. The integral in (4) remains finite because the effective volume is $O(h^{K/2})$. A Monte Carlo average resolves that integral only if enough preterminal draws enter this shrinking region.

Definition 2.1 (Effective endpoint simulation size)**.** For a K-dimensional endpoint estimator with Euler step $h = 1/M$, define

$$R_{M,S} = Sh^{K/2} = \frac{S}{M^{K/2}}. \tag{6}$$

This is the expected-order number of simulations falling in an $O(\sqrt{h})$ neighbourhood of an endpoint with a regular positive preterminal density.

The next sections make this interpretation exact.

3 Exact Brownian analysis

Consider standard Brownian motion in $\mathbb{R}^K$,

$$dY_t = dW_t, \tag{7}$$

observed over a unit interval. Euler discretisation is exact. This benchmark is not an adversarial construction outside the scope of the published theory: with $\mu \equiv 0$ and $\Sigma \equiv I_K$, all coefficient derivatives vanish and are therefore continuous and bounded of every order, so Assumption 1 of Brandt and Santa-Clara (2002) holds, and $\Sigma\Sigma^\top = I_K$ is positive definite, so Assumption 2 holds. Lemmas 1–3 of that paper are stated under exactly these two assumptions. Any failure exhibited here is therefore a failure inside the stated hypotheses, and cannot be repaired by strengthening the regularity conditions. Set $Y_0 = x$ and evaluate the transition density at $Y_1 = y$. With $h = 1/M$,

$$Z_h \sim \mathcal{N}(x, (1-h)I_K)$$

and a single endpoint summand is

$$G_h = (2\pi h)^{-K/2} \exp\left[-\frac{\|y - Z_h\|^2}{2h}\right]. \tag{8}$$

3.1 Exact moments

Proposition 3.1 (Exact Brownian moments)**.** *Let $r > 0$. For the summand* (8)*,*

$$\begin{aligned}\mathbb{E}[G_h^r] &= (2\pi h)^{-rK/2}\left(1 + \frac{r(1-h)}{h}\right)^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{h + r(1-h)\}}\right] \\ &= (2\pi)^{-rK/2} h^{-(r-1)K/2}\{r - (r-1)h\}^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{r-(r-1)h\}}\right].\end{aligned} \tag{9}$$

In particular,

$$\mathbb{E}[G_h] = (2\pi)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2}\right] = p(y \mid x), \tag{10}$$

and

$$\mathbb{E}[G_h^2] = (2\pi)^{-K} h^{-K/2}(2-h)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2-h}\right]. \tag{11}$$

Proof. For $Z \sim \mathcal{N}(x, (1-h)I_K)$ and $c > 0$, the Gaussian identity

$$\mathbb{E}\left[e^{-c\|Z-y\|^2}\right] = \{1 + 2c(1-h)\}^{-K/2} \exp\left[-\frac{c\|x-y\|^2}{1+2c(1-h)}\right]$$

follows by completing the square. Apply it with $c = r/(2h)$ and multiply by the prefactor $(2\pi h)^{-rK/2}$. The cases $r = 1$ and $r = 2$ follow directly. □

Corollary 3.2 (Exact variance)**.** *For S independent simulations,*

$$\operatorname{Var}(\widehat{q}_{M,S}) = \frac{1}{S}\left[(2\pi)^{-K} h^{-K/2}(2-h)^{-K/2} e^{-\|y-x\|^2/(2-h)} - p(y \mid x)^2\right]. \tag{12}$$

Hence

$$\operatorname{Var}(\widehat{q}_{M,S}) \sim \frac{\tau_K^2(x,y)}{S h^{K/2}}, \qquad \tau_K^2(x,y) = (2\pi)^{-K} 2^{-K/2} e^{-\|y-x\|^2/2}. \tag{13}$$

The variance is not uniformly $O(1/S)$. It is $O(M^{K/2}/S)$. The estimator is unbiased in this example for every M, but its distribution becomes increasingly dominated by rare, very large contributions when $R_{M,S}$ is small.

3.2 A direct joint-limit counterexample

A variance calculation alone does not prove failure of convergence in probability because a non-uniformly integrable sequence can converge in probability while retaining large or divergent moments. The next theorem gives a direct failure result. It is a statement about the simulated transition density at a single pair of points; Section 7.7 explains why it does not transfer automatically to the parameter estimator.

Theorem 3.3 (Collapse along $M = S$)**.** *Let $K > 2$, let $x, y \in \mathbb{R}^K$ be arbitrary, and choose $M_n = S_n = n$. Then the Brownian endpoint estimator satisfies*

$$\widehat{q}_{n,n}(y \mid x) \longrightarrow 0 \qquad \textit{in probability,} \tag{14}$$

although the true density is strictly positive at every such pair,

$$p(y \mid x) = (2\pi)^{-K/2} e^{-\|y-x\|^2/2} > 0.$$

In every dimension the sequence satisfies the rate condition

$$\frac{\sqrt{S_n}}{M_n} = n^{-1/2} \longrightarrow 0$$

used in Theorems 1 and 2 of Brandt and Santa-Clara (2002), and it satisfies without any rate restriction the hypotheses of their Lemma 2, which requires only $M \to \infty$ and $S \to \infty$.

Proof. Write $h_n = 1/n$ and let $Z_{n,1}, \ldots, Z_{n,n}$ be the preterminal draws, each distributed as $\mathcal{N}(x, (1 - 1/n)I_K)$. Fix a constant $c > K$ and define

$$r_n = \sqrt{\frac{c \log n}{n}}.$$

Let A_n be the event that at least one draw lies in the ball $B(y, r_n)$. For $n \geq 2$ the covariance satisfies $1 - 1/n \geq \frac{1}{2}$, so the density of $Z_{n,1}$ is bounded by $\pi^{-K/2}$ everywhere, uniformly in n and in the pair (x, y). Hence, with ω_K the volume of the unit ball and $C_K = \pi^{-K/2}\omega_K$,

$$\mathbb{P}(A_n) \leq n\mathbb{P}(\|Z_{n,1} - y\| \leq r_n) \leq C_K n r_n^K = C_K c^{K/2} n^{1-K/2} (\log n)^{K/2} \longrightarrow 0$$

because $K > 2$. Note that this bound does not involve x or y: displacing the evaluation point only moves the ball, and the Gaussian density bound is global.

On A_n^c, every summand is bounded by

$$\left(\frac{n}{2\pi}\right)^{K/2} \exp\left(-\frac{n r_n^2}{2}\right) = (2\pi)^{-K/2} n^{(K-c)/2},$$

which tends to zero because $c > K$. The sample average obeys the same bound. Therefore, for every $\varepsilon > 0$,

$$\mathbb{P}(\widehat{q}_{n,n} > \varepsilon) \leq \mathbb{P}(A_n) + \mathbf{1}\left\{(2\pi)^{-K/2} n^{(K-c)/2} > \varepsilon\right\} \longrightarrow 0. \qquad \square$$

Remark 3.4 (The collapse is not an artefact of the evaluation point)**.** Neither bound in the proof depends on x or y: the first is a global Gaussian density bound applied to a ball centred at y, and the second uses only that every draw lies outside that ball. The collapse is therefore not a feature of evaluating at the mode, where one might suspect the estimator is stressed unusually, nor of the coincident case $x = y$. It holds at every pair of points, and the limit is zero at all of them while $p(y \mid x)$ is strictly positive at all of them. What varies with (x, y) is only the constant in Proposition 3.1, not the rate or the conclusion.

The same sequence refutes a second published statement. Lemma 3 of Brandt and Santa-Clara (2002) asserts that, under $\sqrt{S}/M \to 0$,

$$\sqrt{S}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\}$$

converges in distribution to a normal law. That cannot hold along $M_n = S_n = n$.

Corollary 3.5 (Failure of the published joint central-limit statement)**.** *Let $K > 2$ and $x = y = 0$, and take $M_n = S_n = n$, a sequence satisfying $\sqrt{S_n}/M_n \to 0$. Then*

$$\sqrt{S_n}\,\{\widehat{q}_{n,n}(0 \mid 0) - p(0 \mid 0)\} \longrightarrow -\infty \qquad \textit{in probability,} \tag{15}$$

so the sequence converges in distribution to no finite limit, normal or otherwise. The joint central-limit statement of Lemma 3 is therefore false along an admissible sequence, whatever variance is assigned to the limit.

Proof. Write $p = p(0 \mid 0) = (2\pi)^{-K/2} > 0$ and fix $\Lambda > 0$. By Theorem 3.3, $\widehat{q}_{n,n} \to 0$ in probability, so with $\varepsilon = p/2$,

$$\mathbb{P}\left(\widehat{q}_{n,n} > \tfrac{p}{2}\right) \longrightarrow 0.$$

On the complementary event, $\widehat{q}_{n,n} - p \le -\frac{p}{2}$, hence $\sqrt{S_n}(\widehat{q}_{n,n} - p) \le -\frac{p}{2}\sqrt{n}$. Since $\frac{p}{2}\sqrt{n} > \Lambda$ for all n large,

$$\mathbb{P}\left(\sqrt{S_n}\,\{\widehat{q}_{n,n} - p\} > -\Lambda\right) \le \mathbb{P}\left(\widehat{q}_{n,n} > \tfrac{p}{2}\right) \longrightarrow 0,$$

which is (15). A sequence diverging to $-\infty$ in probability is not tight, so it has no weak limit in $\mathbb{R}$. $\square$

Remark 3.6 (What is and is not refuted)**.** It is worth stating the hierarchy in one place, since the four statements are easy to run together.

(i) The joint transition-density consistency of Lemma 2 is *directly false* (Theorem 3.3).

(ii) The joint central-limit statement of Lemma 3 is *directly false* along the same sequence (Corollary 3.5). Note also that the limiting variance it states is the variance of a summand that itself depends on M, so the statement is ill-posed as an asymptotic claim even before the counterexample is applied.

(iii) The published proofs of Theorems 1 and 2 *fail*, since Lemmas 4 to 6 take Lemmas 2 and 3 as inputs.

(iv) The conclusions of the parameter-estimator theorems are *not* disproved. The construction says nothing directly about the simulated-likelihood argmax; see Section 7.7 and Section H.

Remark 3.7 (How an unbiased estimator converges to zero)**.** For every n, $\mathbb{E}[\widehat{q}_{n,n}] = p(0 \mid 0)$. The convergence in probability to zero is sustained by increasingly rare simulations with increasingly large endpoint densities. The sequence is not uniformly integrable. This is precisely the rare-event geometry hidden by a fixed-M application of the strong law.

Corollary 3.8 (Contradiction in four dimensions)**.** *For $K = 4$, mean-square consistency requires*

$$\frac{S}{M^2} \longrightarrow \infty,$$

whereas the published condition $\sqrt{S}/M \to 0$ is equivalent to

$$\frac{S}{M^2} \longrightarrow 0.$$

The two conditions are mutually exclusive. Moreover, Theorem 3.3 shows that the published condition permits a sequence along which the density estimator converges to the wrong value.

Remark 3.9 (Where the conflict is unconditional)**.** Two words are worth separating before this is read. A sequence is *admissible* when it satisfies the published assumptions, in particular $\sqrt{S}/M \to 0$; it is *density-consistent* when $S/M^{K/2} \to \infty$, which is what mean-square consistency of the simulated transition density requires. These are different properties, and the point below is that in four dimensions no sequence has both, not that either class is empty. Admissible sequences abound, $S = M$ among them.

The incompatibility in Corollary 3.8 is a property of dimensions $K \geq 4$. Mean-square consistency requires $S/M^{K/2} \to \infty$ and the published condition requires $S/M^2 \to 0$; for $K \geq 4$ the first demands $S \gtrsim M^{K/2} \geq M^2$ and the second demands $S \ll M^2$, so no sequence satisfies both. For $K \leq 3$ the two are compatible: with $K = 3$ and $S = M^{7/4}$ one has $S/M^{3/2} \to \infty$ and $\sqrt{S}/M \to 0$ simultaneously. Theorem 3.3 is nevertheless not confined to $K \geq 4$. It shows that even when the published condition admits consistent sequences, as at $K = 3$, it also admits inconsistent ones, so the condition cannot be what makes the theorem work. The empirical application of Brandt and Santa-Clara (2002) has $K = 4$, where no admissible sequence is also density-consistent.

3.3 Correct triangular-array central limit theorem

Theorem 3.10 (Brownian endpoint CLT)**.** *Let $M \to \infty$ and $S \to \infty$ such that*

$$Sh^{K/2} = \frac{S}{M^{K/2}} \longrightarrow \infty. \tag{16}$$

Then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}\Big(0, \tau_K^2(x,y)\Big), \tag{17}$$

where $\tau_K^2(x,y)$ is defined in (13)*.*

Proof. Index the array by n, with $h_n = 1/M_n$ and $s = 1, \ldots, S_n$; within row n the summands $G_{n,s}$ are independent and identically distributed. Proposition 3.1 at $r = 2$ gives $h_n^{K/2}\mathbb{E}[G_{n,1}^2] \to \tau_K^2$, and $\mathbb{E}G_{n,1} = p(y \mid x)$ exactly for every M, so

$$h_n^{K/2}\,\mathrm{Var}(G_{n,1}) = h_n^{K/2}\mathbb{E}[G_{n,1}^2] - h_n^{K/2}p(y \mid x)^2 \longrightarrow \tau_K^2.$$

At $r = 3$ the same proposition gives $\mathbb{E}[G_{n,1}^3] = O(h_n^{-K})$, a bound on the *raw* moment. Since $G_{n,1} \geq 0$ and $\mathbb{E}G_{n,1} = p(y \mid x)$ is constant in n, the inequality $|a-b|^3 \leq 4(a^3 + b^3)$ for $a, b \geq 0$ gives the centred bound

$$\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3 \leq 4\left\{\mathbb{E}[G_{n,1}^3] + p(y \mid x)^3\right\} = O(h_n^{-K}).$$

The Lyapunov ratio at order three is therefore

$$\frac{S_n\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3}{\{S_n\,\mathrm{Var}(G_{n,1})\}^{3/2}} = \frac{S_n O(h_n^{-K})}{S_n^{3/2}\tau_K^3 h_n^{-3K/4}\{1 + o(1)\}} = O\Big(\big(S_n h_n^{K/2}\big)^{-1/2}\Big) \longrightarrow 0$$

by (16). The triangular-array Lyapunov central limit theorem applies. Because the Euler scheme is exact for Brownian motion, $q_M = p$ for every M and no bias term appears, so the statement may be centred at p directly. □

The correct normalisation depends on dimension. A fixed-M $\sqrt{S}$ central limit theorem is valid for each M separately, but it is not stable as $M \to \infty$: see Remark 4.10.

4 General multidimensional diffusions

The Brownian calculation reflects a local property of Gaussian endpoint kernels. We now derive the generic moment law. We state it under a clean global framework, so that every step of the proof is checkable and no condition is phrased in terms of the conclusion it is meant to deliver; Remark 4.4 explains how localisation would weaken it.

Throughout this section the model is time homogeneous, so that $\mu(z)$ and $V(z)$ carry no time argument. The one-step kernel (3) is then evaluated at the preterminal state alone, and no ambiguity arises about whether coefficients are taken at time $1-h$ or at time 1. Remark 4.5 records what changes otherwise.

Let $G_h = g_h(y \mid Z_h)$ be one summand in (5), where Z_h is the Euler chain at time $1-h$. Fix x and y.

Assumption 4.1 (Global regularity)**.** *(A1) $\mu : \mathbb{R}^K \to \mathbb{R}^K$ is bounded and continuous, with $\|\mu\|_\infty = \bar{\mu} < \infty$.*

(A2) $V : \mathbb{R}^K \to \mathbb{R}^{K\times K}$ is continuous and globally uniformly elliptic: there exist constants $0 < \underline{v} \le \overline{v} < \infty$ with

$$\underline{v}\, I_K \preceq V(z) \preceq \overline{v}\, I_K \qquad \text{for every } z \in \mathbb{R}^K. \tag{18}$$

(A3) For each $h \in (0, \frac{1}{2}]$ the preterminal Euler state Z_h, that is the chain (2) *at time $1-h$, has a density $r_h(\cdot \mid x)$ with respect to Lebesgue measure.*

(A4) The family is uniformly bounded: $\sup_{0<h\le 1/2} \sup_{z\in\mathbb{R}^K} r_h(z \mid x) = \bar{r} < \infty$.

(A5) There is a continuous density $p(\cdot \mid x)$ with $p(y \mid x) > 0$ such that $r_h(\cdot \mid x) \to p(\cdot \mid x)$ uniformly on some closed ball $\overline{B}(y, \delta_0)$, $\delta_0 > 0$, as $h \downarrow 0$.

Two comments on the content of these conditions. (A5) is a joint requirement in time and space: the preterminal state lives at time $1-h$, not at time 1, so the stated convergence combines convergence of the Euler law to the diffusion law with continuity of $t \mapsto p_t(\cdot \mid x)$ at $t = 1$ in the topology of uniform convergence near y. The limit $p(\cdot \mid x)$ is the diffusion transition density at time 1. And (A4) is not implied by (A3): it is a uniform-in-h statement, which for uniformly elliptic coefficients follows from the Gaussian upper bounds for Euler densities established by Bally and Talay (1996b), since the preterminal time $1-h \ge \frac{1}{2}$ is bounded away from zero.

Assumption 4.1 is compatible with the regular setting used to justify Euler density expansions in Bally and Talay (1996a) and Bally and Talay (1996b). It is not compatible with the square-root boundaries in the later financial application, where (A2) fails at the origin; that distinction is discussed in Section 8.

The following theorem proves the dimensional moment law under Assumption 4.1. It concerns the simulated transition density.

Theorem 4.2 (Local moment law)**.** *Under Assumption 4.1, for every fixed $r > 1$,*

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow c_{r,K}\, p(y \mid x)\, |V(y)|^{-(r-1)/2}, \tag{19}$$

where

$$c_{r,K} = (2\pi)^{-(r-1)K/2} r^{-K/2}. \tag{20}$$

In particular,

$$h^{K/2}\mathbb{E}[G_h^2] \longrightarrow (4\pi)^{-K/2} p(y \mid x)|V(y)|^{-1/2}. \tag{21}$$

Proof. Write $m_h(z) = y - z - h\mu(z)$, so that by (3)

$$g_h(y \mid z)^r = (2\pi h)^{-rK/2}|V(z)|^{-r/2} \exp\left[-\frac{r}{2h} m_h(z)^\top V(z)^{-1} m_h(z)\right], \tag{22}$$

and fix $\delta \in (0, \delta_0]$. Split

$$
\begin{aligned}
h^{(r-1)K/2}\mathbb{E}[G_h^r] &= I_h(\delta) + T_h(\delta),\\
I_h(\delta) &= h^{(r-1)K/2}\int_{B(y,\delta)} g_h(y \mid z)^r\, r_h(z \mid x)\, dz,\\
T_h(\delta) &= h^{(r-1)K/2}\int_{B(y,\delta)^c} g_h(y \mid z)^r\, r_h(z \mid x)\, dz.
\end{aligned}
$$

Tail term. For $z \notin B(y,\delta)$ and $h \le \delta/(4\bar{\mu}+1)$, (A1) gives

$$
\|m_h(z)\| \ge \|y-z\| - h\bar{\mu} \ge \tfrac{1}{2}\|y-z\| \qquad \text{and} \qquad \|m_h(z)\| \ge \tfrac{1}{2}\delta.
$$

By (A2) we have $V(z)^{-1} \succeq \overline{v}^{-1} I_K$ and $|V(z)|^{-r/2} \le \underline{v}^{-rK/2}$, so

$$
g_h(y \mid z)^r \le (2\pi h)^{-rK/2}\underline{v}^{-rK/2}\exp\left[-\frac{r\|m_h(z)\|^2}{2\overline{v}h}\right].
$$

Splitting the exponent in half and using the two lower bounds separately,

$$
g_h(y \mid z)^r \le (2\pi h)^{-rK/2}\underline{v}^{-rK/2}\exp\left[-\frac{r\delta^2}{16\overline{v}h}\right]\exp\left[-\frac{r\|y-z\|^2}{16\overline{v}h}\right].
$$

Using (A4) and $\int_{\mathbb{R}^K} e^{-r\|w\|^2/(16\overline{v}h)}dw = (16\pi\overline{v}h/r)^{K/2}$, and collecting powers as $h^{(r-1)K/2}\cdot h^{-rK/2}\cdot h^{K/2} = 1$,

$$
T_h(\delta) \le \bar{r}\,\underline{v}^{-rK/2}(2\pi)^{-rK/2}(16\pi\overline{v}/r)^{K/2}\exp\left[-\frac{r\delta^2}{16\overline{v}h}\right] \xrightarrow[h\downarrow 0]{} 0. \tag{23}
$$

Local term. Substitute $z = z_h(u) = y + \sqrt{h}\,u$, so that $dz = h^{K/2}du$ and the domain becomes $\|u\| < \delta/\sqrt{h}$. Since $h^{(r-1)K/2}(2\pi h)^{-rK/2}h^{K/2} = (2\pi)^{-rK/2}$, every power of h cancels and

$$
I_h(\delta) = (2\pi)^{-rK/2}\int_{\mathbb{R}^K}\mathbf{1}\{\|u\| < \delta/\sqrt{h}\}\,\Psi_h(u)\,du,
$$

$$
\Psi_h(u) = |V(z_h(u))|^{-r/2}r_h(z_h(u) \mid x)\exp\left[-\frac{r}{2h}m_h(z_h(u))^{\top}V(z_h(u))^{-1}m_h(z_h(u))\right].
$$

Here $m_h(z_h(u)) = -\sqrt{h}\,u - h\mu(z_h(u))$, so

$$
\frac{1}{h}m_h(z_h(u))^{\top}V(z_h(u))^{-1}m_h(z_h(u)) = \big(u + \sqrt{h}\,\mu(z_h(u))\big)^{\top}V(z_h(u))^{-1}\big(u + \sqrt{h}\,\mu(z_h(u))\big).
$$

Fix u. Then $z_h(u) \to y$, so continuity of μ and V gives $V(z_h(u)) \to V(y)$ and $\sqrt{h}\,\mu(z_h(u)) \to 0$, and the quadratic form converges to $u^{\top}V(y)^{-1}u$. For h small enough that $\sqrt{h}\|u\| \le \delta_0$, (A5) and continuity of $p(\cdot \mid x)$ give $r_h(z_h(u) \mid x) \to p(y \mid x)$. Hence, pointwise in u,

$$
\Psi_h(u) \longrightarrow |V(y)|^{-r/2}p(y \mid x)\exp\left[-\tfrac{r}{2}u^{\top}V(y)^{-1}u\right].
$$

Domination. Let $h \le \min\{\delta_0, (2\bar{\mu})^{-2}\}$. If $\|u\| \ge 2\sqrt{h}\,\bar{\mu}$ then $\|u + \sqrt{h}\,\mu(z_h(u))\| \ge \frac{1}{2}\|u\|$, so by (A2) and (A4)

$$
\Psi_h(u) \le \underline{v}^{-rK/2}\,\bar{r}\,\exp\left[-\frac{r\|u\|^2}{8\overline{v}}\right],
$$

while if $\|u\| < 2\sqrt{h}\,\bar{\mu} \le 1$ then $\Psi_h(u) \le \underline{v}^{-rK/2}\bar{r}$. The dominating function

$$\Lambda(u) = \underline{v}^{-rK/2}\bar{r}\left\{\exp\left[-\frac{r\|u\|^2}{8\bar{v}}\right] + \mathbf{1}\{\|u\| \le 1\}\right\}$$

is integrable and does not depend on h. Dominated convergence therefore gives

$$\lim_{h\downarrow 0} I_h(\delta) = (2\pi)^{-rK/2}|V(y)|^{-r/2}p(y \mid x)\int_{\mathbb{R}^K}\exp\Big[-\tfrac{r}{2}u^{\mathsf{T}}V(y)^{-1}u\Big]\,du.$$

The Gaussian integral equals $(2\pi/r)^{K/2}|V(y)|^{1/2}$, so

$$\lim_{h\downarrow 0} I_h(\delta) = (2\pi)^{-rK/2}(2\pi/r)^{K/2}p(y \mid x)|V(y)|^{-(r-1)/2} = c_{r,K}\,p(y \mid x)|V(y)|^{-(r-1)/2}.$$

This limit does not depend on δ, and $T_h(\delta) \to 0$ by (23), which proves (19). Taking $r = 2$ and using $(2\pi)^{-K/2}2^{-K/2} = (4\pi)^{-K/2}$ gives (21). □

Remark 4.3 (Verification in exactly solvable cases)**.** The constant can be checked against closed forms. For standard Brownian motion $V \equiv I_K$, and substituting $p(y \mid x) = (2\pi)^{-K/2}e^{-\|y-x\|^2/2}$ into (19) reproduces the exact limit of Proposition 3.1 term by term. For a constant-coefficient diffusion $dY = \mu\,dt + \Sigma\,dW$ the same computation with $V = \Sigma\Sigma^{\mathsf{T}}$ reproduces (19) including the determinant factor. For a multidimensional Ornstein–Uhlenbeck process the preterminal law is Gaussian and the second moment is again available in closed form; Figure 4 confirms convergence to (21) numerically. The cases $r = 2$ and $r = 3$ supply exactly the second and third moments used in Theorem 4.9.

Remark 4.4 (Localisation)**.** Only three features of Assumption 4.1 are used: ellipticity and boundedness of μ near y, which drive the local term; the uniform bound (A4), which drives domination; and enough global control to make the tail negligible. Global uniform ellipticity is therefore stronger than necessary. It can be replaced by ellipticity on $B(y, \delta_0)$ together with any condition forcing the contribution of $\{Z_h \notin B(y, \delta)\}$ to vanish after multiplication by $h^{(r-1)K/2}\sup_z g_h(y \mid z)^r$, for instance a Gaussian upper bound of Aronson type on r_h. We state the global version because it keeps the proof short and because the content of the theorem is the dimensional constant, not the weakest hypotheses.

We do not claim the assumption set is directly verifiable, and an earlier version did. Conditions (A1) to (A4) are checkable from the coefficients, but (A5) is not a primitive condition: it requires uniform convergence on a ball of the *Euler* density at time $1 - h$ to the *diffusion* density at time 1, which bundles a discretisation limit together with continuity of $t \mapsto p_t$ in the uniform topology. Results of Bally and Talay (1996a) and Bally and Talay (1996b) deliver statements of this kind, but under hypotheses of their own that are stronger than (A1) and (A2), so appealing to them to justify (A5) while citing (A1) and (A2) as the primitive conditions would be mildly circular. The honest description is that (A5) is an assumption imported from the Euler-approximation literature, not one the reader can check from μ and V alone.

Remark 4.5 (Time-inhomogeneous coefficients)**.** If μ and V depend on time, the one-step kernel (3) evaluates them at time $1 - h$, and the proof goes through verbatim provided μ and V are jointly continuous at $(y, 1)$ and (A1) and (A2) hold uniformly in t on a neighbourhood of 1. The limit constant is unchanged and $V(y)$ is read as $V(y, 1)$. We suppress the time argument in the statement because carrying it obscures the cancellation of powers of h, which is where the dimensional constant comes from.

Corollary 4.6 (Generic variance scale)**.** *Let $q_M(y \mid x) = \mathbb{E}[G_h]$ and suppose $q_M(y \mid x) \to p(y \mid x)$. Then*

$$\operatorname{Var}(\widehat{q}_{M,S}) \sim \frac{\tau^2(x,y)}{Sh^{K/2}}, \qquad \tau^2(x,y) = (4\pi)^{-K/2}p(y \mid x)|V(y)|^{-1/2}. \tag{24}$$

The central limit theorem below centres at the Euler density $q_M(y \mid x) = \mathbb{E}[G_h]$, and its proof needs that quantity to be bounded uniformly in h. That is not automatic: the summand G_h itself is unbounded, since a Gaussian of covariance hV has height of order $h^{-K/2}$ at its centre. What makes the *expectation* bounded is that the preterminal state has a uniformly bounded density, so the mass the kernel can concentrate on is limited. We record this separately rather than assume it.

Lemma 4.7 (The Euler density is bounded uniformly in the step)**.** *Under Assumption 4.1,*

$$\sup_{0<h\leq 1/2} q_M(y \mid x) \;\leq\; \bar{r}\left(\frac{4\bar{v}}{\underline{v}}\right)^{K/2} + C_K\,\bar{r}\,\underline{v}^{-K/2}\bar{\mu}^K \;<\; \infty, \tag{25}$$

where C_K depends only on K. In particular $q_M(y \mid x) = O(1)$.

Proof. Writing Z_h for the preterminal state and substituting $u = y - z$, which has unit Jacobian,

$$q_M(y \mid x) = \int \varphi_{hV(y-u)}(u - \mu(y-u)h)\, r_h(y - u \mid x)\, du.$$

By (A4) the density factor is at most $\bar{r}$, and by (A2),

$$\varphi_{hV(z)}(w) \leq (2\pi h)^{-K/2}\underline{v}^{-K/2}\exp\Big(-\frac{\|w\|^2}{2h\bar{v}}\Big).$$

Split the integral at $\|u\| = 2\bar{\mu}h$. On $\|u\| \geq 2\bar{\mu}h$, (A1) gives $\|u-\mu(y-u)h\| \geq \|u\| - \bar{\mu}h \geq \|u\|/2$, so that part is at most

$$\bar{r}\,(2\pi h)^{-K/2}\underline{v}^{-K/2}\int_{\mathbb{R}^K}\exp\Big(-\frac{\|u\|^2}{8h\bar{v}}\Big)du = \bar{r}\,(2\pi h)^{-K/2}\underline{v}^{-K/2}(8\pi h\bar{v})^{K/2} = \bar{r}\left(\frac{4\bar{v}}{\underline{v}}\right)^{K/2},$$

which is free of h. On $\|u\| < 2\bar{\mu}h$ the integrand is at most $\bar{r}(2\pi h)^{-K/2}\underline{v}^{-K/2}$ and the region has volume $\omega_K(2\bar{\mu}h)^K$, so that part is at most $C_K\bar{r}\underline{v}^{-K/2}\bar{\mu}^K h^{K/2}$ with $C_K = \omega_K 2^K(2\pi)^{-K/2}$, and $h \leq 1/2$ bounds it. Adding the two gives (25). □

Remark 4.8 (Boundedness, not convergence, is what is used)**.** Lemma 4.7 gives only $q_M = O(1)$, and that is all Theorem 4.9 requires: the theorem centres at q_M rather than at p, so no rate of approach is needed. The stronger statement $q_M(y \mid x) \to p(y \mid x)$ also holds under Assumption 4.1, by the approximate-identity argument of Theorem 4.2 applied at $r = 1$ together with (A5), but we do not use it here. Where the distinction matters is Corollary 4.11, which recentres at p and there assumes a first-order Euler density expansion separately.

Theorem 4.9 (Generic endpoint CLT)**.** *Let (M_n, S_n) satisfy $M_n \to \infty$ and $S_n \to \infty$, write $h_n = 1/M_n$, and suppose the effective endpoint size diverges:*

$$R_n = S_n h_n^{K/2} = \frac{S_n}{M_n^{K/2}} \longrightarrow \infty. \tag{26}$$

Then under Assumption 4.1,

$$\sqrt{R_n}\,\{\widehat{q}_{M_n,S_n}(y \mid x) - q_{M_n}(y \mid x)\} \Longrightarrow \mathcal{N}\Big(0, \tau^2(x,y)\Big), \tag{27}$$

where, as in (24),

$$\tau^2(x,y) = (4\pi)^{-K/2}p(y \mid x)|V(y)|^{-1/2}. \tag{28}$$

This is a statement about the simulated transition density, centred at the Euler density q_{M_n} and not at the diffusion density p; Corollary 4.11 treats the latter. It is not a statement about any parameter estimator.

Proof. Define the triangular array

$$G_{n,s} = g_{h_n}(y \mid Z_{h_n,s}), \qquad s = 1, \dots, S_n,$$

where $Z_{h_n,1}, \dots, Z_{h_n,S_n}$ are the independent preterminal draws in (5). Within row n these are independent and identically distributed with common mean $\mathbb{E}G_{n,1} = q_{M_n}(y \mid x)$, and

$$\widehat{q}_{M_n,S_n} - q_{M_n} = \frac{1}{S_n} \sum_{s=1}^{S_n} (G_{n,s} - \mathbb{E}G_{n,s}) \,.$$

Variance. Theorem 4.2 with $r = 2$ gives $h_n^{K/2}\mathbb{E}[G_{n,1}^2] \to \tau^2$. Since $\mathbb{E}G_{n,1} = q_{M_n}$ is bounded uniformly in n by Lemma 4.7, $h_n^{K/2}(\mathbb{E}G_{n,1})^2 \to 0$, so

$$h_n^{K/2}\,\mathrm{Var}(G_{n,1}) \longrightarrow \tau^2, \qquad \mathrm{Var}(\widehat{q}_{M_n,S_n}) = \frac{\mathrm{Var}(G_{n,1})}{S_n} \sim \frac{\tau^2}{R_n}. \tag{29}$$

Third centred moment. Theorem 4.2 with $r = 3$ bounds the *raw* moment, $\mathbb{E}[G_{n,1}^3] = O(h_n^{-K})$. The centred moment does not follow from this automatically, and is obtained as follows. The summand is a Gaussian density value, hence nonnegative, so the elementary inequality $|a - b|^3 \le 4(a^3 + b^3)$ for $a, b \ge 0$ applies and

$$\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3 \le 4\left\{\mathbb{E}[G_{n,1}^3] + (\mathbb{E}G_{n,1})^3\right\} = O(h_n^{-K}) + O(1) = O(h_n^{-K}), \tag{30}$$

using Lemma 4.7 once more. Nonnegativity of the summand is what makes this step elementary; without it the centred third moment would require a separate argument.

Lyapunov condition. Write $\sigma_n^2 = \mathrm{Var}(G_{n,1})$, so $\sigma_n^2 \sim \tau^2 h_n^{-K/2}$ by (29). The Lyapunov ratio at order $2 + \delta$ with $\delta = 1$ is

$$L_n = \frac{S_n\,\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3}{(S_n\sigma_n^2)^{3/2}} = \frac{S_n\,O(h_n^{-K})}{S_n^{3/2}\,\tau^3 h_n^{-3K/4}\{1 + o(1)\}} = O\!\left(S_n^{-1/2}h_n^{-K/4}\right) = O\!\left(R_n^{-1/2}\right),$$

because $S_n^{-1/2}h_n^{-K/4} = (S_n h_n^{K/2})^{-1/2}$. Condition (26) gives $L_n \to 0$.

The Lyapunov central limit theorem for triangular arrays with row-wise independent summands therefore applies to $(S_n\sigma_n^2)^{-1/2}\sum_s(G_{n,s} - \mathbb{E}G_{n,s})$. Multiplying by $\sigma_n h_n^{K/4} \to \tau$ converts the normalisation $\sqrt{S_n}\,\sigma_n$ into $\sqrt{R_n}$ and yields (27). ☐

Remark 4.10 (Why a fixed-M central limit theorem cannot be reused)**.** For fixed M the summands do not change with S, their variance is a constant, and the classical $\sqrt{S}$ central limit theorem applies. The array above is genuinely triangular: the law of $G_{n,s}$ changes with n through h_n, and its variance diverges at rate $h_n^{-K/2}$. The correct normalisation is $\sqrt{R_n}$, which coincides with $\sqrt{S_n}$ only when $K = 0$. Substituting a fixed-M central limit theorem and then letting M grow, as in Lemma 3 of Brandt and Santa-Clara (2002), is exactly the step that fails; the limiting variance stated there is the variance of a summand that itself depends on M.

Corollary 4.11 (CLT centred at the diffusion density)**.** *Suppose in addition that*

$$q_M(y \mid x) - p(y \mid x) = b(x, y)h + o(h). \tag{31}$$

If

$$Sh^{K/2} \to \infty, \qquad \sqrt{Sh^{K/2}}\,h \to 0, \tag{32}$$

then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}(0, \tau^2(x, y)). \tag{33}$$

In terms of M and S, the second condition in (32) *is*

$$\frac{\sqrt{S}}{M^{1+K/4}} \longrightarrow 0. \tag{34}$$

For $K = 4$, the two conditions together define the feasible regime

$$M^2 \ll S \ll M^4. \tag{35}$$

Proof. Decompose

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S} - p\} = \sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S} - q_M\} + \sqrt{Sh^{K/2}}\,\{q_M - p\}\,.$$

The first term converges to $\mathcal{N}(0, \tau^2)$ by Theorem 4.9. By (31) the second term equals

$$\sqrt{Sh^{K/2}}\,\{b(x, y)h + o(h)\}\,,$$

which vanishes precisely under the second condition in (32). Slutsky's theorem gives (33).

For the restatement, $\sqrt{Sh^{K/2}}\,h = \sqrt{S}\,h^{K/4+1} = \sqrt{S}/M^{1+K/4}$, since $h = 1/M$. This is (34), and squaring gives the equivalent form $S/M^{2+K/2} \to 0$.

For $K = 4$ the first condition in (32) reads $S/M^2 \to \infty$ and the second reads $S/M^4 \to 0$, which is (35). The regime is nonempty: $S = M^3$ satisfies both. □

Remark 4.12 (Bias negligibility is not variance consistency)**.** The two conditions in (32) do different work and pull in opposite directions. The first is $S/M^{K/2} \to \infty$: it makes the Monte Carlo error vanish and is what Theorem 4.9 needs. It is a *lower* bound on S, requiring $S \gg M^{K/2}$. The second is $S/M^{2+K/2} \to 0$, equivalently $\sqrt{S}/M^{1+K/4} \to 0$: it makes the Euler bias negligible *relative to the Monte Carlo standard deviation*, and because that standard deviation shrinks as S grows, it is an *upper* bound on S, requiring $S \ll M^{2+K/2}$. At $K = 4$ the pair is $M^2 \ll S \ll M^4$. Conflating the two is the source of the difficulty in Table 2: the published condition $\sqrt{S}/M \to 0$, that is $S \ll M^2$, has the form of the second requirement but at the wrong power, and it then contradicts the first when $K \geq 4$.

Remark 4.13 (Dependency structure)**.** The density-level results depend on one another as follows. Assumption 4.1 supports Theorem 4.2, which is used only through its cases $r = 2$ and $r = 3$. The case $r = 2$ gives the variance scale Corollary 4.6; the two cases together give the Lyapunov condition in Theorem 4.9. Corollary 4.11 adds the Euler bias expansion (31), which is an assumption imported from Bally and Talay (1996a) and Bally and Talay (1996b) rather than proved here, and combines it with Theorem 4.9. Proposition 5.1 uses Corollary 4.6 and (31) but no central limit theorem. In the Brownian benchmark, Proposition 3.1 replaces Theorem 4.2 with an exact computation and the bias expansion is unnecessary because the bias is zero, so Theorem 3.10 rests on no imported result at all. Theorem 3.3 is independent of every central limit theorem above; it uses only the exact Gaussian tail bound.

Remark 4.14 (The dimensional conflict)**.** The condition $\sqrt{S}/M \to 0$ suppresses an $O(M^{-1})$ Euler bias under an incorrectly fixed Monte Carlo scale. Once the variance inflation $M^{K/2}$ is included, the bias must be compared with the standard deviation $M^{K/4}/\sqrt{S}$, producing (34). In four dimensions, the lower consistency requirement and the upper bias requirement leave the nonempty interval (35). The published condition places S below the lower boundary of that interval.

5 Bias–variance trade-off and dimensional rate

Suppose the Euler density admits (31). Because (5) is unbiased for q_M,

$$\mathbb{E}\Big[(\widehat{q}_{M,S} - p)^2\Big] = (q_M - p)^2 + \mathrm{Var}(\widehat{q}_{M,S})$$
$$= b(x,y)^2 h^2 + \frac{\tau^2(x,y)}{Sh^{K/2}} + o\Big(h^2 + \frac{1}{Sh^{K/2}}\Big). \tag{36}$$

Proposition 5.1 (Pointwise optimal discretisation)**.** *If $b(x,y) \neq 0$, the leading terms in* (36) *are balanced by*

$$h \asymp S^{-2/(K+4)}, \qquad M \asymp S^{2/(K+4)}. \tag{37}$$

The resulting optimal pointwise mean squared error is

$$\mathrm{MSE}_{\mathrm{opt}} \asymp S^{-4/(K+4)}. \tag{38}$$

Proof. Differentiate the leading expression $Bh^2 + C/(Sh^{K/2})$ with respect to h or equate its two terms. This gives $h^{K/2+2} \asymp S^{-1}$ and hence (37). Substitution yields (38). □

The rate is the familiar structure of a kernel-density problem: reducing the bandwidth controls approximation bias but decreases the effective number of simulations. The dimension enters through the volume of the final Gaussian kernel.

Remark 5.2 (The optimal design is not a design at which the CLT centres at p)**.** The mean-squared-error optimum of Proposition 5.1 and the recentred central limit theorem of Corollary 4.11 pull in opposite directions, and the tension is worth naming rather than leaving for a reader to trip over. At $M \asymp S^{2/(K+4)}$ the two error sources are balanced by construction, so the bias-negligibility condition required for a limit centred at p, namely $\sqrt{S}/M^{1+K/4} \to 0$, fails exactly there. The consequence is the usual one in nonparametric estimation: at the mean-squared-error optimum the limiting distribution acquires a mean shift, and a confidence interval built from the centred limit needs undersmoothing, that is a smaller h than the optimal one, to remain valid. Neither statement is wrong; they answer different questions, one about point accuracy and one about coverage.

Some illustrative exponents are

Dimension K	M_{opt}	Optimal MSE
1	$S^{2/5}$	$S^{-4/5}$
2	$S^{1/3}$	$S^{-2/3}$
4	$S^{1/4}$	$S^{-1/2}$
8	$S^{1/6}$	$S^{-1/3}$

This does not mean that M should always be chosen by pointwise MSE. Likelihood estimation requires accuracy over many endpoints and parameter values, and higher-order or bridge-based methods can change both constants and rates. It does mean that increasing M while holding S fixed, or while letting S grow too slowly relative to dimension, can make the simulation approximation worse rather than better.

Table 2: Principal endpoint-density rate conditions.

Property	General dimension K	Four dimensions
Monte Carlo mean-square consistency	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Density CLT centred at q_M	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Euler bias negligible in corrected CLT	$\sqrt{S}/M^{1+K/4} \to 0$	$S/M^4 \to 0$
Condition in Brandt and Santa-Clara (2002)	$\sqrt{S}/M \to 0$	$S/M^2 \to 0$

6 From density error to log-likelihood error

For observations $Y_0, \ldots, Y_N$, the simulated log likelihood is a sum of terms

$$\log \widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta).$$

Even when $\widehat{q}_{M,S}$ is unbiased for q_M, its logarithm is not unbiased. The effective endpoint size $R_{M,S} = Sh^{K/2}$ controls this nonlinearity.

The next result is conditional. Its first two statements follow from Theorem 4.9, but the third holds only under an additional uniform-integrability and lower-tail condition, stated in the proposition and assumed rather than verified.

Proposition 6.1 (Log-density delta expansion)**.** *Fix an endpoint with $q_M \to p > 0$ and suppose the conditions of Theorem 4.9 hold. If $R_{M,S} = Sh^{K/2} \to \infty$, then*

$$\sqrt{R_{M,S}}\,\{\log \widehat{q}_{M,S} - \log q_M\} \Longrightarrow \mathcal{N}\left(0, \frac{\tau^2}{p^2}\right), \tag{39}$$

and

$$\log \widehat{q}_{M,S} - \log q_M = \frac{\widehat{q}_{M,S} - q_M}{q_M} - \frac{(\widehat{q}_{M,S} - q_M)^2}{2q_M^2} + o_{\mathbb{P}}(R_{M,S}^{-1}). \tag{40}$$

Whenever the second-order expansion is uniformly integrable, it follows that

$$\mathbb{E}[\log \widehat{q}_{M,S}] - \log q_M = -\frac{\tau^2}{2p^2 R_{M,S}} + o(R_{M,S}^{-1}). \tag{41}$$

Proof. The first statement is the delta method applied to Theorem 4.9. Since $(\widehat{q} - q_M)/q_M = O_{\mathbb{P}}(R^{-1/2})$ and q_M is bounded away from zero, Taylor expansion of $\log(1+u)$ gives (40). Taking expectations requires uniform integrability of the remainder and lower-tail control; under that additional condition, the first-order term has zero expectation and the second-order term gives (41). □

The leading nonlinear scale is

$$R_{M,S}^{-1} = \frac{M^{K/2}}{S}, \tag{42}$$

not uniformly $1/S$. The statement in Brandt and Santa-Clara (2002, p. 171) that the logarithmic transformation induces a bias of order $1/S$ is valid with M fixed. It is not a joint-M, S order.

Suppose simulations are independent across observations and the local constants are uniformly bounded. For the average simulated criterion

$$\widehat{Q}_{N,M,S}(\theta) = \frac{1}{N} \sum_{n=0}^{N-1} \log \widehat{q}_{n,M,S}(\theta),$$

the stochastic average of the first-order simulation error has standard deviation of order

$$(NR_{M,S})^{-1/2},$$

while the average nonlinear bias remains of order $R_{M,S}^{-1}$. The simulation bias does not average out across observations.

A full estimator theorem requires uniform control in θ of densities, scores, and Hessians. The following statement is an abstract *local* perturbation result. It is not a theorem about the simulated-likelihood estimator: it assumes both maximisers are localised near θ_0 and assumes the uniform score and Hessian bounds, none of which we verify for that estimator. It is included to separate what the endpoint calculations deliver from what an estimator theorem would additionally require. Every hypothesis below is probabilistic, so the conclusion is a statement in probability, not a deterministic implication.

Proposition 6.2 (Abstract local equivalence of exact and simulated maximisers)**.** *Let $U \subseteq \mathbf{\Theta}$ be a fixed open neighbourhood of θ_0, let Q_N be an exact average log likelihood and $\widehat{Q}_{N,M,S}$ a simulated criterion, both twice continuously differentiable on U. Let $\widehat{\theta}_N$ be an interior local maximiser of Q_N in U and $\widetilde{\theta}_{N,M,S}$ an interior local maximiser of $\widehat{Q}_{N,M,S}$ in U. Assume:*

(i) both maximisers are consistent: $\widehat{\theta}_N \to \theta_0$ and $\widetilde{\theta}_{N,M,S} \to \theta_0$, in probability;

(ii) with probability tending to one both first-order conditions hold, $\nabla Q_N(\widehat{\theta}_N) = 0$ and $\nabla \widehat{Q}_{N,M,S}(\widetilde{\theta}_{N,M,S}) = 0$;

(iii) the scores agree uniformly on U at the parametric rate,

$$\sup_{\theta \in U} \|\nabla \widehat{Q}_{N,M,S}(\theta) - \nabla Q_N(\theta)\| = o_{\mathbb{P}}(N^{-1/2});$$

(iv) the Hessians agree uniformly on U,

$$\sup_{\theta \in U} \|\nabla^2 \widehat{Q}_{N,M,S}(\theta) - \nabla^2 Q_N(\theta)\| = o_{\mathbb{P}}(1);$$

(v) the exact Hessian converges uniformly on U in probability to a continuous matrix function H, with $H(\theta_0)$ nonsingular;

(vi) with probability tending to one the line segment joining $\widehat{\theta}_N$ and $\widetilde{\theta}_{N,M,S}$ lies in U.

Then

$$\sqrt{N}(\widetilde{\theta}_{N,M,S} - \widehat{\theta}_N) \to 0 \quad \textit{in probability.}$$

Proof. Work on the event, of probability tending to one, on which (ii) and (vi) both hold. Write $\Delta_N = \widetilde{\theta}_{N,M,S} - \widehat{\theta}_N$. Since $\widehat{Q}_{N,M,S}$ is twice continuously differentiable on U and the segment lies in U, the fundamental theorem of calculus applied to $s \mapsto \nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N + s\Delta_N)$ gives

$$0 = \nabla \widehat{Q}_{N,M,S}(\widetilde{\theta}_{N,M,S}) = \nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N) + \bar{H}_N \Delta_N, \qquad \bar{H}_N = \int_0^1 \nabla^2 \widehat{Q}_{N,M,S}(\widehat{\theta}_N + s\Delta_N)\, ds.$$

The integral form is used rather than a coordinatewise mean-value theorem, which would supply a different intermediate point in each row and no single matrix.

For the right-hand side, the exact first-order condition $\nabla Q_N(\widehat{\theta}_N) = 0$ gives

$$\|\nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N)\| = \|\nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N) - \nabla Q_N(\widehat{\theta}_N)\| \le \sup_{\theta \in U} \|\nabla \widehat{Q}_{N,M,S}(\theta) - \nabla Q_N(\theta)\| = o_{\mathbb{P}}(N^{-1/2})$$

by (iii).

For the matrix, decompose

$$\|\bar{H}_N - H(\theta_0)\| \le \sup_{\theta\in U}\|\nabla^2\widehat{Q}_{N,M,S}(\theta) - \nabla^2 Q_N(\theta)\| + \sup_{\theta\in U}\|\nabla^2 Q_N(\theta) - H(\theta)\| + \sup_{0\le s\le 1}\|H(\widehat{\theta}_N + s\Delta_N) - H(\theta_0)\|.$$

The first term is $o_{\mathbb{P}}(1)$ by (iv) and the second by (v). For the third, (i) makes both endpoints of the segment converge to θ_0 in probability, so $\sup_{0\le s\le 1}\|\widehat{\theta}_N + s\Delta_N - \theta_0\| \to 0$ in probability; continuity of H at θ_0 then makes the term $o_{\mathbb{P}}(1)$. Hence $\bar{H}_N \to H(\theta_0)$ in probability. Because $H(\theta_0)$ is nonsingular and matrix inversion is continuous at a nonsingular point, with probability tending to one $\bar{H}_N$ is invertible and $\bar{H}_N^{-1} = O_{\mathbb{P}}(1)$.

Combining, $\Delta_N = -\bar{H}_N^{-1}\nabla\widehat{Q}_{N,M,S}(\widehat{\theta}_N)$, so

$$\sqrt{N}\,\Delta_N = -\bar{H}_N^{-1}\cdot\sqrt{N}\,\nabla\widehat{Q}_{N,M,S}(\widehat{\theta}_N) = O_{\mathbb{P}}(1)\cdot o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1). \qquad \square$$

Remark 6.3 (Where the localisation would have to come from)**.** Assumption (i) assumes consistency of the simulated maximiser rather than deriving it, and it is the hypothesis that does the most work: without it the segment need not shrink to θ_0, $\bar{H}_N$ need not approach a nonsingular limit, and the argument fails. It could be replaced by the weaker requirement $\mathbb{P}(\widetilde{\theta}_{N,M,S} \in U) \to 1$ if H were additionally assumed nonsingular throughout U, with smallest singular value bounded away from zero there. Either way the localisation must be supplied, and the usual routes are global uniform convergence of the simulated criterion to a limit criterion uniquely maximised at θ_0, or an argmax theorem. We prove neither here, and Section 7.7 explains why the transition-density counterexample does not by itself settle that question. Proposition 6.2 should therefore be read as a conditional abstract result, not as an asymptotic theorem for the endpoint simulated-likelihood estimator.

One of those requirements can be made exact rather than plausible, and it is the one that matters most. Hypothesis (iii) of Proposition 6.2 is about the *score*, not the density, and by Proposition H.1 the simulated density is a kernel estimate of bandwidth $\sqrt{h}$; differentiating a kernel estimate costs one further power of the bandwidth. For the Brownian benchmark the cost can be computed in closed form, and the answer is clean.

Proposition 6.4 (Exact score second moment)**.** *For K-dimensional Brownian motion at $x = y = 0$, let G_h be the endpoint summand* (8) *and let ∂_j denote differentiation in the j-th coordinate of the evaluation point. Then for every $h \in (0,1)$ and every j,*

$$\mathbb{E}\big[(\partial_j G_h)^2\big] = (1-h)(2\pi)^{-K}h^{-K/2-1}(2-h)^{-K/2-1} = \frac{1-h}{h(2-h)}\,\mathbb{E}\big[G_h^2\big]. \tag{43}$$

The ratio to the density second moment is therefore free of K, and as $h \downarrow 0$,

$$\mathbb{E}\big[(\partial_j G_h)^2\big] \;\sim\; (2\pi)^{-K}2^{-(K+2)/2}\,h^{-(K+2)/2}. \tag{44}$$

Proof. At $y = x = 0$ the preterminal draw is $Z = \sqrt{1-h}\,\xi$ with $\xi \sim \mathcal{N}(0, I_K)$, and $\partial_j G_h = (Z_j/h)G_h$. Hence

$$\mathbb{E}\big[(\partial_j G_h)^2\big] = h^{-2}\,\mathbb{E}\big[Z_j^2 G_h^2\big] = h^{-2}(1-h)(2\pi h)^{-K}\,\mathbb{E}\big[\xi_j^2 e^{-a\|\xi\|^2}\big], \qquad a = \frac{1-h}{h}.$$

Writing $c = a + \frac{1}{2} = (2-h)/(2h)$ and integrating, $\mathbb{E}[\xi_j^2 e^{-a\|\xi\|^2}] = (2c)^{-K/2-1} = (h/(2-h))^{K/2+1}$. Substituting gives (43); the second equality follows from $\mathbb{E}[G_h^2] = (2\pi)^{-K}h^{-K/2}(2-h)^{-K/2}$, which is Proposition 3.1 at $r = 2$ and $y = x$. Letting $h \downarrow 0$ gives (44). $\square$

The consequence is a *score-level* effective simulation size, one power of M more demanding than the density-level one:

$$R^{\text{score}}_{M,S} = S\,h^{(K+2)/2} = \frac{S}{M^{(K+2)/2}}, \qquad \text{against} \qquad R_{M,S} = \frac{S}{M^{K/2}}. \tag{45}$$

Corollary 6.5 (The published condition fails more decisively at score level)**.** *In $K = 4$, hypothesis (iii) of Proposition 6.2 requires $S/M^3 \to \infty$, whereas the rate condition of Brandt and Santa-Clara (2002), equivalent to $S/M^2 \to 0$, requires $S \ll M^2$. Since $M^3 \gg M^2$, no sequence satisfies both, and the gap is wider by a factor M than the density-level incompatibility of Corollary 3.8.*

This bears directly on Lemma 4 of Brandt and Santa-Clara (2002), which asserts $\ln \widehat{\mathcal{L}} - \ln \mathcal{L} = o(N/S^{1/2})$ under $S^{1/2}/M \to 0$ and derives it from Lemma 3, a statement about the density. The score is not the density, and Proposition 6.4 quantifies the difference exactly: at $K = 4$ the score's Monte Carlo variance is larger by a factor of order M. Whether the argmax conclusion survives is still open, for the reasons set out in Section H; what is not open is that the rate condition under which the published proof operates is insufficient for the score even to converge.

Under uniform analogues of the pointwise density expansion, the remaining requirements for first-order equivalence include

$$Sh^{(K+2)/2} \to \infty, \qquad \frac{\sqrt{N}}{Sh^{K/2}} \to 0, \qquad \sqrt{N}\,h \to 0, \tag{46}$$

corresponding respectively to vanishing score simulation noise, vanishing nonlinear simulation bias at the $N^{-1/2}$ scale, and negligible Euler bias. The first condition is now the score-level one of (45) rather than the density-level $Sh^{K/2} \to \infty$ that an earlier version of this display carried; Proposition 6.4 is what fixes the exponent. Therefore (46) is a design implication, not a claimed complete theorem for arbitrary diffusions.

The broader Monte Carlo likelihood literature treats joint data and simulation limits through empirical-process, importance-sampling, or exact-simulation arguments rather than by combining two pointwise limits; see, among others, Lee (1995), Sung and Geyer (2007), Beskos, Papaspiliopoulos, and Roberts (2009), and Miasojedow et al. (2016).

7 What fails in the published asymptotic argument

This section distinguishes statements refuted by the Brownian example from steps that are merely unproved under the stated assumptions. Two published statements are directly refuted: the joint transition-density consistency of Lemma 2 and the joint central-limit statement of Lemma 3, by Theorem 3.3 and Corollary 3.5 respectively. Everything else below is a gap in a proof, not a demonstrated falsehood. Remark 3.6 states the hierarchy compactly.

That the distinction is worth drawing carefully is confirmed by how the results were received. Stramer and Yan (2007b), proposing the modified Brownian bridge as an alternative sampler, write that consistency and asymptotic normality for it "is conjectured to be true", but that "a rigorous proof, which would extend the results of Pedersen (1995) and Brandt and Santa-Clara (2002b), is not straightforward and is worth further investigation". The published theorems were thus taken as an established base to be extended rather than as statements to be checked, which is the ordinary and reasonable default. The purpose of this section is to identify precisely which parts of that base do not carry the weight.

7.1 The published statements

Because everything below turns on the quantifier structure of the published lemmas, we reproduce the two that are refuted, together with the sentence that fixes their role. All are from Appendix A of Brandt and Santa-Clara (2002, pp. 202–203), in their notation, with $\widehat{q}_{M,S}$ their simulated density, q_M the Euler density, and p the diffusion density.

The appendix opens: "Lemmas 1–6 combine into a proof of Theorem 1 and Lemma 7 establishes Theorem 2."

> **Lemma 2.** *Given Assumptions 1 and 2, as $M \to \infty$ and $S \to \infty$,*
>
> $$\widehat{q}_{M,S}(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta) \to p(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta) \quad \textit{almost surely.}$$
>
> **Lemma 3.** *Given Assumptions 1 and 2, as $M \to \infty$ and $S \to \infty$, with $S^{1/2}/M \to 0$,*
>
> $$\begin{aligned} S^{1/2}[&\widehat{q}_{M,S}(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta) - p(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta)] \\ &\sim N\big(0, \mathrm{var}[\phi(Y_{t_{n+1}}; z_s + \mu(z_s)h, V(z_s)h)]\big). \end{aligned}$$

Three features of these statements are what the results below engage with, and each can be read off the display.

First, **Lemma 2 carries no rate condition**. Its hypotheses are Assumptions 1 and 2 and the requirement that both indices diverge. Nothing restricts how they diverge relative to one another, so any sequence with $M_n \to \infty$ and $S_n \to \infty$ falls under it, $M = S$ among them. Theorem 3.3 exhibits such a sequence along which the stated conclusion fails.

Second, **Lemma 3 does carry the rate condition** $S^{1/2}/M \to 0$, and the sequence $M = S = n$ satisfies it, since $S^{1/2}/M = n^{-1/2} \to 0$. Corollary 3.5 therefore refutes Lemma 3 under its own stated hypothesis rather than outside it.

Third, the **dependency is explicit in the published proofs**, so the consequences propagate. Lemma 2's proof establishes $\widehat{q}_{M,S} \to q_M$ as $S \to \infty$ and then writes "Use of Lemma 1 completes the proof". Lemma 4's proof ends "The last equality follows from Lemma 3". Lemma 5's proof expands the simulated log likelihood and invokes Lemma 4 for the two differences. With the appendix's own summary sentence, the chain is

$$\text{Lemma 1+Lemma 2} \longrightarrow \text{Lemma 3} \longrightarrow \text{Lemma 4} \longrightarrow \text{Lemma 5} \xrightarrow{\text{+ Lemma 6}} \text{Theorem 1},$$

with Lemma 7 establishing Theorem 2. Refuting Lemmas 2 and 3 therefore removes inputs that the published proofs of Theorems 1 and 2 use, which is the precise sense in which those proofs fail. It is not the sense in which their conclusions are false; see Section 7.7.

7.2 Lemma 2: sequential convergence is used as joint convergence

The proof of Lemma 2 in Brandt and Santa-Clara (2002) applies the strong law to the Monte Carlo average and then invokes convergence of the Euler density. For every fixed M,

$$\widehat{q}_{M,S} \to q_M \quad \text{as } S \to \infty,$$

and separately

$$q_M \to p \quad \text{as } M \to \infty.$$

This gives the iterated limit

$$\lim_{M\to\infty} \lim_{S\to\infty} \widehat{q}_{M,S} = p.$$

It does not give convergence along every sequence $(M_n, S_n) \to (\infty, \infty)$. The summand changes with M and its variance diverges as $M^{K/2}$. Theorem 3.3 directly refutes the stated joint conclusion.

Two features of Lemma 2 are worth isolating. It is stated as a joint limit in M and S with no rate restriction whatsoever, so the refutation does not depend on the separate discussion of $\sqrt{S}/M \to 0$ below; and it is stated under Assumptions 1 and 2 alone, which standard Brownian motion satisfies. The counterexample therefore meets the lemma on its own terms.

Because Lemmas 3 to 6 take Lemma 2 as an input, and Theorem 1 is stated as the summary of Lemmas 1 to 6, the published proofs of Theorems 1 and 2 fail. That is a statement about the proofs. It does not establish that the conclusions of those theorems are false, and the subsections that follow identify further steps that are independently unproved.

7.3 Lemma 3: the joint central-limit statement is directly false

The paper uses a $\sqrt{S}$ central limit theorem and requires $\sqrt{S}/M \to 0$ to suppress the Euler bias. Two objections apply, and they are of different strengths.

The weaker objection is that the statement is ill-posed. The variance of the triangular-array summand is not asymptotically constant; it is of order $M^{K/2}$, so the limiting variance written in Lemma 3, namely the variance of $\phi(Y_{t_{n+1}}; z_s + \mu h, Vh)$, is the variance of a quantity that itself depends on M and diverges. The correct standardisation is $\sqrt{S/M^{K/2}}$, and the Euler bias must then be compared with that scale, producing (34).

The stronger objection is that the statement is false, not merely ill-posed. Corollary 3.5 shows that along $M_n = S_n = n$, a sequence satisfying the stated rate condition, $\sqrt{S_n}(\widehat{q}_{n,n} - p) \to -\infty$ in probability. The sequence is not tight, so it converges in distribution to no limit at all, and no choice of limiting variance rescues the statement. This refutation is independent of the objection about ill-posedness and does not rely on identifying what the intended limiting variance was.

7.4 Lemma 4: an $O_{\mathbb{P}}$ term is treated as $o_{\mathbb{P}}$

Let $x_n = \widehat{q}_n - p_n$. A nondegenerate central limit theorem implies $x_n = O_{\mathbb{P}}(S^{-1/2})$ for fixed M, not $o_{\mathbb{P}}(S^{-1/2})$. The Taylor expansion

$$\log\left(1 + \frac{x_n}{p_n}\right) = \frac{x_n}{p_n} + O_{\mathbb{P}}(x_n^2)$$

retains a first-order stochastic term. In the joint limit, the correct order is $O_{\mathbb{P}}\{(Sh^{K/2})^{-1/2}\}$. Moreover, summing pointwise errors over N observations requires lower bounds or tail control for p_n, dependence control, and a triangular-array argument. These are not supplied.

7.5 Lemma 5: convergence of objective values does not control maximisers

An argmax result requires uniform convergence of the criterion and separation or local curvature around the maximiser. A bounded Hessian gives an upper quadratic bound. To infer that two maximisers are close from a small objective difference, one needs a lower quadratic bound or a negative Hessian whose smallest eigenvalue is bounded away from zero after normalisation. Pointwise convergence and identification alone are insufficient.

The structure of the argument makes this explicit. Summing the two second-order expansions leaves a difference of objective values on the left and a term proportional to $(\widehat{\theta}_N - \widehat{\theta}_{N,M,S})^2$ on the right, whose coefficient is an average of two Hessians. Assumption 4 of Brandt and Santa-Clara (2002) bounds that coefficient above. Dividing a small left-hand side by a coefficient that is only bounded above yields no bound at all on the squared distance; what is required is a bound below, that is, a uniformly negative definite Hessian in a neighbourhood. The conclusion drawn, $\widehat{\theta}_{N,M,S} - \widehat{\theta}_N = o(N^{1/2}/S^{1/4})$, also inherits whatever is wrong with the $o(N/S^{1/2})$ input from Lemma 4.

7.6 Lemmas 6 and 7: exact-MLE asymptotics require additional structure

Consistency of the exact diffusion MLE ordinarily requires a uniform law of large numbers, global or local identification, and stability assumptions such as stationarity and ergodicity when $N \to \infty$. A score expansion around θ_0 is not by itself a proof of consistency because the intermediate Hessian is evaluated between the estimator and θ_0.

Similarly, the martingale central limit theorem invoked from Hall and Heyde (1980) requires convergence of the realised conditional quadratic variation and a conditional Lindeberg condition. Defining Fisher information as an expectation does not make the realised conditional variance condition automatic. Boundedness of $\partial p/\partial\theta$ is also not a bound on the score $(\partial p/\partial\theta)/p$ when p can be small; strict positivity of p at each point is weaker than the uniform lower bound the argument needs.

These points do not imply that no correct exact-MLE theorem can be written. They imply that the short proof in Appendix A does not establish the stated result, even apart from the false transition-density joint limit.

7.7 What a counterexample to the argmax estimator would require

It is worth being explicit about the limit of the negative results above. Theorem 3.3 refutes a statement about the simulated transition density at one pair of points. It does not refute the conclusion of Theorem 1, which concerns the maximiser of the simulated log likelihood. Failure of a criterion pointwise does not imply failure of its argmax, and three obstacles separate the two.

First, the collapse is driven by a rare-event geometry that is not uniform over the state space or over the parameter. Theorem 3.3 fixes $x = y = 0$. An argmax argument needs control of $\widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta)$ simultaneously over N observed pairs and over a neighbourhood of θ_0, and the sets on which individual densities collapse need not align.

Second, the simulated criterion is evaluated with common random numbers across θ, as the published implementation requires for a smooth objective. The summands are therefore strongly dependent across parameter values. A criterion that is badly behaved at each fixed θ may still have a well-behaved maximiser if the errors move together; conversely, common random numbers can create a random limit criterion whose argmax is random. Which happens is a property of the coupling, not of the pointwise limit.

Third, the logarithm is unbounded below. Collapse towards zero at a single observation drives one log-density term to $-\infty$, which may push the maximiser away from the region where collapse occurs rather than towards a wrong value. The direction of that effect depends on how the collapse region varies with θ.

A complete argmax counterexample would need to exhibit a parametric family, an admissible sequence (M_n, S_n), and either a limit criterion whose maximiser differs from θ_0, or a demonstration that the simulated criterion fails to converge uniformly on a neighbourhood in a way that the maximiser inherits. We record in Section H the approaches we attempted and where each fails. Until such a construction exists, the correct statement is that the published proofs of Theorems 1 and 2 do not establish their conclusions, not that those conclusions are false.

8 Mathematical status of the exchange-rate application

The empirical application estimates a four-dimensional system consisting of two interest rates, the log exchange rate, and exchange-rate volatility. It also reconstructs a market-incompleteness state. This section states the findings that bear on one question: whether the generic theorem of Section 4 covers the model that was actually estimated. The supporting analysis is in Section B to Section G, and every claim below is proved there.

None of what follows is an asymptotic theorem. These are calculations about one published model, reported as diagnostics.

8.1 Summary of findings

(i) *The coefficients are outside the assumptions, but not equally so.* The interest rates follow square-root processes, which violate (A2) at the origin. At the reported estimates the Feller ratios range from 6.4 to 48.5, so the exact origin is unattainable, and at the benchmark states of Table 5 the one-step Euler negativity probabilities, though strictly positive, are of order 10^{-2667} or smaller at the long-run means. Unattainability does not mean the process cannot come arbitrarily close to zero, and without the observed rate path we do not establish a maximum over the empirical sample; what we claim is a gap in proof coverage together with negligible probabilities at the states displayed. The incompleteness state is different. Its Feller ratio is exactly 1/2 for every admissible parameter value, so no estimate can move it away from its boundary, and the worst-case one-step Euler negativity probability equals $\Phi(-\sqrt{1-2\alpha h})$ exactly, which decreases to $\Phi(-1) \approx 0.16$ as the step is refined rather than to zero. See Section B.

(ii) *The estimator does not simulate the state whose boundary is at issue.* The implementation eliminates the incompleteness state and simulates (r, r^*, e, v), recovering ϵ_t from a square root of the variance identity. Every Euler substep therefore requires that radicand to stay nonnegative at a *simulated* state, not merely at the observed data, and a violation makes the next substep's coefficients complex rather than merely inaccurate. See Section C.

(iii) *The two state representations are not interchangeable.* Taking volatility as primitive, which is what the estimator does, the variance identity reconstructs incompleteness uniquely. Taking incompleteness as primitive, recovering positive volatility requires solving a quadratic, and at the reported U.S.–U.K. loadings the inverse map is not globally single-valued: the quadratic has either no positive root or two distinct ones, coinciding only on a repeated-root boundary. A branch-selection condition would be needed to read the specification that way, and none is supplied. See Section D.

(iv) *Correlation validity holds wherever we can check it; only the global guarantee is missing.* The reported matrices are positive definite at every algebraically feasible reference point examined, with minimum eigenvalues from 0.68 down to 0.013 at rates a hundred times their long-run means, and no violation arose in 5,000 perturbation draws per system. We find no algebraically feasible point at which the matrix is indefinite, in any examined specification, that is models A, B and C across both currency systems, once the time-varying cases are evaluated by solving the variance identity for volatility rather than reading it from data. What remains is a gap in the specification rather than a defect in the estimates. The variance identity automatically bounds the two implied, state-dependent entries, so the entrywise constraint is not what secures those; the four free entries of R_t and the fifth primitive parameter ρ_{zz^*}, which enters through C_t rather than through R_t, still require their own restrictions, and the joint semidefiniteness condition is nowhere enforced. In the degenerate configuration the joint condition splits into a compatibility relation fixing ρ_{xy} and a separate magnitude condition on ρ_{ww^*}, ρ_{wy} and ρ_{w^*y}, and neither is imposed. A separate finding emerges from the same sweep: on 37,536 of the 56,784 baseline candidate grid points, the U.S.–U.K. model-C inverse identity admits no positive volatility solution, and no point on any grid admits exactly one. See Section E.

(v) *Identification is by a corner rule, not by the likelihood.* The minimum-incompleteness objective is affine in the selected scalar parameter, so over any compact feasible set its optimum is the minimum or maximum feasible value of that parameter. For the published affine constraints this is a one-dimensional linear programme and the optimum is a vertex; once positive-semidefinite requirements are added the feasible set need not be polyhedral, and the optimum is a boundary point rather than a vertex. Which constraint is active depends on the full feasible set. See Section F.

(vi) *The final estimator is not the one the theorem describes.* It combines simulated likelihood, auxiliary least squares, a corner selection rule, and inequality constraints that the paper reports as binding. See Section G.

Taken together these say that the estimator analysed in Section 4 and the estimator implemented in the application are different objects, quite apart from the rate conditions. The remainder of this section reports the one diagnostic that concerns the simulation design itself.

8.2 Finite-sample scaling diagnostics for the implementation

Before turning to the structure of the model it is worth recording how the implemented simulation design sits relative to the asymptotic ordering that the published theorems describe. Theorem 2 of Brandt and Santa-Clara (2002) imposes two conditions jointly,

$$\frac{\sqrt{S}}{M} \longrightarrow 0, \qquad \frac{N}{S^{1/4}} \longrightarrow 0, \tag{47}$$

the second entering through the $o(N^{1/2}/S^{1/4})$ bound of their Lemma 5. A sequence satisfying both must eventually have $N^4 \ll S \ll M^2$.

What follows is a set of diagnostics, not a test. A single finite triple (N, M, S) cannot satisfy or violate a limit; only a sequence can. The question a diagnostic can answer is narrower and still useful: are the finite ratios appearing in (47) small, so that the implementation resembles a point along such a sequence, and does the theorem supply any error bound at the implemented sizes? The answer to the second question is no in any case, because (47) is a statement about limits and carries no quantitative finite-sample content.

The estimation uses $N = 544$ weekly observations from January 1990 to May 2000, $M = 10$ Euler substeps, and $S = 5{,}000$ simulations together with 5,000 antithetic variates. Table 3 reports the ratios. At the implemented values $\sqrt{S}/M = 7.07$ and $N/S^{1/4} = 64.7$. Neither is small, so the implementation does not resemble a design drawn from a sequence along which (47) is operating. That is the whole of the claim; nothing here shows that the estimates are inaccurate.

The antithetic variates deserve a separate word, and a careful one. The paired innovation vectors ε and $-\varepsilon$ are deterministic opposites, but the endpoint density contributions are nonlinear functions of them, so their correlation is not -1 and need not even be negative. It depends on the model, the state, the parameter and the endpoint. For the Brownian benchmark it can be computed exactly, and the answer is not the favourable one.

Because S is fixed throughout Section 2 to Section 6 as the number of independent endpoint-density contributions, we do not reuse it here. For this subsection only, let

$$P = \text{number of independent antithetic pairs}, \quad n_{\text{eval}} = 2P = \text{number of density evaluations},$$

For pair i write G_i^+ and G_i^- for the two contributions and $A_i = \frac{1}{2}(G_i^+ + G_i^-)$ for the pair average, so the antithetic estimator is

$$\widehat{q}_{\text{ant}} = \frac{1}{P}\sum_{i=1}^{P} A_i = \frac{1}{2P}\sum_{i=1}^{P}(G_i^+ + G_i^-). \tag{48}$$

Proposition 8.1 (Antithetic pairing at a coincident endpoint)**.** *For K-dimensional Brownian motion started at x, let G^+ denote the endpoint summand* (8) *generated from a preterminal draw Z and let G^- denote its antithetic partner, generated from the reflected innovations. If $y = x$ then*

$$G^- = G^+ \qquad \textit{almost surely}, \tag{49}$$

so the pair is perfectly positively *correlated and provides no variance reduction whatsoever. More generally, if* $\text{Var}(G_i^\pm) = \sigma^2$ *and* $\text{Corr}(G_i^+, G_i^-) = \rho_A$*, then*

$$\text{Var}(A_i) = \frac{\sigma^2(1+\rho_A)}{2}, \qquad \text{Var}(\widehat{q}_{\text{ant}}) = \frac{\sigma^2(1+\rho_A)}{2P}. \tag{50}$$

At $\rho_A = +1$ this equals σ^2/P, the variance of only P independent evaluations, although $n_{\text{eval}} = 2P$ were computed. At $\rho_A = -1$ it is zero.

Definition 8.2 (Variance-equivalent evaluation count)**.** For an antithetic design of P pairs and *eval* $= 2P$ evaluations with pair correlation $\rho_A \in (-1, 1]$, the *variance-equivalent number of independent evaluations* is the number N_{eff} of independent draws whose average would have the same variance:

$$N_{\text{eff}}(\rho_A) = \frac{n_{\text{eval}}}{1+\rho_A} = \frac{2P}{1+\rho_A}. \tag{51}$$

Two ratios follow from (51), and they must not be confused, because they move in opposite directions:

$$\frac{n_{\text{eval}}}{N_{\text{eff}}} = 1 + \rho_A, \qquad \frac{N_{\text{eff}}}{P} = \frac{2}{1+\rho_A}. \tag{52}$$

The first says how far the raw evaluation count n_{eval} misstates the effective count; the second compares the effective count with the number of independent units. At $\rho_A = +1$ the first is 2 and the second is 1.

The interpretation in the four cases is as follows.

- $\rho_A = +1$: $N_{\text{eff}} = P = n_{\text{eval}}/2$. The *eval* density evaluations have the variance of only P independent evaluations, so counting *eval* as independent overstates the variance-equivalent count by a factor of two.
- $\rho_A = 0$: $N_{\text{eff}} = n_{\text{eval}}$. The antithetic evaluations perform exactly like *eval* independent ones.
- $-1 < \rho_A < 0$: $N_{\text{eff}} > n_{\text{eval}}$. Pairing delivers variance reduction beyond what *eval* independent evaluations would give, which is the case antithetic sampling is designed for.
- $\rho_A \to -1$: N_{eff} diverges, because the idealised pair average has variance tending to zero. This is a statement about a variance-equivalence convention, not a literal count of independent observations; no finite design produces infinitely many of those.

In short, n_{eval} overstates the effective count by the factor $1 + \rho_A$ when $\rho_A > 0$, equals it when $\rho_A = 0$, and understates it when $\rho_A < 0$.

Proof. Reflecting every innovation maps the preterminal state $Z = x + \sqrt{1-h}\,\xi$ to $Z^- = x - \sqrt{1-h}\,\xi$, so $Z^- - x = -(Z - x)$. The summand depends on Z only through $\|y - Z\|^2$, and at $y = x$ this is $\|x - Z\|^2 = \|Z - x\|^2$, which is invariant under $Z - x \mapsto -(Z - x)$. Hence $G^- = G^+$ pointwise, giving (49) and $\rho_A = +1$. For (50), $\mathrm{Var}(G_i^+ + G_i^-) = 2\sigma^2(1 + \rho_A)$, so $\mathrm{Var}(A_i) = \frac{1}{4} \cdot 2\sigma^2(1 + \rho_A) = \sigma^2(1 + \rho_A)/2$; the pairs are independent across i, so the average of P of them divides this by P. □

Proposition 8.1 is the case $\rho_A = +1$, where the n_{eval} evaluations buy exactly what P would have bought. Numerically, at $h = 1/10$ and $K = 4$, ρ_A is $+1$ at $\|y - x\| = 0$, $+0.063$ at $\|y - x\| = 0.5$ and -0.012 at $\|y - x\| = 1.5$; at $K = 1$ and $\|y - x\| = 0.5$ it is -0.479. These values are illustrative of the mechanism at particular endpoints, not summaries of the estimator's behaviour over a sample.

That last point needs emphasis, because it is what stops N_{eff} from being a single number for a given design. The pair correlation ρ_A depends on the endpoint y, the initial state x, the state dimension K, the discretisation level h, the diffusion coefficients and the parameter value. Every transition in the likelihood therefore carries its own ρ_A, and a single variance-equivalent count for a 544-term log likelihood does not exist without evaluating the pair covariance transition by transition. What can be said without that evaluation is that $\rho_A \le 1$ always, so $N_{\mathrm{eff}} \ge E/2 = P$, and the coincident-endpoint case attains the bound.

Accordingly we report two distinct scaling diagnostics rather than one effective size:

$$R_{\mathrm{pair}} = \frac{P}{M^{K/2}}, \qquad R_{\mathrm{var}} = \frac{N_{\mathrm{eff}}}{M^{K/2}} = \frac{2P}{(1 + \rho_A)M^{K/2}}. \tag{53}$$

R_{pair} counts independent units and requires no knowledge of ρ_A, and it is the conservative diagnostic in the sense that $R_{\mathrm{var}} \ge R_{\mathrm{pair}}$ always. Lemma 8.3 shows that independent pair averages preserve the dimensional consistency scale $P/M^{K/2}$; it does not carry the theory of Section 4 across wholesale. A complete antithetic central limit theorem would additionally require the limiting joint second and third moments of each pair, none of which the lemma supplies, as Remark 8.5 records. R_{var} is the variance-equivalent diagnostic and is known only once ρ_A is. Neither quantity is an exact variance equivalence for the antithetic design taken as a whole: Definition 2.1 is stated for independent contributions, so R_{pair} is its literal instance under the substitution $S \mapsto P$, and R_{var} is a per-transition quantity that varies with ρ_A.

For the reported design, $P = 5{,}000$ independent pairs and $n_{\mathrm{eval}} = 10{,}000$ endpoint-density evaluations. Table 3 reports both, with R_{pair} and with R_{var} evaluated at two reference values of ρ_A rather than at a single assumed one.

Finally, none of this touches the dimensional exponent. Pairing changes the variance constant and can change it in either direction, but the $h^{-K/2}$ growth of the second moment survives it.

Lemma 8.3 (Antithetic pairing preserves the endpoint moment exponent)**.** *Let G_h^+ and G_h^- be nonnegative with identical marginal laws and let $A_h = \frac{1}{2}(G_h^+ + G_h^-)$, with no assumption on their dependence. Then*

$$\tfrac{1}{2}\,\mathbb{E}[G_h^2] \;\le\; \mathbb{E}[A_h^2] \;\le\; \mathbb{E}[G_h^2], \tag{54}$$

where G_h denotes the common marginal. Consequently, if $\mathbb{E}[G_h^2] \asymp h^{-K/2}$ then $\mathbb{E}[A_h^2] \asymp h^{-K/2}$; and if in addition $\mathbb{E}[A_h] = O(1)$, then $\mathrm{Var}(A_h) \asymp h^{-K/2}$.

Proof. Write $a = G_h^+$ and $b = G_h^-$, both nonnegative. Then $4A_h^2 = (a + b)^2 = a^2 + b^2 + 2ab$. Since $ab \ge 0$, we get $4A_h^2 \ge a^2 + b^2$; and since $2ab \le a^2 + b^2$, we get $4A_h^2 \le 2(a^2 + b^2)$. Hence

$$\frac{a^2 + b^2}{4} \;\le\; A_h^2 \;\le\; \frac{a^2 + b^2}{2},$$

where nonnegativity is used only for the lower bound. Taking expectations and using $\mathbb{E}[a^2] = \mathbb{E}[b^2] = \mathbb{E}[G_h^2]$ gives (54). Both bounds are constants free of h and of the dependence structure, so the two second moments have the same order, and $\mathbb{E}[A_h^2] \asymp h^{-K/2}$ follows from $\mathbb{E}[G_h^2] \asymp h^{-K/2}$.

For the variance statement, $\mathrm{Var}(A_h) = \mathbb{E}[A_h^2] - \mathbb{E}[A_h]^2$. The subtracted term is $O(1)$ by hypothesis while $\mathbb{E}[A_h^2]$ diverges at rate $h^{-K/2}$, so it is negligible and $\mathrm{Var}(A_h) \asymp h^{-K/2}$. The hypothesis $\mathbb{E}[A_h] = O(1)$ holds in the present setting because $\mathbb{E}[A_h] = \mathbb{E}[G_h]$, which converges to the transition density. □

So the dimensional law of Theorem 4.2 is untouched by the coupling: only the constant in front of $h^{-K/2}$ responds to ρ_A, and Lemma 8.3 bounds that response within a factor of two of the independent case.

Nonnegativity says something sharper about ρ_A itself, and it is not encouraging for the design.

Proposition 8.4 (Antithetic pairing cannot help at leading order)**.** *Suppose $G_h^+, G_h^- \geq 0$ have the common marginal law of G_h, with $\mathbb{E}[G_h] = O(1)$ and $\mathrm{Var}(G_h) \asymp h^{-K/2}$. Then*

$$\rho_A \;\geq\; -\frac{\mathbb{E}[G_h]^2}{\mathrm{Var}(G_h)} \;=\; O(h^{K/2}), \tag{55}$$

so $\liminf_{h\downarrow 0} \rho_A \geq 0$. Consequently $N_{\mathrm{eff}} \leq n_{\mathrm{eval}}\,(1 + o(1))$: relative to n_{eval} independent evaluations, antithetic pairing cannot reduce *the leading rare-event variance constant as the mesh is refined.*

Proof. Since both variables are nonnegative, $\mathbb{E}[G_h^+ G_h^-] \geq 0$, so

$$\mathrm{Cov}(G_h^+, G_h^-) = \mathbb{E}[G_h^+ G_h^-] - \mathbb{E}[G_h]^2 \;\geq\; -\mathbb{E}[G_h]^2.$$

Dividing by $\mathrm{Var}(G_h)$ gives (55). The numerator is $O(1)$ by hypothesis and the denominator diverges at rate $h^{-K/2}$, so the bound tends to zero and $\liminf \rho_A \geq 0$. The statement about $N_{\mathrm{eff}} = n_{\mathrm{eval}}/(1 + \rho_A)$ follows immediately. □

The mechanism is worth naming. A negative ρ_A requires the pair to be jointly small when one member is large, but both members are nonnegative densities whose variance is driven by a rare event in which one of them is enormous; on that event the product is nonnegative, and the covariance cannot be more negative than $-\mathbb{E}[G_h]^2$, which does not grow.

The direction of the conclusion should be read carefully, because Proposition 8.4 is one-sided. It says antithetic pairing cannot reduce the leading $h^{-K/2}$ variance constant below the independent-evaluation constant. It does not say the constant is unchanged: if $\rho_A \to c > 0$, the constant is multiplied by $1 + c$, and at a coincident endpoint $\rho_A = 1$ doubles it relative to n_{eval} independent evaluations. So the possibilities the proposition leaves open are that pairing leaves the leading constant alone or makes it worse, never that it improves it. Antithetic sampling can still reduce variance at a *fixed* h and a favourable endpoint, as the value $\rho_A = -0.479$ above shows, and that is a real finite-sample gain. In the theoretical sections S continues to mean the number of independent contributions.

Remark 8.5 (What Lemma 8.3 does not give)**.** The lemma fixes the order of the second moment and, with $\mathbb{E}[A_h] = O(1)$, of the variance. It does not give the exact local variance constant, the third centred-moment constant, the Lyapunov ratio for a pair-level triangular array, or a central limit theorem at either the pair level or the likelihood level. Establishing an antithetic analogue of Theorem 4.9 would require the limiting joint second and third moments of (G_h^+, G_h^-), which depend on the coupling and are not supplied by a sandwich bound.

Table 3: Finite-sample scaling ratios for the implemented design and three variations, at $N = 544$ and $K = 4$. Here $P = S$ is the number of independent units, equal to the number of antithetic pairs, and $n_{\text{eval}} = 2P$ the number of endpoint-density evaluations. $R_{\text{pair}} = P/M^2$ counts independent units and needs no knowledge of ρ_A; it is Definition 2.1 under the substitution $S \mapsto P$, and is a conservative diagnostic rather than an exact variance equivalence. $R_{\text{var}} = N_{\text{eff}}/M^2$ with $N_{\text{eff}} = n_{\text{eval}}/(1 + \rho_A)$ is the variance-equivalent diagnostic, and it is reported at *two reference values* of ρ_A because no single value applies to a 544-term likelihood: $\rho_A = 1$ is the worst case, attained at a coincident endpoint, and $\rho_A = 0$ the neutral case in which pairing neither helps nor hurts. No separate N_{eff} column is needed at these two values, because $N_{\text{eff}}(1) = P$ and $N_{\text{eff}}(0) = n_{\text{eval}}$: the variance-equivalent count is read off the P and n_{eval} columns already shown. Any $\rho_A < 0$ gives R_{var} larger than both columns shown, and $R_{\text{var}} \geq R_{\text{pair}}$ always. These are diagnostics: the columns record how the implementation is placed, not whether any limit holds.

					$R_{\text{var}} = N_{\text{eff}}/M^2$			
Scenario	M	$P = S$	$n_{\text{eval}} = 2P$	R_{pair}	$\rho_A = 1$	$\rho_A = 0$	$\sqrt{P}/M$	$N/P^{1/4}$
Implemented	10	5,000	10,000	50.0	50.0	100.0	7.07	64.7
Double M and P	20	10,000	20,000	25.0	25.0	50.0	5.00	54.4
Double M, P fixed	20	5,000	10,000	12.5	12.5	25.0	3.54	64.7
Double M, quadruple P	20	20,000	40,000	50.0	50.0	100.0	7.07	45.7

The dimensional diagnostic of Definition 2.1 gives the complementary reading. With $K = 4$, $M = 10$ and $P = 5{,}000$ independent pairs,

$$R_{\text{pair}} = \frac{P}{M^{K/2}} = \frac{5{,}000}{100} = 50. \tag{56}$$

Each transition density in a 544-term log likelihood is therefore resolved by an independent-pair count of order 10^2, not the 10^4 that the raw evaluation total suggests. The variance-equivalent figure R_{var} lies between 50 and 100 wherever $\rho_A \geq 0$ and above 100 wherever $\rho_A < 0$, so the order of magnitude is 10^2 on any reading short of near-perfect negative pairing. Again this is a scaling statement and not a verdict: a count of 50 may well be adequate at these parameter values, and the exact Brownian formulas of Section 3 would be the way to check that for a given target density.

The last three rows of Table 3 record how $R_{M,S}$ responds to refinement, because the arithmetic is easy to get wrong. In four dimensions $R_{M,S} = S/M^2$, so

$$R_{2M,2S} = \frac{2S}{4M^2} = \frac{R_{M,S}}{2}, \qquad R_{2M,S} = \frac{R_{M,S}}{4}, \qquad R_{2M,4S} = R_{M,S}. \tag{57}$$

The paper reports a final round of optimisation in which both the simulation count and M are doubled. In four dimensions that halves the independent-pair scaling diagnostic $R_{\text{pair}} = P/M^2$ rather than increasing it. The corresponding change in the variance-equivalent diagnostic R_{var} cannot be determined without the antithetic pair correlation at each transition, since ρ_A may itself vary with M; what can be said is that $R_{\text{var}} \geq R_{\text{pair}}$ throughout. The reported stability of the estimates under this refinement is therefore not evidence that the simulation is converging in the endpoint dimension; it is a comparison between two designs, the second of which has half the independent-pair scaling diagnostic of the first. Its variance-equivalent size may change by a different factor, because the pair correlation may change with M. Holding R_{pair} fixed while doubling M requires quadrupling the number of pairs, which is the last row of the table.

9 Numerical evidence

All numerical results in this section are generated by the accompanying Python script. They use exact Gaussian benchmarks where possible and are intended to illustrate the theorems rather than replace them.

9.1 Second-moment scaling

Figure 1 plots

$$h^{K/2}\mathbb{E}[G_h^2]$$

divided by its limiting value for $K \in \{1, 2, 4, 8\}$. The exact formula (11) converges to the constant predicted by Theorem 4.2. The unscaled second moment grows as $M^{K/2}$.

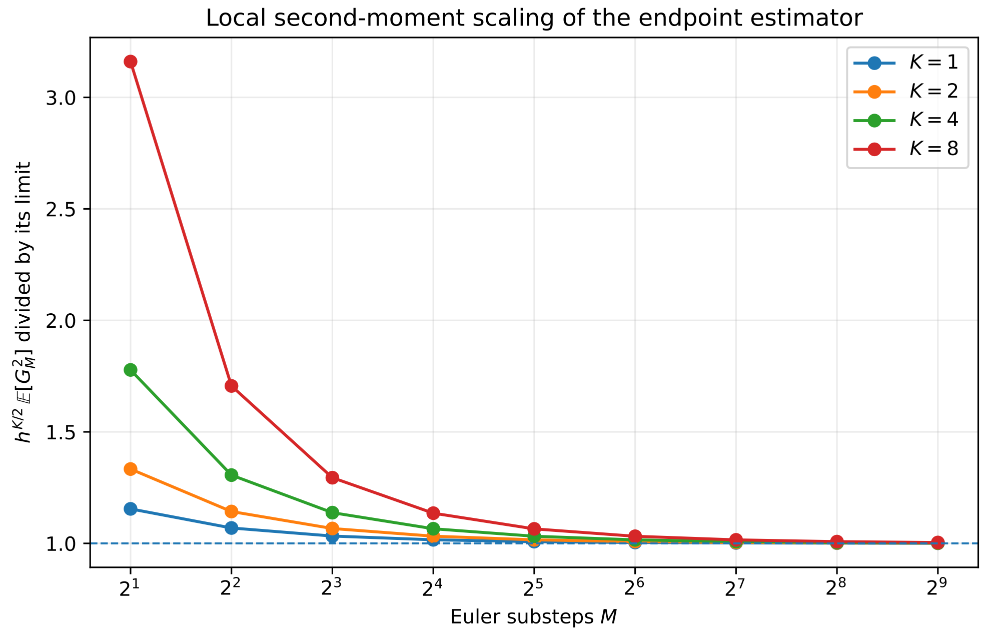

Figure 1: Exact local second-moment scaling for standard Brownian motion. The plotted quantity converges to one after division by its theoretical limit.

9.2 Three joint asymptotic paths

For $K = 4$, Figure 2 compares the exact relative root mean squared error under $S = M$, $S = M^2$, and $S = M^3$. The first path satisfies the published condition but has increasing RMSE. The second keeps the effective endpoint size constant and does not become consistent. The third satisfies $S/M^2 \to \infty$ and has decreasing RMSE.

9.3 Typical estimates collapse while the mean remains correct

Figure 3 simulates the counterexample path $M = S$. The sample median falls rapidly towards zero, and by $M = 512$ more than 95% of estimates are below half the true density. The sample mean remains near one after normalisation because a small number of very large realisations carry the expectation.

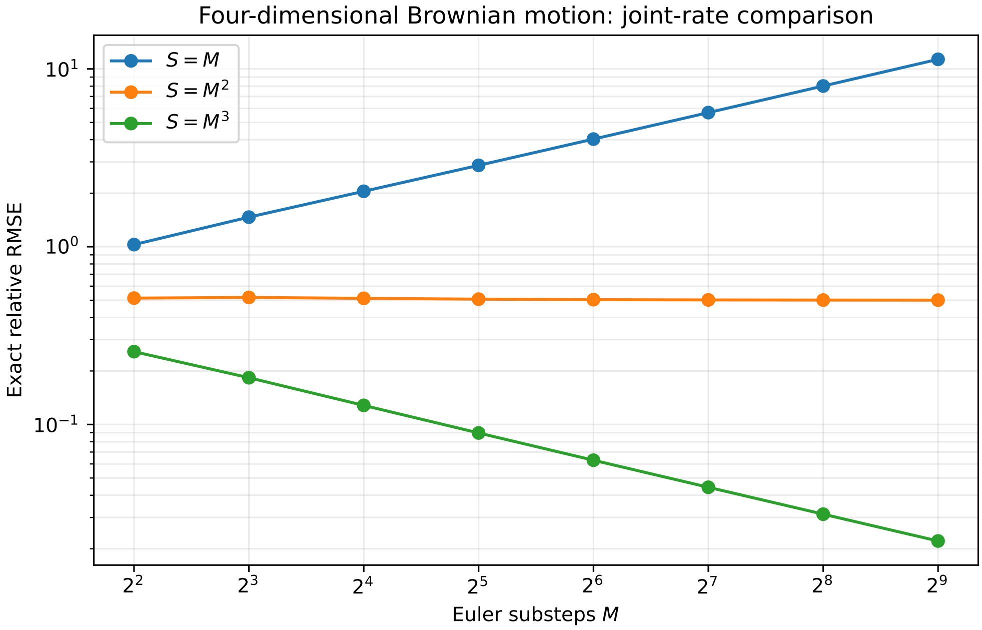

Figure 2: Exact relative RMSE at the origin for four-dimensional Brownian motion.

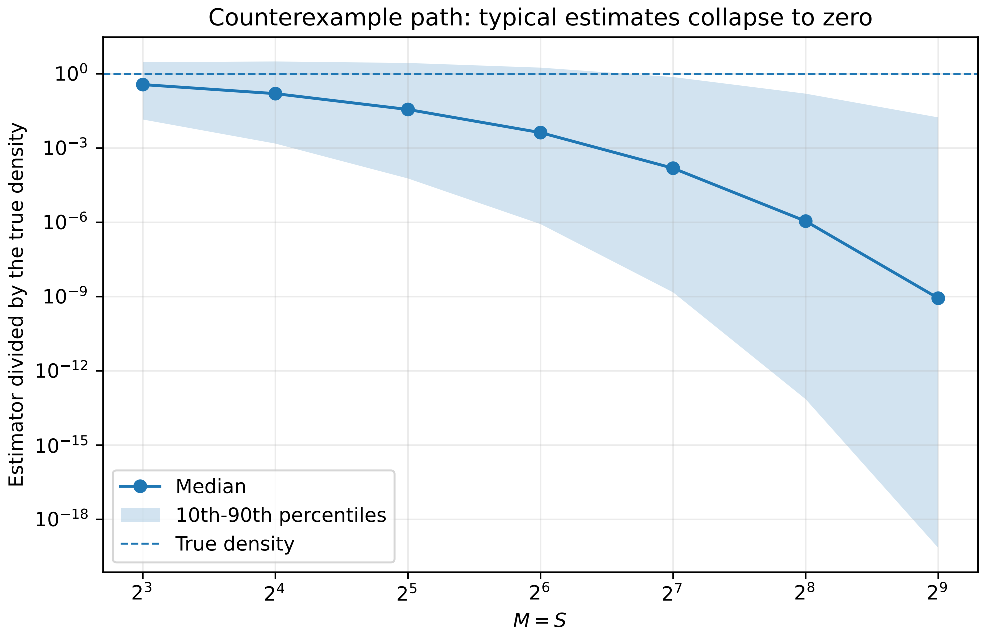

Figure 3: Monte Carlo distribution along $M = S$ for four-dimensional Brownian motion. Values are divided by the true density. The dashed line is one.

Table 4: Simulation of the four-dimensional Brownian counterexample path $M = S$. The estimator is divided by the true density. Replication counts follow $\min\{20{,}000, \max(1{,}500, 5 \times 10^6/M)\}$, a fixed total draw budget, which is why they fall from 20,000 to 9,765 at the largest steps. The parenthesised figure is the Monte Carlo standard error of the mean, and it is the point of the column rather than a decoration: the mean is carried by rare large draws, so at $M = 512$ the entry reads 0.998 ± 0.121 and is consistent with unbiasedness without being evidence of precision. The mean stays at one because Proposition 3.1 makes the estimator exactly unbiased at every M; the median and lower quantiles are where the collapse of Theorem 3.3 is visible.

M	Replications	Mean	(s.e.)	Median	10th pct.	90th pct.	$\mathbb{P}(\widehat{q} < 0.5p)$
8	20000	1.001	(0.010)	0.3674	1.427×10^{-2}	2.940	0.560
16	20000	1.009	(0.015)	0.1582	1.520×10^{-3}	3.161	0.665
32	20000	1.027	(0.021)	0.03628	5.925×10^{-5}	2.746	0.754
64	20000	1.002	(0.028)	0.004223	8.507×10^{-7}	1.777	0.827
128	20000	1.042	(0.042)	1.537×10^{-4}	1.487×10^{-9}	0.7409	0.886
256	19531	0.941	(0.054)	1.118×10^{-6}	7.168×10^{-14}	0.1579	0.929
512	9765	0.998	(0.121)	8.604×10^{-10}	7.076×10^{-20}	0.01738	0.955

9.4 A nonzero drift does not remove the dimensional law

Consider the diagonal Ornstein–Uhlenbeck process

$$dY_t = -Y_t dt + dW_t$$

in four dimensions. Its exact transition density is known, and the preterminal Euler state is Gaussian, so the second moment of the endpoint estimator can again be computed exactly. Figure 4 shows convergence to the local constant of Theorem 4.2. The phenomenon is therefore not specific to driftless Brownian motion.

One point of bookkeeping should be stated rather than glossed. The drift $\mu(y) = -y$ is *not* bounded, so this process does not satisfy (A1) and is not covered by Theorem 4.2 as that theorem is stated. The agreement in Figure 4 is therefore evidence that the conclusion extends beyond the hypotheses under which it is proved, not a verification of the theorem on one of its own instances. It is the localised form sketched in Remark 4.4, which needs ellipticity and drift boundedness only on a ball around y together with a Gaussian upper bound on the preterminal density, that would cover an Ornstein–Uhlenbeck process; we have not written that version out, and until it is written the figure should be read as an illustration rather than as a confirmation.

9.5 Euler boundary violations in the incompleteness process

Figure 5 evaluates (65) at the reported estimates $(\alpha, \beta) = (0.320, 0.088)$ and $(0.338, 0.101)$, using the step sizes $h = 1/520$ and $h = 1/1040$ that correspond to weekly data with $M = 10$ and $M = 20$, with a coarser value shown only as a labelled illustration. The horizontal scale is logarithmic so that the region near the boundary is visible; on a linear scale the entire phenomenon is compressed into a sliver next to the origin.

The figure makes Proposition C.1 visible: refining the step translates the curve to the left and lowers its peak only slightly, towards the strictly positive limit $\Phi(-1)$. As explained in Section C, this calculation applies to the incompleteness specification (61) treated as a state in its own right. The implemented chain simulates (r, r^*, e, v) and recovers ϵ_t from (64), so the boundary that must not be crossed is the complete-markets surface rather than $\{H = 0\}$;

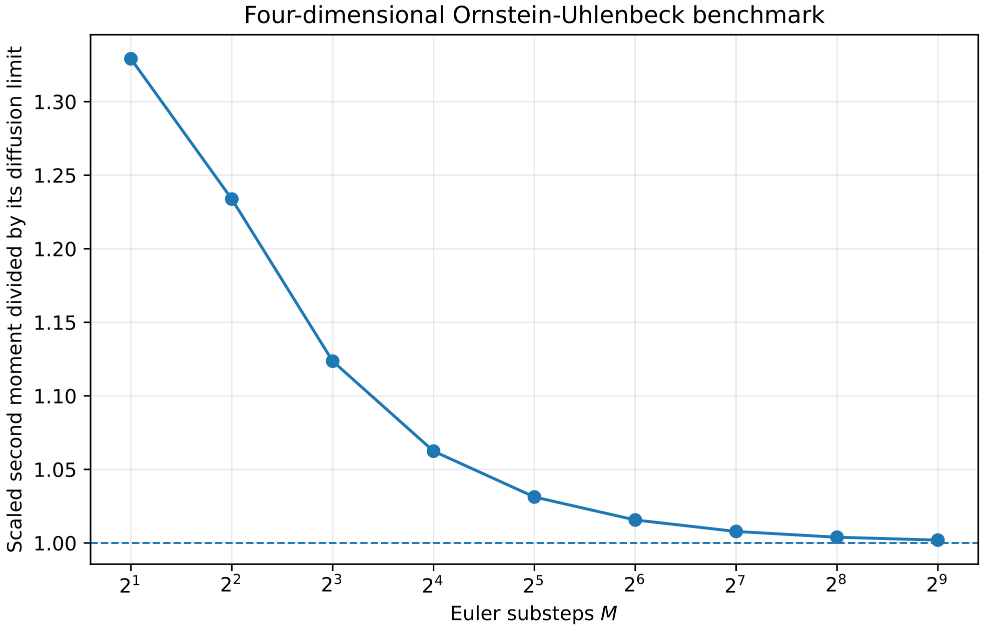

Figure 4: Scaled second moment for a four-dimensional Ornstein–Uhlenbeck diffusion.

the figure is an exact illustration of the mechanism rather than of the implemented chain. In particular, the curves give one-step probabilities conditional on the current state; they are not occupation frequencies, and we make no claim about how often either chain visits the region where they peak. The contrast with the interest rates in Table 6 is the substantive content: for r and r^* the Feller ratios exceed one by an order of magnitude, so the exact origin is unattainable and the displayed benchmark-state Euler probabilities in Table 5 are negligible, whereas for the incompleteness state the ratio is exactly $1/2$ for all parameter values and zero is attainable. Unattainable is not the same as unapproachable, and we do not reconstruct the observed rate path, so no sample-wide bound is claimed for the interest rates either.

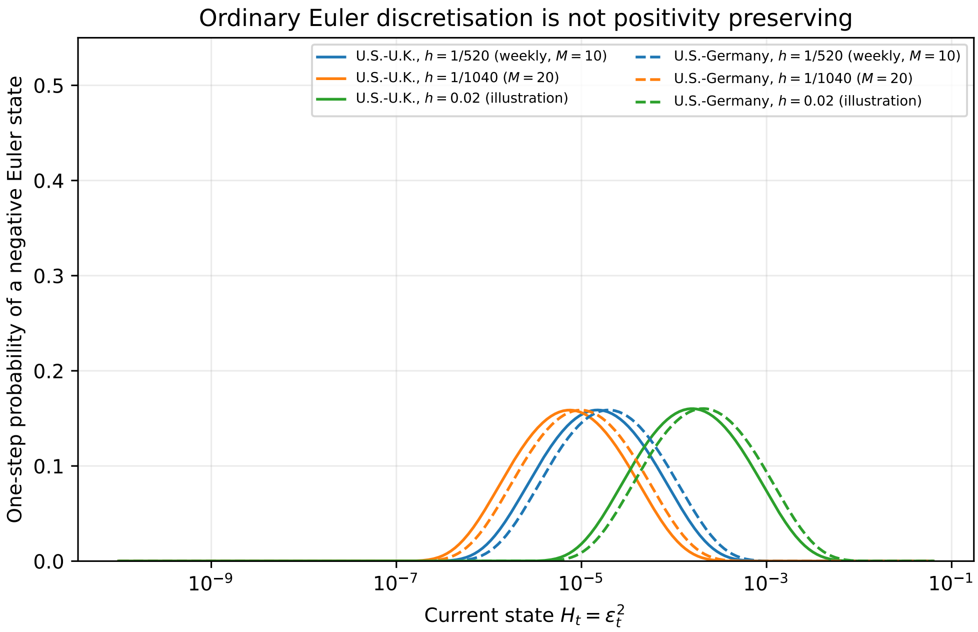

Figure 5: One-step probability that ordinary Euler discretisation of the incompleteness specification $H_t = \epsilon_t^2$ in (61) produces a negative state, at the estimates in Table 3 of Brandt and Santa-Clara (2002). Solid lines are the U.S.–U.K. estimates and dashed lines the U.S.–Germany estimates. The application step sizes $h = 1/520$ and $h = 1/1040$ are shown together with a coarser illustrative value. Halving h shifts each curve left and lowers its peak slightly, the maximum decreasing monotonically to $\Phi(-1) \approx 0.1587$ from above rather than to zero; see Proposition C.1. Three points about scope. The figure concerns *direct* Euler discretisation of H; it is not a simulation of the implemented transformed chain, which substitutes ϵ_t out and simulates (r, r^*, e, v); and it illustrates the boundary mechanism only. Each curve is a one-step probability conditional on the current state, so no point on it is a frequency with which any chain visits that state. See Section C.

10 Remedies and alternative constructions

The endpoint variance problem is not an unavoidable property of likelihood inference for diffusions. It is a consequence of drawing preterminal states from the unconditioned forward Euler law and asking a shrinking final Gaussian kernel to connect them to a fixed observed endpoint.

10.1 Bridge and importance proposals

A bridge proposal guides simulated paths towards the observed endpoint. In importance-sampling language, it increases the probability of drawing paths in the region that contributes to the transition-density integral and corrects the proposal through a likelihood ratio. This is the direction taken by the acceleration methods studied by Durham and Gallant (2002), the path-augmentation schemes of Elerian et al. (2001), conditioned-diffusion methods such as Delyon and Hu (2006), the regularised proposal of Lindström (2012), and later guided proposals for multidimensional bridges (Schauer et al., 2017; Meulen and Schauer, 2017). The nonparametric simulation approach of Nicolau (2002) replaces the endpoint Gaussian kernel by a density estimate, which changes the bandwidth problem rather than removing it.

A useful proposal should be assessed through the moments of its importance weights, and this is the respect in which bridge proposals differ in kind rather than in degree. Stramer and Yan (2007a) show that the modified Brownian bridge has importance-weight variance bounded uniformly in M, which is exactly what fails here: Corollary 4.6 gives $\mathrm{Var}(G_h) \asymp h^{-K/2}$ for the endpoint summand. Their numerical comparison makes the contrast visible, the endpoint estimator's error increasing with M at fixed N while the bridge's does not, and their accuracy for the bridge is dimension-free where the endpoint rate is not. Conditioning on the endpoint therefore addresses the rare-event geometry identified here at its source, rather than merely reducing a constant.

10.2 Exact or discretisation-free likelihood estimators

For restricted classes of diffusions, exact simulation and unbiased transition-density estimators remove Euler bias entirely. Beskos, Papaspiliopoulos, Roberts, and Fearnhead (2006) and Beskos, Papaspiliopoulos, and Roberts (2009) develop likelihood-based methods of this kind. Their applicability is model dependent, but they illustrate the correct theoretical structure: unbiasedness, continuity in parameters, finite-moment conditions, and joint control of data and Monte Carlo sample sizes are stated explicitly.

10.3 Analytical density expansions

Closed-form expansions avoid endpoint Monte Carlo variance entirely but introduce an approximation-order problem of their own. For the four-dimensional system at issue here the relevant reference is the multivariate expansion of Aït-Sahalia (2008) rather than the univariate one of Aït-Sahalia (2002): it constructs the expansion for a multivariate diffusion after a reducibility transformation, and it is the natural competitor to a simulated likelihood at $K = 4$. Its applicability turns on whether the model can be transformed to satisfy the required regularity conditions, which for square-root states with a Feller ratio of one half is not automatic. Approximate martingale estimating functions, as in Kessler (1997), are a further discretisation-bias-free alternative that avoids the transition density altogether at some cost in efficiency.

10.4 Positivity-preserving discretisation

For square-root components, ordinary Euler should be replaced by an exact transition where available or by a specified positivity-preserving approximation. The choice is part of the estimator, not an implementation detail, because different truncation and reflection rules define different transition densities.

10.5 Practical design rule

If the unconditioned endpoint estimator is nevertheless used, the practical content of this paper reduces to four points, which we gather here because they are otherwise distributed across Section 8 and the results above.

One: monitor the right quantity. The diagnostic is not S but the effective endpoint size

$$R_{M,S} = \frac{S}{M^{K/2}},$$

and for parameter estimation rather than density evaluation it is the score-level version $S/M^{(K+2)/2}$ of (45), which is more demanding by one power of M. Increasing M without increasing these is not an accuracy improvement.

Two: refining both at once makes things worse, not better. This is the point most likely to be missed in practice, because refining M and S together looks like unambiguous progress. In $K = 4$,

$$R_{2M,2S} = \frac{2S}{(2M)^2} = \frac{R_{M,S}}{2}, \tag{58}$$

so doubling the substeps and the simulation count halves the effective size. Holding $R_{M,S}$ fixed while doubling M requires *quadrupling* S; and in K dimensions, holding it fixed requires $S \propto M^{K/2}$. Stability of estimates under a joint refinement is therefore not evidence of convergence, since the refinement has moved the design in the unfavourable direction.

Three: know where the implemented design sits. At the values used by Brandt and Santa-Clara (2002), $K = 4$, $M = 10$ and 5,000 independent simulation units, $R_{M,S} = 50$. Each transition density in a 544-term log likelihood is resolved by an effective sample of order 10^2, not 10^4. That may be adequate; the point is that it is the number to check, and it is not the one the design appears to supply.

Four: prefer a proposal that uses the endpoint. The dimensional penalty comes from the unconditioned proposal, and bridge or guided proposals remove it rather than mitigating it: Stramer and Yan (2007a) show the modified Brownian bridge has importance-weight variance bounded uniformly in M, with an optimal allocation that is dimension free. If the endpoint construction is retained, monitor the variance and lower tail of every estimated transition density, use log-sum-exp evaluation, and verify stability along a path that holds $R_{M,S}$ fixed or increasing rather than along $M = S$.

11 Conclusion

The endpoint simulated-likelihood estimator has a simple but consequential dimensional structure. Its final Gaussian transition is a shrinking kernel. In K dimensions, the second moment of one simulated contribution grows as $M^{K/2}$, so the Monte Carlo variance is of order $M^{K/2}/S$. The relevant simulation size is $S/M^{K/2}$.

This leads to four conclusions about Brandt and Santa-Clara (2002), stated in decreasing order of strength. First, two published statements are directly false: the joint transition-density consistency of Lemma 2 and the joint central-limit statement of Lemma 3. An exact Brownian sequence satisfying the paper's own Assumptions 1 and 2, and satisfying its rate condition, makes the simulated density converge in probability to zero rather than to the positive true density, and makes $\sqrt{S}$ times the centred density diverge to $-\infty$. Second, because Lemmas 4 to 6 take those results as inputs, the published proofs of Theorems 1 and 2 fail. That is a statement about the proofs; we do not show that their conclusions are false, and Section H explains why the density counterexample does not transfer to the maximiser. Third, the rate condition used in those theorems is incompatible with pointwise mean-square consistency of the simulated transition density in the four-dimensional application. Fourth, and independently of all of the above, the ratios corresponding to the paper's own conditions $\sqrt{S}/M \to 0$ and $N/S^{1/4} \to 0$ are not small at the implemented design $N = 544$, $M = 10$, $S = 5{,}000$, so that design is not in a regime resembling the asymptotic ordering the theorems describe.

The corrected transition-density analysis also supplies constructive results. The density estimator has a triangular-array CLT under $S/M^{K/2} \to \infty$. If the Euler density bias is first order, the MSE-optimal discretisation is $M \asymp S^{2/(K+4)}$. The nonlinear log-likelihood scale is $M^{K/2}/S$, subject to the usual lower-tail conditions required for expectations of logarithms. These results explain why bridge proposals and other importance-sampling methods are not merely computational refinements: they address the rare-event geometry of the estimator.

The exchange-rate application raises separate questions. Its square-root states fall outside the smooth uniformly elliptic theory, and for the interest-rate components the principal issue is failure of the theorem's global smoothness assumptions, the displayed benchmark calculations giving no indication of a practical positivity problem at those states. The incompleteness state is different: its Feller ratio is exactly one half for every admissible parameter value, so zero is attainable. Whether the identification rule places the state on that boundary depends on the specification, and Table 13 is the authoritative summary. In the constant-risk-price specification the corner formula gives an exact observed-date zero under its assumptions; in the time-varying specifications the original authors report that the nonnegativity constraint binds, which we repeat rather than derive. Because the estimator substitutes that state out and simulates volatility instead, the requirement becomes that a recovered square root remain real along every simulated Euler path, and no convention for handling a violation is documented. We do not show that the implemented chain fails: the exact worst-case calculation applies to the directly discretised incompleteness state, and transferring it would require the simulated state distribution, which is not available. The volatility equation is implicit when risk prices depend on volatility, and at the reported U.S.–U.K. estimates the inverse mapping from incompleteness to positive volatility is not globally single-valued without a branch-selection condition; that is an algebraic-domain statement about a grid of candidate states, not a statement about how often the process visits them. The Brownian correlation matrix is positive definite everywhere we could check, including at the feasible minimum volatility. The residual point is that a joint positive-semidefinite restriction is required and entrywise bounds do not deliver it: in the degenerate configuration it splits into a compatibility relation fixing ρ_{xy} and a separate magnitude condition on the free correlations, and the specification imposes neither. The final empirical estimator also contains a minimum-incompleteness corner rule and constraints that the original authors report as binding, which are not covered by ordinary interior SML asymptotics.

None of these points, individually or collectively, proves that the finite-sample economic conclusions are false. They show that those conclusions cannot inherit the claimed maximum-likelihood asymptotics from the mathematical argument given. The appropriate post-publication response is a corrected theorem, a clear statement of the estimator actually implemented, and a re-examination of numerical stability under dimensionally valid simulation rates or bridge-based alternatives.

A Additional proof details

A.1 The Brownian collapse for more general subcritical paths

The proof of Theorem 3.3 extends well beyond $M = S$, and it is worth stating the general version as a result rather than as a sketch, since the sketch previously carried two vague clauses that turn out to be unnecessary or explicit.

Proposition A.1 (Collapse along any subcritical path)**.** *Let $K \geq 1$, let $x, y \in \mathbb{R}^K$, and let $h_n \downarrow 0$ with $h_n \leq \frac{1}{2}$. Suppose the simulation counts satisfy*

$$S_n\, h_n^{K/2} \{\log(1/h_n)\}^{K/2} \longrightarrow 0. \tag{59}$$

Then the Brownian endpoint estimator satisfies $\widehat{q}_{M_n,S_n}(y \mid x) \to 0$ in probability, while $p(y \mid x) > 0$.

Proof. Fix any $c > K$ and set $r_n = \sqrt{c\, h_n \log(1/h_n)}$. Let A_n be the event that some preterminal draw lies in $B(y, r_n)$. As in Theorem 3.3, the preterminal density is bounded by

$\pi^{-K/2}$ uniformly, so with $C_K = \pi^{-K/2}\omega_K$,

$$\mathbb{P}(A_n) \leq C_K S_n r_n^K = C_K c^{K/2}\, S_n h_n^{K/2}\{\log(1/h_n)\}^{K/2} \longrightarrow 0$$

by (59). On A_n^c every summand is at most

$$(2\pi h_n)^{-K/2} \exp\Big(-\frac{r_n^2}{2h_n}\Big) = (2\pi h_n)^{-K/2} h_n^{c/2} = (2\pi)^{-K/2} h_n^{(c-K)/2} \longrightarrow 0,$$

since $c > K$; the average obeys the same bound. Combining the two displays as in Theorem 3.3 gives the conclusion. □

Two clauses in the earlier sketch can now be dropped. "For sufficiently large c" is simply $c > K$, and no growth restriction on S_n beyond (59) is needed: the bound on each summand does not involve S_n at all, so the average inherits it whatever S_n does. Condition (59) is the effective-size condition $R_{M,S} \to 0$ of Definition 2.1 weakened by a logarithmic factor, so the proposition says that collapse occurs on the whole subcritical side of the threshold and not merely along the diagonal. Theorem 3.3 uses $M = S$ because that sequence is transparent and, in four dimensions, satisfies the published rate condition directly.

B Square-root coefficients violate the generic assumptions

The interest rates are modelled as correlated Cox–Ingersoll–Ross processes (Cox et al., 1985),

$$dr_t = \lambda(\vartheta - r_t)dt + \sigma\sqrt{r_t}\, dW_t, \qquad dr_t^* = \lambda^*(\vartheta^* - r_t^*)dt + \sigma^*\sqrt{r_t^*}\, dW_t^*. \tag{60}$$

The generic theory assumes infinitely differentiable coefficients with bounded derivatives and a positive-definite covariance matrix. The function $r \mapsto \sqrt{r}$ is not differentiable at zero, has unbounded derivatives near zero, is not defined on the negative half-line as a real coefficient, and makes the covariance degenerate at the boundary. The same singularity appears explicitly in the transformed system of Appendix B of Brandt and Santa-Clara (2002), whose drift for $\sqrt{r_t}$ contains the term $-\frac{1}{8}\sigma^2 r_t^{-1/2}$.

It is important to separate this structural violation from its practical consequences, and the two point in opposite directions for the interest rates. Table 6 reports the Feller ratio $2\lambda\vartheta/\sigma^2$ at the estimates of Table 3 of Brandt and Santa-Clara (2002). It ranges from 6.4 to 48.5, so in every case the ratio comfortably exceeds one and the origin is unattainable for the exact processes. Unattainable is not the same as unapproachable: the Feller condition prevents the process from reaching zero, not from coming arbitrarily close to it, and it says nothing about how close the observed path came.

The corresponding Euler negativity probabilities are strictly positive, because a Gaussian increment has full support, and they should not be called zero. At the displayed benchmark states they are, however, numerically negligible. Table 5 reports them at $h = 1/520$, corresponding to weekly data with $M = 10$. At the long-run means the standardised distance to the boundary ranges from 111 to 203 standard deviations, so the one-step probabilities are of order 10^{-2667} to 10^{-8986}. Even from the implausibly low state $r = 0.001$ the smallest distance is 13.8 standard deviations and the largest probability is about 2×10^{-43}. Many of these tails underflow to exactly zero in binary64 arithmetic; Table 5 records the result state by state, and that is why the table reports base-ten logarithms rather than the probabilities themselves.

The observed rate path is not reconstructed here, so none of this is a sample-wide bound. What the calculation supports is that at the states displayed the one-step positivity problem is numerically negligible, not that no simulated or observed state could behave otherwise.

The assumptions of the generic theorem are genuinely violated by (60), and the estimator applied is therefore not the estimator analysed. But for r and r^* that gap is a matter of proof coverage rather than of demonstrated numerical damage: at the displayed benchmark states, where the one-step violation probability is of order 10^{-2667}, such a violation is numerically negligible for any realistically sized simulation.

Table 5: One-step Euler negativity diagnostics for the square-root interest-rate processes at $h = 1/520$, using the estimates of Table 3 of Brandt and Santa-Clara (2002). Definitions: from a current state $r > 0$ the Euler update has mean $r + \lambda(\vartheta - r)h$ and standard deviation $\sigma\sqrt{rh}$, so the negativity probability is $\Phi(-z)$ with $z = [r + \lambda(\vartheta - r)h]/(\sigma\sqrt{rh})$; the table reports z and $\log_{10}\Phi(-z)$, computed through a numerically stable log-CDF rather than by taking the logarithm of an evaluated tail. "Underflow" records whether direct IEEE-754 binary64 evaluation of $\Phi(-z)$ returns exactly zero. Binary64 represents positive values down to about 5×10^{-324}, using subnormal numbers below the smallest normal value of about 2.2×10^{-308}, but a particular cumulative-distribution routine may return zero earlier: the implementation used here does so at about 6×10^{-311}, inside the subnormal range. The flag therefore reports what that routine returns, not a property of the mathematics, and the probability is strictly positive at every state. These are the displayed benchmark states only, not a sample-wide bound.

Process	State	r	$2\lambda\vartheta/\sigma^2$	z	$\log_{10}\mathbb{P}(\text{negative})$	Underflow
U.S. (vs. U.K.)	long-run mean	0.053	38.4	187.5	−7636	yes
	half the mean	0.0265		132.6	−3823	yes
	1 per cent	0.01		81.6	−1449	yes
	0.1 per cent	0.001		26.5	−154	no
U.K.	long-run mean	0.074	22.9	110.8	−2667	yes
	half the mean	0.037		78.4	−1337	yes
	1 per cent	0.01		41.0	−366	yes
	0.1 per cent	0.001		13.8	−42.6	no
U.S. (vs. Germany)	long-run mean	0.058	48.5	203.4	−8986	yes
	0.1 per cent	0.001		27.6	−167	no
Germany	long-run mean	0.064	6.4	137.4	−4099	yes
	0.1 per cent	0.001		17.4	−67.0	no

The market-incompleteness state behaves differently. Write $H_t = \epsilon_t^2$ for it; the letter X is reserved for the Brownian motion driving the log exchange rate in (78). Equation (37) of Brandt and Santa-Clara (2002) specifies

$$dH_t = (\beta^2 - 2\alpha H_t)dt + 2\beta\sqrt{H_t}\,dY_t, \tag{61}$$

which is a CIR process with $\kappa = 2\alpha$, long-run mean $\beta^2/(2\alpha)$, and diffusion coefficient 2β. Its Feller ratio is

$$\frac{2\kappa\vartheta_X}{\sigma_X^2} = \frac{2\beta^2}{4\beta^2} = \frac{1}{2}$$

for every $\alpha > 0$ and every $\beta \neq 0$.

The parameter-independence is not a coincidence, and identifying its source changes how the rest of this appendix should be read. Suppose ϵ_t itself follows the Ornstein–Uhlenbeck process

$$d\epsilon_t = -\alpha\epsilon_t\,dt + \beta\,dY_t. \tag{62}$$

Then Itô's lemma applied to $H = \epsilon^2$ gives $dH = 2\epsilon\,d\epsilon + (d\epsilon)^2 = (\beta^2 - 2\alpha H)\,dt + 2\beta\epsilon\,dY$, which is (61) exactly, with $2\beta\epsilon = 2\beta\sqrt{H}$ on $\epsilon > 0$. The incompleteness specification is therefore the *square of an Ornstein–Uhlenbeck process*, and its Feller ratio is one half for the same reason that a squared Gaussian has a chi-square distribution with one degree of freedom: it is a property of the squaring map, not of α or β.

Three consequences follow, and they cut in different directions.

First, the attainability of zero is demystified rather than alarming. An Ornstein–Uhlenbeck process crosses zero almost surely; H touching zero is that crossing, seen through the square. Nothing pathological is happening to the model.

Second, the natural state is ϵ, not H. In the coordinate ϵ there is no boundary, no positivity constraint, and an exact Gaussian transition density requiring no Euler scheme at all. Moreover ϵ_t enters the exchange-rate diffusion only through $\epsilon_t \, dU_t$, so its sign is immaterial to the model. A competent implementation would simulate (62) exactly and never meet a square root.

Third, and consequently, the calculation in the remainder of this appendix concerns a discretisation that the natural parameterisation makes unnecessary, and which Section C establishes the published implementation did not use. We retain it because it is exact and because it isolates the mechanism cleanly, but it should be read as a worked illustration of how a shrinking-bandwidth boundary problem behaves, not as a diagnosis of anything in Brandt and Santa-Clara (2002). The criticism that does bear on their implementation is the radicand condition of Section C, which no change of coordinates removes because the estimator eliminates ϵ in favour of v.

Returning to the direct discretisation: no admissible parameter estimate can make the zero boundary of H unattainable, and the instantaneous covariance degenerates there. That is a statement about attainability, not about where the state typically sits: it does not say the process is usually near zero. This is a structural feature of the specification, not a property of a particular estimate. Whether the boundary is not merely attainable in principle but attained in the reported results depends on the specification, and Table 13 records the three different statuses. In brief: under the conditions of Proposition F.1(iii) the constant-risk-price optimisation yields an exact zero; for the time-varying specifications Brandt and Santa-Clara (2002) report that the nonnegativity constraint binds, which we repeat rather than verify.

Table 6: Feller ratios at the estimates reported in Table 3 of Brandt and Santa-Clara (2002). For a process $dZ = \kappa(\vartheta_Z - Z)dt + \sigma_Z \sqrt{Z}\, dB$ the ratio is $2\kappa\vartheta_Z/\sigma_Z^2$, and the origin is unattainable if and only if it is at least one. For the incompleteness state the ratio equals $1/2$ for all admissible parameters.

System	State	κ	ϑ_Z	σ_Z	$2\kappa\vartheta_Z/\sigma_Z^2$
U.S. vs. U.K.	r (U.S.)	0.284	0.053	0.028	38.4
	r^* (U.K.)	0.486	0.074	0.056	22.9
	$H = \epsilon^2$	0.640	0.0121	0.176	0.5
U.S. vs. Germany	r (U.S.)	0.305	0.058	0.027	48.5
	r^* (Germany)	0.088	0.064	0.042	6.4
	$H = \epsilon^2$	0.676	0.0151	0.202	0.5

C The simulated state and the reality of the incompleteness process

The previous subsection treats (61) as a state in its own right, which is how the model presents it. The estimator does not. Equation (31) and Appendix B of Brandt and Santa-Clara (2002) eliminate ϵ_t using the variance identity and simulate the four-dimensional state

$$(r_t,\ r_t^*,\ e_t,\ v_t), \tag{63}$$

where e_t is the log exchange rate and v_t is its instantaneous volatility, proxied by a one-week at-the-money implied volatility. The incompleteness state is therefore never discretised directly. This relocates the boundary problem rather than removing it, and the relocated version is the one that binds.

The identity that eliminates ϵ_t is

$$\epsilon_t = \sqrt{v_t^2 - (\phi_t^2 + \phi_t^{*2} - 2\rho_{ww^*}\phi_t\phi_t^*) - (\psi_t^2 + \psi_t^{*2} - 2\rho\,\psi_t\psi_t^*)}, \tag{64}$$

where $\phi_t = \phi\sqrt{r_t}$ and $\phi_t^* = \phi^*\sqrt{r_t^*}$ are the market prices of domestic and foreign interest-rate risk and ψ_t, ψ_t^* are the market prices of pure currency risk. The state ϵ_t appears in the diffusion of the log exchange rate, through $v_t\,dX_t = \phi_t dW_t - \phi_t^* dW_t^* + \psi_t dZ_t - \psi_t^* dZ_t^* + \epsilon_t dU_t$, and in the coefficients of the v_t equation obtained from Itô's lemma. Every Euler substep of (63) therefore requires the radicand of (64) to be nonnegative at the simulated state, not merely at the observed data. Ordinary Euler discretisation of a system whose coefficients are defined by (64) offers no such guarantee. The mechanism is not that the Gaussian increment of v_t is unbounded below, which would be the wrong reading: the radicand contains v_t^2, so a large negative v_t makes the radicand large and positive, not negative. What matters is that the update is not constrained to preserve the algebraic domain $v_t^2 \geq Q_t + C_t$. A nondegenerate Gaussian update assigns positive probability to every neighbourhood of any point, including neighbourhoods in which $|v_t|$ is small, while the subtracted terms are continuous and need not be small there. The discretisation therefore supplies no positivity-preserving guarantee for the reconstructed radicand.

We stop at that. Turning it into a violation probability would require the joint conditional covariance of the full state update, since Q_t and C_t move with the other components rather than staying fixed while v_t wanders, and that calculation is not available to us.

This is the operative form of the degeneracy, and three features make it sharp rather than hypothetical.

First, the constraint is not generically slack, though what can be asserted differs by specification and Table 13 is the reference. In the constant specification, the corner formula (92) implies an observed-date zero under its stated assumptions. In the time-varying specifications, Brandt and Santa-Clara (2002) report that the nonnegativity constraint binds; that is their statement, not our derivation. In either case an observed-date zero means one sample date at which the reconstructed incompleteness state sits on the boundary, and the transition simulated *out of* that date starts there, unless the active date is the terminal observation, from which no transition is simulated.

Three things do not follow, and we assert none of them. A binding observed-date constraint does not show that every simulation starts from the boundary; it does not show that any intermediate simulated state reaches the boundary; and it does not show that the original code ever evaluated coefficients at an exact floating-point zero, which is a question about the implementation rather than about the optimisation. The minima of 0.0000 in Table 4 of Brandt and Santa-Clara (2002) are rounded to four decimal places and corroborate rather than prove.

Second, the failure is not a recoverable numerical event such as a negative variance that can be floored at zero. It makes the drift and diffusion coefficients of the next substep complex-valued, so the chain is undefined rather than inaccurate.

Third, the paper reports assigning zero likelihood to parameter values that violate $\epsilon_t^2 \geq 0$ at the observed dates. That is a restriction on parameters, not a rule for what the simulator does when an *intermediate* Euler state violates it, and the two are different objects: the constraint is checked at N observation dates, whereas the chain visits NMS intermediate states.

As with the interest rates, a truncation, reflection, absorption, or exact-transition rule would resolve the issue. Each such rule defines a *different* approximating chain, with a different transition density and therefore a different estimator, and each requires its own convergence analysis; they are not interchangeable implementation details. The published paper does not specify one.

The size of the problem can be computed exactly on the incompleteness specification (61) taken at face value, that is on the state $H_t = \epsilon_t^2$ discretised directly. Because the implemented estimator does not simulate that state, it is worth fixing in advance what the calculation below does and does not deliver.

It establishes four things about the *directly discretised* process: that the exact CIR specification for H has an attainable boundary; that ordinary Euler applied directly to H is not positivity preserving; that the worst-case one-step negativity probability has the closed form (67); and that refining the step moves the worst-case location towards zero while lowering the maximum only to the strictly positive limit $\Phi(-1)$.

It establishes none of the following, and no claim in this paper rests on it as though it did: the probability that the implemented v-state Euler chain produces a negative radicand in (64); the probability of complex coefficients arising in the published code; the frequency of intermediate-state violations along a simulated path; the probability of an invalid likelihood evaluation; or the finite-sample accuracy of the reported estimates. Analysing the implemented chain would require the explicit drift and diffusion coefficients for v_t, the numerical convention actually used, simulation of the intermediate states, monitoring of the radicand, probability or moment bounds for it, and a stated treatment of failed paths in the likelihood. The first is available from Appendix B of Brandt and Santa-Clara (2002); the rest are not.

For an Euler step of size h the probability of a negative next state from $H_t = x > 0$ is

$$P_h(x) = \Phi\left(-\frac{x + (\beta^2 - 2\alpha x)h}{2\beta\sqrt{xh}}\right). \tag{65}$$

This vanishes at $x = 0$, where the diffusion term vanishes while the drift $\beta^2 h$ is strictly positive, and again for large x, so it has an interior maximum. That maximum has a closed form, and it is not what one would hope.

Proposition C.1 (Exact worst case of the one-step violation probability)**.** *Let $\alpha > 0$, $\beta > 0$ and $0 < h < 1/(2\alpha)$. Then P_h attains its supremum on $(0, \infty)$ at*

$$x_h^* = \frac{\beta^2 h}{1 - 2\alpha h}, \tag{66}$$

and the supremum equals

$$\sup_{x>0} P_h(x) = \Phi\left(-\sqrt{1 - 2\alpha h}\right). \tag{67}$$

Consequently $x_h^* \sim \beta^2 h$, and $h \mapsto \sup_x P_h(x)$ is strictly increasing on $(0, 1/(2\alpha))$, so as $h \downarrow 0$ the worst case decreases *to*

$$\Phi(-1) \approx 0.1587$$

from above. The limit depends on neither α nor β.

The sign convention matters here. We take $\beta > 0$, which matches the reported estimates; for $\beta < 0$ the process is unchanged in law and every occurrence of β in the standard deviation should be read as $|\beta|$, after which the statement is identical with $\beta^2 = |\beta|^2$ throughout.

Proof. Write $a = 1 - 2\alpha h > 0$ and $b = \beta^2 h > 0$. The argument of Φ in (65) is $-g(x)$ with

$$g(x) = \frac{x + (\beta^2 - 2\alpha x)h}{2\beta\sqrt{xh}} = \frac{ax + b}{2\beta\sqrt{h}\,\sqrt{x}},$$

so, Φ being strictly increasing, maximising P_h is minimising g over $x > 0$. Differentiating the numerator of g with respect to $\sqrt{x}$, or directly,

$$g'(x) = \frac{1}{2\beta\sqrt{h}} \cdot \frac{a\sqrt{x} - (ax + b)/(2\sqrt{x})}{x} = \frac{ax - b}{4\beta\sqrt{h}\,x^{3/2}},$$

which is negative for $x < b/a$, zero at $x = b/a$ and positive for $x > b/a$. Hence g has a unique minimum at $x_h^* = b/a$, which is (66). There

$$g(x_h^*) = \frac{a(b/a) + b}{2\beta\sqrt{h}\,\sqrt{b/a}} = \frac{2b}{2\beta\sqrt{h}\,\sqrt{b/a}} = \frac{\sqrt{ab}}{\beta\sqrt{h}} = \frac{\beta\sqrt{ah}}{\beta\sqrt{h}} = \sqrt{a},$$

giving (67).

For the monotonicity, write $f(h) = \Phi(-\sqrt{1 - 2\alpha h})$ on $(0, 1/(2\alpha))$. Then

$$\frac{d}{dh}\Big(-\sqrt{1 - 2\alpha h}\Big) = \frac{\alpha}{\sqrt{1 - 2\alpha h}} > 0, \qquad \text{so} \qquad f'(h) = \varphi\big(\sqrt{1 - 2\alpha h}\big)\,\frac{\alpha}{\sqrt{1 - 2\alpha h}} > 0,$$

with φ the standard normal density. Hence f is strictly increasing in h, so refining the mesh strictly decreases the worst case; and since $a = 1 - 2\alpha h \uparrow 1$ as $h \downarrow 0$, the limit $\Phi(-1)$ is approached from above. $\square$

The practical reading needs care in both directions. Refining the Euler step does reduce the worst-case one-step violation probability, but it does not drive it to zero: the maximum decreases monotonically to the strictly positive limit $\Phi(-1) \approx 0.1587$, and by the time h is small the reduction is in the fourth decimal place. Refinement also moves the dangerous state towards zero in proportion to h. So mesh refinement alone does not eliminate the local positivity problem; it relocates it and leaves its severity essentially unchanged. Table 7 reports (66) and (67) at the reported estimates, and its final column is strictly decreasing as h falls, from 0.1602 at $h = 0.02$ to 0.1588 at $h = 1/520$ and 0.1587 at $h = 1/1040$.

The scope of this must be stated carefully, because a shrinking region invites an inference it does not support. The maximising state x_h^* is of order h, so the region in which the one-step violation probability is large moves toward zero as the mesh is refined. This geometric shrinkage alone does not determine how often the discretised chain occupies that region. Occupation probabilities depend on the transition law, the initial condition, the boundary behaviour and, where relevant, the invariant distribution of the numerical scheme; near an attainable boundary the state distribution may itself concentrate towards zero as h falls, and nothing in Proposition C.1 excludes that. The proposition establishes a local worst-case result, not an integrated pathwise violation probability.

Five quantities must be kept apart, and only the first two are computed here:

(a) the pointwise one-step probability $P_h(x)$ at a given current state x;
(b) its supremum $\sup_x P_h(x)$ over current states, which is (67);
(c) the distribution π_h of the current state under the scheme;
(d) the integrated one-step failure probability $\int P_h(x)\,\pi_h(dx)$, which requires π_h;
(e) the cumulative pathwise probability of any failure over m steps.

Proposition C.1 bounds (d) above by (b) and says nothing else about it, because it does not analyse π_h.

Quantity (e) deserves a precise statement, and the events need care before it can be written down. The naive choice $\{H_{j+1}^{\mathrm{Euler}} < 0\}$ is not well defined once an earlier step has already gone negative, because the square root in the next update is then undefined and the chain does not continue. We therefore work with the chain stopped at its first invalid step and use *first-failure* events,

$$F_j = \{H_0^{\mathrm{E}} \geq 0,\ \ldots,\ H_j^{\mathrm{E}} \geq 0,\ H_{j+1}^{\mathrm{E}} < 0\}, \tag{68}$$

which are well defined on the stopped chain and, being indexed by the step at which failure first occurs, are pairwise disjoint. A union bound is then available and requires no independence:

$$\mathbb{P}\Big(\bigcup_{j=0}^{m-1} F_j\Big) = \sum_{j=0}^{m-1} \mathbb{P}(F_j) \ \leq\ m \sup_{x\geq 0} P_h(x). \tag{69}$$

The first relation is an equality rather than an inequality precisely because the events are disjoint; the inequality comes from bounding each $\mathbb{P}(F_j) = \mathbb{E}\big[\mathbf{1}\{\text{survived to } j\} P_h(H_j^{\mathrm{E}})\big]$ by the supremum. What is *not* available is any product formula such as $1-(1-p)^m$, which would require the failure events to be independent, or at least a conditional structure that makes the product a valid bound. Successive Euler states are dependent, so no such formula may be used here.

The union bound is therefore correct but uninformative at the probabilities in question. Since $\sup_x P_h(x) \approx 0.1587$ by (67), the right-hand side of (69) reaches one at $m = 7$, and beyond that the bound is the trivial statement $\mathbb{P}(\cdot) \leq 1$. At $m = 10$ it gives 1.59, at $m = 20$ it gives 3.17, and at the $m = 520$ steps of one year at $h = 1/520$ it gives 82.5; all three clip to 1. A useful pathwise statement would need the distribution of H_j, the conditional transition probabilities, the dependence across steps, and probably occupation estimates near the boundary. What Proposition C.1 rules out is the comfortable assumption that refinement makes the per-step boundary problem itself disappear.

Figure 5 plots (65) at these step sizes. It is an exact illustration of the class of failure, not a simulation of the implemented chain, whose boundary is (64) and whose exact crossing probability depends on the drift and diffusion of v_t derived in Appendix B of Brandt and Santa-Clara (2002).

Table 7: Worst-case one-step Euler negativity probability for the incompleteness specification (61), at the estimates of Table 3 of Brandt and Santa-Clara (2002). The step $h = 1/520$ corresponds to weekly data with $M = 10$ and $h = 1/1040$ to $M = 20$; the third row is a coarser illustration. The maximum decreases strictly as h is refined, from 0.1602 through 0.1588 to 0.1587, converging to $\Phi(-1) \approx 0.15866$ from above rather than to zero, in accordance with Proposition C.1; the maximising state scales with h.

System	Step	H_h^*	$\epsilon_h^* = \sqrt{H_h^*}$	$\max_x P_h(x)$
U.S. vs. U.K.	$h = 1/520$	1.49×10^{-5}	0.00386	0.1588
	$h = 1/1040$	7.45×10^{-6}	0.00273	0.1587
	$h = 0.02$	1.57×10^{-4}	0.01253	0.1602
U.S. vs. Germany	$h = 1/520$	1.96×10^{-5}	0.00443	0.1588
	$h = 1/1040$	9.82×10^{-6}	0.00313	0.1587
	$h = 0.02$	2.07×10^{-4}	0.01438	0.1603

D Invertibility of the volatility–incompleteness map

The model can be written with either volatility or incompleteness as the primitive state, and the two representations are equivalent only if the map between them is well defined in both directions. This subsection analyses that map. It is not a claim about what the implemented estimator computes: as described in Section C, the implementation takes v_t as the primitive simulated state and uses the variance identity in the easy direction, to reconstruct ϵ_t. No quadratic is solved at any Euler step.

The time-varying pure-currency risk prices have the form

$$\psi_t = a_t + bv_t, \qquad \psi_t^* = a_t^* + b^* v_t, \tag{70}$$

where a_t and a_t^* depend on the interest-rate differential and the log exchange rate. The variance identity is

$$v_t^2 = D_t + \psi_t^2 + (\psi_t^*)^2 - 2\rho\psi_t\psi_t^*, \tag{71}$$

where D_t collects the interest-rate risk term and ϵ_t^2, and $\rho = \rho_{zz^*}$.

Substituting (70) into (71) and collecting terms defines

$$F(v; \text{state}) = (1 - \kappa_b)v^2 - 2\eta_t v - \zeta_t, \tag{72}$$

so that the identity reads $F(v_t; \text{state}) = 0$, where

$$\kappa_b = b^2 + (b^*)^2 - 2\rho bb^*, \tag{73}$$

$$\eta_t = a_t b + a_t^* b^* - \rho(a_t b^* + a_t^* b), \tag{74}$$

$$\zeta_t = D_t + a_t^2 + (a_t^*)^2 - 2\rho a_t a_t^*. \tag{75}$$

Both κ_b and the state-dependent part of ζ_t are quadratic forms in the two-dimensional correlation matrix $\left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, evaluated at $(b, -b^*)$ and $(a_t, -a_t^*)$ respectively. When that matrix is positive semidefinite and $D_t \geq 0$ one therefore has $\kappa_b \geq 0$ and $\zeta_t \geq 0$.

The two directions of the map are not symmetric.

Volatility primitive. Given v_t and the remaining states, (71) determines ϵ_t^2 by subtraction, uniquely and without any branch condition. This is the direction the estimator uses, and it is unproblematic apart from the reality requirement discussed in Section C.

Incompleteness primitive. Given ϵ_t^2 and the remaining states, recovering a positive v_t requires solving $F(v; \text{state}) = 0$. This is the direction implied by reading equations (30) and (37) of Brandt and Santa-Clara (2002) as the primitive specification, with the volatility dynamics of their equation (31) derived from it by Itô's lemma. The implicit function theorem gives local solvability wherever

$$\frac{\partial F}{\partial v}(v; \text{state}) = 2(1 - \kappa_b)v - 2\eta_t \neq 0, \tag{76}$$

that is away from the singular set

$$v = \frac{\eta_t}{1 - \kappa_b} \qquad (\kappa_b \neq 1), \tag{77}$$

at which the two roots coalesce and the local inverse fails to exist. Local solvability is not global uniqueness, and the following proposition describes the global branch structure.

The following proposition concerns the state transformation between volatility and incompleteness. It is not a statement about the implemented optimisation algorithm, which uses the identity only in the direction described above.

Proposition D.1 (Branch structure of the volatility identity)**.** *Let* $\zeta_t > 0$*.*

(i) If $\kappa_b < 1$, equation $F(v; state) = 0$ has exactly one positive root and one negative root, so at every state satisfying the standing hypothesis $\zeta_t > 0$ the positive branch is unique and the inverse map is single valued on the positive half-line.

(ii) If $\kappa_b > 1$, write $\Delta_t = \eta_t^2 - (\kappa_b - 1)\zeta_t$. Then exactly one of the following holds:

(a) $\eta_t < 0$ and $\Delta_t > 0$: two distinct *positive roots,*

$$v_t^{\pm} = \frac{-\eta_t \pm \sqrt{\Delta_t}}{\kappa_b - 1};$$

(b) $\eta_t < 0$ and $\Delta_t = 0$: one positive root of multiplicity two, $v_t = -\eta_t/(\kappa_b - 1)$, which is exactly the singular set (77)*;*

(c) otherwise, that is $\Delta_t < 0$ or $\eta_t \geq 0$: no positive root.

Remark D.2 (Counting roots)**.** Case (b) is why the classification is stated this way rather than as "no positive root or exactly two". That shorter phrasing is consistent only if roots are counted with multiplicity, and the distinction matters for the inverse map: at a repeated root the two branches coincide, so the map is single valued there, but it is single valued on a set where the two branches meet rather than on a region where only one exists. Away from the singular set the inverse map is either undefined or two-valued, never genuinely single valued.

Case (b) is also a codimension-one condition on the state, so a grid of candidate points will not meet it except by construction. That is precisely what the sweep of Section E.1 reports: across every grid design, no candidate point yields exactly one positive root, which is case (b) failing to occur rather than case (b) being impossible.

Proof. Rewrite $F(v; \text{state}) = 0$ as $(\kappa_b - 1)v^2 + 2\eta_t v + \zeta_t = 0$.

(i) If $1 - \kappa_b > 0$ the product of the roots is $-\zeta_t/(1 - \kappa_b) < 0$, so the real roots have opposite signs. The discriminant $4\{\eta_t^2 + (1 - \kappa_b)\zeta_t\}$ is strictly positive, so both roots are real and exactly one is positive.

(ii) If $\kappa_b - 1 > 0$ the product of the roots is $\zeta_t/(\kappa_b - 1) > 0$ and their sum is $-2\eta_t/(\kappa_b - 1)$. A positive product excludes roots of opposite signs and excludes a zero root, so real roots are either both positive or both negative, and both are positive exactly when the sum is positive, that is when $\eta_t < 0$. Real roots exist when the discriminant $4\{\eta_t^2 - (\kappa_b - 1)\zeta_t\}$ is nonnegative, and the stated formula is the quadratic formula. The roots coincide when $\eta_t^2 = (\kappa_b - 1)\zeta_t$, in which case the common value is $-\eta_t/(\kappa_b - 1) = \eta_t/(1 - \kappa_b)$, which is (77). $\square$

Table 8 evaluates κ_b at the model-C estimates of Table 4 of Brandt and Santa-Clara (2002). The two systems fall on opposite sides of the threshold.

For the U.S.–Germany system, $\kappa_b \approx 0.0702 < 1$, case (i) applies, and the inverse map is single valued on the states the proposition covers: at any state with $\zeta_t > 0$ there is exactly one positive volatility consistent with the identity. The restriction to $\zeta_t > 0$ is the proposition's own hypothesis and is not removed here. Establishing $\zeta_t > 0$ throughout the model domain would need a separate argument, which we do not give, so this is uniqueness on admissible states rather than global uniqueness.

For the U.S.–U.K. system, $\kappa_b \approx 1.9982 > 1$ and case (ii) applies. The reported coefficients therefore do not define a globally unique inverse mapping from the incompleteness state to positive volatility without a branch-selection condition. At states where positive roots exist there are two distinct ones, except on the repeated-root boundary $\Delta_t = 0$ of case (b), and the specification does not say which is intended; at states where $\eta_t \geq 0$ or $\Delta_t < 0$ there is none. Because $\zeta_t \geq 0$ follows from positive semidefiniteness of the two-dimensional correlation matrix together with $D_t \geq 0$, both of which hold by construction, this is a consequence of $\kappa_b > 1$ alone and not of a knife-edge parameter value.

Two qualifications matter for how much weight this carries. It does not affect the reconstruction of ϵ_t from an observed implied volatility, which is the direction the estimation uses and which is unique. And it does not by itself show that the reported estimates are wrong. What it shows is that the two representations of the model are not interchangeable at the U.S.–U.K. estimates, so that reading the specification as a generative diffusion in the incompleteness state requires an additional branch-selection rule that the paper does not supply. Appendix B of Brandt and Santa-Clara (2002) defines the drift and diffusion of v_t only implicitly, with those coefficients appearing inside the formulas used to define them, so a well-posedness condition is required there in any case.

Table 8: Quadratic coefficient implied by the model-C volatility loadings reported in Table 4 of Brandt and Santa-Clara (2002), where $\kappa_b = b^2 + (b^*)^2 - 2\rho b b^*$ with $b = c_3$, $b^* = c_3^*$ and $\rho = \rho_{zz^*}$. The sign of $1 - \kappa_b$ determines which case of Proposition D.1 applies.

Market	b	b^*	ρ	κ_b	$1-\kappa_b$	Branch structure
U.S. vs. U.K.	1.514	2.581	0.89	1.9982	−0.9982	zero or two positive roots
U.S. vs. Germany	−0.142	0.127	0.94	0.0702	0.9298	unique positive root

E Feasibility of the Brownian correlation matrix

The application specifies an instantaneous correlation matrix for the four Brownian motions (W_t, W_t^*, X_t, Y_t), where W and W^* drive the two interest rates, X drives the log exchange rate and Y drives the incompleteness state. Write it in block form as

$$R_t = \begin{pmatrix} A_t & r_t \\ r_t^\top & 1 \end{pmatrix}, \tag{78}$$

where A_t is the 3×3 block for (W_t, W_t^*, X_t) and $r_t = (\rho_{wy}, \rho_{w^*y}, \rho_{xy})^\top$ collects the three correlations with Y_t.

A symmetric 4×4 correlation matrix has six off-diagonal entries, and it is worth listing them, because the count is easy to get wrong. R_t contains *four* free entries, namely ρ_{ww^*} together with the three components of r_t, and *two* state-dependent implied entries, $\rho_{wx,t}$ and $\rho_{w^*x,t}$. The structural model contains one further primitive correlation parameter, ρ_{zz^*}, which enters the currency-risk quadratic form C_t rather than R_t directly, and so reaches R_t only through the volatility v_t. That gives five primitive correlation parameters in the model, four of which are entries of R_t. Table 11 sets this out row by row. From equation (28) of Brandt and Santa-Clara (2002),

$$\rho_{wx,t} = \frac{\phi_t - \rho_{ww^*}\phi_t^*}{v_t}, \qquad \rho_{w^*x,t} = \frac{\rho_{ww^*}\phi_t - \phi_t^*}{v_t}, \tag{79}$$

state dependent through $\phi_t = \phi\sqrt{r_t}$, $\phi_t^* = \phi^*\sqrt{r_t^*}$ and v_t.

Three different questions must be kept apart, and the three subsections below answer them separately. First, is the matrix valid at the reported estimates and at reference points satisfying the reported algebraic restrictions? Second, does the specification guarantee validity for every admissible parameter and state? Third, did the numerical implementation impose safeguards not described in the published paper? The answers are yes, no, and unknown.

Remark E.1 (Algebraic feasibility is not dynamic accessibility)**.** Every statement in this appendix concerns whether a candidate point satisfies the model's algebraic identities: whether the variance identity holds with $\epsilon_t^2 \geq 0$, whether the inverse mapping admits a positive volatility, and whether the resulting matrix is positive semidefinite. None of it concerns whether the diffusion visits such a point. Establishing that would require a support theorem for the process, an accessibility argument, or analysis of the transition or invariant densities, none of which is attempted here. Accordingly a point is called *algebraically feasible* or *infeasible*, never reachable or unreachable, and counts of grid points are reported as counts rather than as proportions of a state domain.

E.1 Algebraic domain of the state transformation

Everything here depends on the algebraic relation between the states, so we begin with the complete variance identity. Equation (29) of Brandt and Santa-Clara (2002) is

$$v_t^2 = Q_t + C_t + \epsilon_t^2, \tag{80}$$

where

$$Q_t = \phi_t^2 + \phi_t^{*2} - 2\rho_{ww^*}\phi_t\phi_t^*, \qquad C_t = \psi_t^2 + \psi_t^{*2} - 2\rho_{zz^*}\psi_t\psi_t^*,$$

are the interest-rate and currency-risk quadratic forms and $\epsilon_t^2 \geq 0$ is the incompleteness state. Both quadratic forms are nonnegative whenever the corresponding two-dimensional correlation matrix is positive semidefinite. A state is feasible precisely when (80) holds with $\epsilon_t^2 \geq 0$, that is when

$$v_t^2 \;\geq\; Q_t + C_t. \tag{81}$$

The weaker bound $v_t^2 \geq Q_t$ is necessary but not sufficient, and it is attained only in the degenerate configuration $C_t = \epsilon_t^2 = 0$.

For the constant-risk-price specification, C_t is a constant that can be read directly from the published estimates. Table 3 of Brandt and Santa-Clara (2002) reports $\psi^2 - \rho_{zz^*}\psi\psi^*$ and $\psi^{*2} - \rho_{zz^*}\psi\psi^*$ separately, and their sum is exactly C. This gives

$$C_{\text{U.K.}} = 0.024 + (-0.021) = 0.003, \qquad C_{\text{Germany}} = (-0.010) + 0.013 = 0.003.$$

Both are strictly positive, so at the reported estimates the degenerate configuration is *algebraically infeasible*, because it requires $C_t = 0$ while the reported constant currency-risk term is $C = 0.003 > 0$: the least volatility satisfying the variance identity exceeds $\sqrt{Q_t}$ by a margin that does not vanish. At the long-run mean rates the feasible minimum is 0.0640 for the U.S.–U.K. system against $\sqrt{Q_t} = 0.0330$, and 0.0622 against 0.0295 for the U.S.–Germany system.

For the time-varying specifications C_t is state dependent, and in model C it depends on v_t itself through (70). That looks like an obstruction and is not one, for two reasons.

First, model B carries no volatility loading: Table 4 reports three coefficients per risk price for that specification against four for model C, so under model B the quantity C_t is an explicit function of the interest-rate differential and the log exchange rate, and v_t follows from (80) by taking a square root.

Second, the apparent circularity in model C is exactly the quadratic of Section D. Substituting (70) into C_t and then into (80) gives

$$(1-\kappa_b)v_t^2 - 2\eta_t v_t - \left(Q_t + \zeta_t^a + \epsilon_t^2\right) = 0, \qquad \zeta_t^a = a_t^2 + a_t^{*2} - 2\rho a_t a_t^*, \tag{82}$$

so v_t is *determined* by $(r_t, r_t^*, e_t, \epsilon_t^2)$ rather than being an independent unknown. The fixed point is solvable in closed form, and by Proposition D.1 it has one positive solution when $\kappa_b < 1$ and either none or two when $\kappa_b > 1$.

What remains genuinely unobserved is only the level of the log exchange rate e_t. Rather than requiring its realised path we sweep it over a grid of candidate values.

What this diagnostic does and does not establish must be stated precisely, because the temptation to read a grid as a state space is strong. The grid evaluates, at each of a finite collection of candidate state combinations, whether the inverse variance identity admits no positive volatility solution, exactly one, or two. That is an algebraic question about the map (82) at a point. It establishes nothing about dynamic accessibility, about the support of the diffusion, about the probability of visiting any point, about occupation times, about the invariant distribution, or about the measure of any subset of the state domain. Counts of grid points are counts of grid points. They are reported below as such, and they are not percentages of a state space.

Table 10 reports the result over a grid of 56,784 candidate points per specification, spanning interest rates from 0.05 to 5 times their long-run means, $e_t \in [-2, 2]$, and $\epsilon_t^2 \in \{0, 0.002, 0.01, 0.05\}$. Because Table 4 does not re-report the interest-rate risk prices or the three free correlations, those are taken from Table 3; that is an assumption, and it is the only one.

For three of the four specifications, no algebraic branch or positive-semidefinite problem was found on the examined grids. For model B in both systems and for model C in the U.S.–Germany system, every candidate point examined admits exactly one positive volatility and the correlation matrix is positive definite at all of them, with worst minimum eigenvalues of 0.070, 0.135 and 0.088 respectively. This does not prove global validity outside the examined grids, and no such claim is made.

The U.S.–U.K. model C specification behaves differently, and the behaviour sharpens Proposition D.1 from a statement about what can happen into a statement about what does happen throughout the grid. On 37,536 of the 56,784 candidate points examined, the inverse identity admits *no positive volatility solution*; on the remaining 19,248 it admits two. Not one candidate point admits exactly one, which is Proposition D.1(ii) confirmed as a computational fact rather than a possibility. Where a branch does exist the correlation matrix is comfortably valid, with minimum eigenvalue at least 0.445, so this is a well-posedness failure of the inverse representation rather than a correlation failure. The no-solution region also grows with incompleteness: at the long-run mean rates a branch exists only for $e_t \gtrsim 0.3$ when $\epsilon_t^2 = 0$, and for no e_t in the swept range once $\epsilon_t^2 = 0.05$.

The stable qualitative conclusion, and the only one we draw, is this: across every grid examined in Table 9, the U.S.–U.K. model-C inverse mapping fails to be globally defined and is never single-valued at any candidate point where it is defined. The grid establishes an algebraic-domain failure of the inverse representation. It does not establish how often, or whether, the corresponding diffusion visits those candidate states.

Establishing that would require a different argument altogether: a support theorem for the diffusion, an accessibility argument for the region in question, analysis of the invariant or transition densities, or direct simulation under a fully specified and well-posed process. None of these is available here, in part because the well-posedness failure is precisely what obstructs the last of them.

Grid sensitivity. Because a single grid could produce an artefact of its own spacing, the diagnostic is repeated under five further designs: a denser grid, a wider and a narrower log-exchange-rate interval, logarithmically spaced rate multipliers, and additional incompleteness levels. Table 9 reports all six. The counts change with the grid, as counts must; the two qualitative findings do not. On no grid does any candidate point yield exactly one positive root, and on every grid a substantial share of points yields none. The methodology, including every tolerance, is recorded in `docs/grid_diagnostic_methodology.md`.

Table 9: Grid sensitivity of the algebraic-domain diagnostic for the U.S.–U.K. model-C specification. Each row is a different candidate grid. "No root" counts points at which (82) admits no positive solution, "one" and "two" count positive roots, and the matrix column counts branch-specific correlation matrices, one per positive root, so it equals twice the two-root count. Roots are counted positive above 10^{-12} and treated as repeated when within 10^{-9}; eigenvalues are called negative below -10^{-10}. No design yields a point with exactly one positive root, and none yields an indefinite matrix.

Design	Rates	Spacing	e values	$\lvert e\rvert \leq$	ϵ^2 levels	Points	No root	One	Two	Matrices	$\min \lambda_{\min}$
baseline	26	linear	21	2.0000	4	56784	37536	0	19248	38496	0.4455
denser	41	linear	33	2.0000	4	221892	148021	0	73871	147742	0.4380
wider fx	26	linear	21	4.0000	4	56784	32804	0	23980	47960	0.4455
narrower fx	26	linear	21	0.5000	4	56784	54542	0	2242	4484	0.4059
log rates	26	log	21	2.0000	4	56784	37065	0	19719	39438	0.4460
more eps	26	linear	21	2.0000	6	85176	56416	0	28760	57520	0.4455

Table 10: Algebraic-domain diagnostic for the time-varying specifications over a grid of candidate states, using the Table 4 loadings together with the Table 3 interest-rate risk prices and correlations. Volatility is obtained from (80), through (82) under model C, so every matrix evaluated satisfies the variance identity by construction. The final column counts *branch-specific* matrices, one for each positive root: a candidate point with two positive roots contributes two matrices, which is why the U.S.–U.K. model-C row records $38{,}496 = 2 \times 19{,}248$ matrices from 19,248 two-root points and none from the 37,536 points with no root. Counts are counts of grid points and carry no probabilistic interpretation. No candidate point yielded an indefinite matrix in any specification.

System	Model	Candidate points	No root	Two roots	Branch-specific matrices	$\min \lambda_{\min}$
U.S. vs. U.K.	B	56,784	0	0	56,784	+0.0699
	C	56,784	37,536	19,248	38,496	+0.4455
U.S. vs. Germany	B	56,784	0	0	56,784	+0.1351
	C	56,784	0	0	56,784	+0.0880

E.2 Validity at reported reference states

Table 12 evaluates R_t across reference states, reporting the complete decomposition (80) so that feasibility can be judged from the identity rather than from the insufficient bound.

At every algebraically feasible reference point examined the matrix is positive definite. At the long-run means with the sample mean volatility the minimum eigenvalue is 0.683 for the U.S.–U.K. system and 0.639 for the U.S.–Germany system, with Schur complements of 0.997 and 0.994. Pushing to the feasible minimum volatility at rates ten times their long-run means still leaves minimum eigenvalues of 0.115 and 0.125. Perturbing the four estimated correlations independently by their reported standard errors, in 5,000 draws per system, produced no violation of positive semidefiniteness and no entrywise violation.

A systematic search reinforces this. Sweeping both interest rates over multiples of their long-run means from zero to one hundred, and the incompleteness state over the values $\epsilon_t^2 \in \{0, 0.005, 0.02, 0.10\}$, always taking v_t from the identity (80), the smallest minimum eigenvalue encountered over the swept candidate points is +0.0128 for the U.S.–U.K. system and +0.0130 for the U.S.–Germany system, both at the extreme corner of the sweep where the interest rates are a hundred times their long-run means. **No algebraically feasible candidate point with an indefinite correlation matrix was found.** We therefore make no claim that an invalid correlation matrix arises at any algebraically feasible point, still less that the process would visit one.

The last row of each block in Table 12 records what happens at $v_t = \sqrt{Q_t}$, which an earlier version of this analysis mistakenly treated as the floor. There the matrix is indeed indefinite, with minimum eigenvalues -0.0009 and -0.0027, while every entry remains inside $[-1, 1]$. But the implied ϵ_t^2 equals $-C = -0.003$, so the state violates (80) and lies outside the model. It is reported here only to mark the distinction, not as evidence.

E.3 Global validity of the specification

That the reported estimates are valid is not the same as the specification guaranteeing validity. Two results bear on this, and both are conditional algebraic statements rather than claims about the calibrated model.

Proposition E.2 (The two implied correlations are automatically bounded)**.** *Suppose $v_t > 0$, $|\rho_{ww^*}| \leq 1$, and $v_t^2 \geq Q_t$, the last being implied by* (80)*. Then the two* implied *correlations of* (79) *satisfy $|\rho_{wx,t}| \leq 1$ and $|\rho_{w^*x,t}| \leq 1$. The proposition says nothing about the free entries ρ_{wy}, ρ_{w^*y}, ρ_{xy} and ρ_{ww^*}, nor about the primitive parameter ρ_{zz^*}.*

Remark E.3 (Why positivity is needed)**.** Both side conditions are load-bearing, and neither follows from $v_t^2 \geq Q_t$. The strict inequality $v_t > 0$ is needed because (79) divides by v_t, and $v_t^2 \geq Q_t$ alone permits $v_t = Q_t = 0$; the correlations of a degenerate Brownian motion are undefined in any case, so nothing is lost by excluding it. The bound $|\rho_{ww^*}| \leq 1$ is needed because the proof uses $1 - \rho_{ww^*}^2 \geq 0$, and without it the conclusion is false as a standalone algebraic statement. That ρ_{ww^*} is a correlation makes the bound true in the model, and Table 11 records that the published parameterisation imposes it entrywise; but it is a hypothesis of the proposition rather than a consequence of the variance identity, and is stated as one. In the empirical application both conditions hold at every reported estimate, so they are stated because the algebra requires them, not because the data threaten them.

Proof. $Q_t - (\phi_t - \rho_{ww^*}\phi_t^*)^2 = \phi_t^{*2}(1 - \rho_{ww^*}^2) \geq 0$, so $(\phi_t - \rho_{ww^*}\phi_t^*)^2 \leq Q_t \leq v_t^2$, giving $|\rho_{wx,t}| \leq 1$. The second bound is symmetric, using $Q_t - (\rho_{ww^*}\phi_t - \phi_t^*)^2 = \phi_t^2(1 - \rho_{ww^*}^2) \geq 0$. $\square$

The scope of this result must be stated carefully, because it is easy to over-read, and so must the inventory of correlations it is being read against. The model contains *five primitive correlation parameters*. Four of them, ρ_{ww^*}, ρ_{wy}, ρ_{w^*y} and ρ_{xy}, are entries of the reduced 4×4 Brownian correlation matrix R_t directly. The fifth, ρ_{zz^*}, is *not* an entry of R_t at all: it enters the currency-risk quadratic form C_t in (80) and so affects R_t only through the volatility v_t that the variance identity determines. Alongside the four primitive entries, R_t carries two state-dependent *implied* entries, $\rho_{wx,t}$ and $\rho_{w^*x,t}$, given by (79). A symmetric 4×4 correlation matrix has six off-diagonal entries, and these are they: four free and two implied.

With that inventory fixed, the variance identity automatically bounds the two implied entries. The four free entries still require admissible parameter restrictions, ρ_{zz^*} requires its own, and entrywise bounds on every one of them remain insufficient to guarantee positive semidefiniteness of R_t. Table 11 sets out which quantity is which.

The consequence is that the entrywise restriction the paper imposes, through a logistic reparameterisation of the correlations, is not doing any work on the two implied entries: the model structure already implies it there. It is not redundant on the five primitive parameters,

Table 11: Primitive and implied correlation quantities in the structural model and in the reduced Brownian system. Five correlations are primitive parameters, but only four of them are entries of the reduced 4×4 matrix R_t; ρ_{zz^*} enters the currency-risk quadratic form C_t instead, and reaches R_t only through v_t. The two implied entries are the ones Proposition E.2 covers. No row is by itself sufficient for positive semidefiniteness of R_t.

Symbol	Status	In R_t	In C_t	State dep.	Entrywise constraint	Joint constraint
ρ_{ww^*}	primitive	yes	no	no	$[-1, 1]$, imposed	in PSD of R_t; also in (85)
ρ_{wy}	primitive	yes	no	no	$[-1, 1]$, imposed	in PSD of R_t; also in (85)
ρ_{w^*y}	primitive	yes	no	no	$[-1, 1]$, imposed	in PSD of R_t; also in (85)
ρ_{xy}	primitive	yes	no	no	$[-1, 1]$, imposed	in PSD; fixed by (84) if A_t singular
ρ_{zz^*}	primitive	**no**	yes	no	$[-1, 1]$, imposed	keeps $C_t \geq 0$; enters R_t only via v_t
$\rho_{wx,t}$	implied	yes	no	yes	automatic, Proposition E.2	in PSD of R_t
$\rho_{w^*x,t}$	implied	yes	no	yes	automatic, Proposition E.2	in PSD of R_t

where it remains the operative entrywise constraint. What no entrywise restriction implies, on any entry, is the joint condition. When $A_t \succ 0$ that condition is the Schur complement

$$1 - r_t^\top A_t^{-1} r_t \geq 0, \tag{83}$$

and constraining each entry of r_t to $[-1, 1]$ is not sufficient for it.

The remaining question is what happens when A_t is singular, where (83) is unavailable because A_t^{-1} does not exist. This occurs exactly in the degenerate configuration identified above. The correct replacement for (83) has two parts, and it is worth stating in general form because the second part is easy to lose.

Lemma E.4 (Positive semidefiniteness with a singular principal block)**.** *Let $A \in \mathbb{R}^{m \times m}$ be symmetric and positive semidefinite, $r \in \mathbb{R}^m$ and $c \in \mathbb{R}$. Then*

$$B = \begin{pmatrix} A & r \\ r^\top & c \end{pmatrix} \succeq 0 \qquad \textit{if and only if} \qquad r \in \operatorname{Range}(A) \quad \textit{and} \quad c \;\geq\; r^\top A^+ r,$$

where A^+ is the Moore–Penrose pseudoinverse. When $A \succ 0$ the first condition is vacuous and the second is the Schur complement (83)*.*

Proof. Write the quadratic form as $q(x, \tau) = x^\top A x + 2\tau r^\top x + c\tau^2$ for $x \in \mathbb{R}^m$ and $\tau \in \mathbb{R}$.

Suppose first that both conditions hold, and choose w with $r = Aw$. Since $AA^+A = A$ we have $r^\top A^+ r = w^\top A A^+ A w = w^\top A w$, and $r^\top x = w^\top A x$. Completing the square,

$$q(x, \tau) = (x + \tau w)^\top A (x + \tau w) + \tau^2 (c - w^\top A w) = (x + \tau w)^\top A (x + \tau w) + \tau^2 (c - r^\top A^+ r),$$

and both terms are nonnegative, so $B \succeq 0$.

Conversely suppose $B \succeq 0$. For the range condition, let $z \in \operatorname{Null}(A)$ and take $x = sz$, $\tau = 1$. Then $q(sz, 1) = 2s\, r^\top z + c$, which is affine in s and therefore unbounded below unless $r^\top z = 0$.

Hence $r \perp \mathrm{Null}(A)$, and since A is symmetric, $\mathrm{Null}(A)^\perp = \mathrm{Range}(A)$, giving $r \in \mathrm{Range}(A)$. For the magnitude condition, write $r = Aw$ as before and take $x = -w$, $\tau = 1$, which gives

$$0 \le q(-w, 1) = w^\top Aw - 2w^\top Aw + c = c - r^\top A^+ r. \qquad \square$$

The two conditions are logically independent: range membership constrains the *direction* of r relative to $\mathrm{Null}(A)$ and says nothing about its length, while the magnitude condition bounds that length. Neither implies the other.

Proposition E.5 (Compatibility and magnitude in the degenerate configuration)**.** *Suppose $C_t = \epsilon_t^2 = 0$ and $Q_t > 0$, so that $v_t^2 = Q_t$, and suppose $|\rho_{ww^*}| < 1$. Then A_t is singular with a one-dimensional kernel, X_t is a deterministic linear combination of W_t and W_t^*, and $R_t \succeq 0$ holds if and only if* both *of the following are satisfied:*

(i) the compatibility relation
$$\rho_{xy} = \frac{\phi_t \rho_{wy} - \phi_t^* \rho_{w^*y}}{v_t}, \tag{84}$$
which fixes ρ_{xy} as a function of the other correlations; and

(ii) the magnitude condition that the three-dimensional correlation matrix of (W_t, W_t^*, Y_t) be positive semidefinite, equivalently
$$\rho_{wy}^2 - 2\rho_{ww^*}\rho_{wy}\rho_{w^*y} + \rho_{w^*y}^2 \;\le\; 1 - \rho_{ww^*}^2. \tag{85}$$

Condition (84) *alone is necessary but* not *sufficient: if it holds while* (85) *fails, R_t is indefinite, however far each individual entry lies inside* $[-1, 1]$.

Proof. If $C_t = \epsilon_t^2 = 0$ then equation (28) reduces to $v_t\, dX_t = \phi_t\, dW_t - \phi_t^*\, dW_t^*$, so X_t lies in the span of (W_t, W_t^*) and A_t is singular. Let $u = (\phi_t, -\phi_t^*, -v_t)^\top$. Then $u^\top A_t u$ is the variance of $\phi_t W_t - \phi_t^* W_t^* - v_t X_t$, which vanishes, and since $A_t \succeq 0$ this forces $A_t u = 0$; the kernel is one dimensional because the leading 2×2 block of A_t is nonsingular when $|\rho_{ww^*}| < 1$.

Necessity of (i). Extend u to $\tilde{u} = (u^\top, 0)^\top \in \mathbb{R}^4$ and let e_4 be the fourth coordinate vector. For $\tau \in \mathbb{R}$,
$$(\tilde{u} + \tau e_4)^\top R_t (\tilde{u} + \tau e_4) = u^\top A_t u + 2\tau\, u^\top r_t + \tau^2 = 2\tau\, u^\top r_t + \tau^2.$$
If $u^\top r_t \neq 0$, choosing τ of small magnitude and of sign opposite to $u^\top r_t$ makes this negative, so $R_t \not\succeq 0$. Writing out $u^\top r_t = \phi_t \rho_{wy} - \phi_t^* \rho_{w^*y} - v_t \rho_{xy} = 0$ gives (84). In the language of Lemma E.4, this is exactly the range condition $r_t \in \mathrm{Range}(A_t)$.

Reduction under (i). Suppose (84) holds and let

$$\Sigma_t = \begin{pmatrix} B & d \\ d^\top & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & \rho_{ww^*} \\ \rho_{ww^*} & 1 \end{pmatrix}, \qquad d = \begin{pmatrix} \rho_{wy} \\ \rho_{w^*y} \end{pmatrix},$$

be the correlation matrix of (W_t, W_t^*, Y_t), and let

$$T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \phi_t/v_t & -\phi_t^*/v_t & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 3},$$

the map sending (W_t, W_t^*, Y_t) to (W_t, W_t^*, X_t, Y_t). A direct computation gives $T\Sigma_t T^\top = R_t$: the $(3,3)$ entry is $(\phi_t^2 - 2\rho_{ww^*}\phi_t\phi_t^* + \phi_t^{*2})/v_t^2 = Q_t/v_t^2 = 1$, the $(1,3)$ and $(2,3)$ entries reproduce (79), and the $(3,4)$ entry is $(\phi_t\rho_{wy} - \phi_t^*\rho_{w^*y})/v_t$, which is ρ_{xy} precisely by (84). Hence $\Sigma_t \succeq 0$ implies $R_t = T\Sigma_t T^\top \succeq 0$; and conversely Σ_t is the principal submatrix of R_t on rows and columns $\{1, 2, 4\}$, so $R_t \succeq 0$ implies $\Sigma_t \succeq 0$. Therefore, given (84), $R_t \succeq 0$ if and only if $\Sigma_t \succeq 0$.

Explicit form of (ii). Since $|\rho_{ww^*}| < 1$, $B \succ 0$ and Lemma E.4 reduces to the ordinary Schur complement $1 - d^\mathsf{T} B^{-1} d \geq 0$. With $B^{-1} = (1-\rho_{ww^*}^2)^{-1} \left(\begin{smallmatrix} 1 & -\rho_{ww^*} \\ -\rho_{ww^*} & 1 \end{smallmatrix}\right)$ this is

$$\frac{\rho_{wy}^2 - 2\rho_{ww^*}\rho_{wy}\rho_{w^*y} + \rho_{w^*y}^2}{1-\rho_{ww^*}^2} \leq 1,$$

which is (85). □

Remark E.6 (The perfectly correlated case)**.** If $|\rho_{ww^*}| = 1$ the kernel of A_t is two dimensional and B is singular, so Proposition E.5 does not apply as stated. Writing $\rho_{ww^*} = \sigma \in \{-1, +1\}$, condition (85) does not become vacuous. Its right-hand side is $1 - \sigma^2 = 0$ and its left-hand side is a perfect square, since $\sigma^2 = 1$ gives

$$\rho_{wy}^2 - 2\sigma\rho_{wy}\rho_{w^*y} + \rho_{w^*y}^2 = \left(\rho_{wy} - \sigma\rho_{w^*y}\right)^2,$$

so (85) reads $(\rho_{wy} - \sigma\rho_{w^*y})^2 \leq 0$ and therefore *forces*

$$\rho_{w^*y} = \sigma\rho_{wy} = \rho_{ww^*}\rho_{wy}.$$

That is exactly the range-compatibility relation of Lemma E.4 applied to B, so in this configuration (85) collapses onto the compatibility condition rather than carrying no information.

What it does *not* retain is the separate magnitude restriction. Once compatibility holds the inequality is satisfied with equality, $0 \leq 0$, and the magnitude condition must be obtained from the Moore–Penrose criterion instead: since $B^+ = \frac{1}{4}\left(\begin{smallmatrix} 1 & \sigma \\ \sigma & 1 \end{smallmatrix}\right)$, a compatible d gives $d^\mathsf{T} B^+ d = \frac{1}{4}(\rho_{wy} + \sigma\rho_{w^*y})^2 = \rho_{wy}^2$, so the requirement is $|\rho_{wy}| \leq 1$. This is the only configuration in which the entrywise bound is the operative magnitude constraint, and it arises because $W_t^* = \sigma W_t$ almost surely, so the model has one interest-rate Brownian motion rather than two.

That the second condition is not implied by the first is worth seeing concretely, because an earlier version of this analysis asserted that it was. Take $\phi_t = 0.03$, $\phi_t^* = 0.02$ and $\rho_{ww^*} = 0.3$, so that $v_t = \sqrt{Q_t} = 0.03066$, and set $\rho_{wy} = \rho_{w^*y} = 0.9$. The implied correlations are $\rho_{wx,t} = 0.7828$ and $\rho_{w^*x,t} = -0.3588$, and (84) gives $\rho_{xy} = 0.2935$, so *every* entry of R_t lies well inside $[-1, 1]$ and the compatibility relation holds exactly. But the left side of (85) is 1.134 against a right side of 0.91, and the spectrum of R_t is $\{-0.133, 0, 1.558, 2.575\}$. The matrix is indefinite. Compatibility bought the zero eigenvalue that singularity requires; it did nothing about the negative one.

Two things follow. First, in the degenerate configuration the correlation ρ_{xy} is not a free parameter but a function of the others, whereas the application estimates it freely. Second, and this is the part that a compatibility condition alone would miss, fixing ρ_{xy} correctly still leaves (85) to be satisfied by ρ_{wy}, ρ_{w^*y} and ρ_{ww^*}, and that is a joint restriction on three free parameters which entrywise bounds do not deliver. It is a genuine gap in the specification: nothing in the model enforces either condition, and no Cholesky, partial-correlation, hyperspherical or Schur-complement parameterisation is stated that would enforce (83) more generally.

It is not, however, a defect in the reported estimates, for the reason established in Section E.1: the configuration requires $C_t = 0$, and at the reported estimates $C = 0.003 > 0$. To see how far the reported values are from satisfying (84), evaluating the right-hand side at three times the long-run rates gives -0.0538 and -0.0696 against reported ρ_{xy} of -0.012 and $+0.006$; but that comparison is made at a point the reported estimates render algebraically infeasible, and it establishes only that the identity would have to be imposed rather than estimated if the specification were used in a regime where C_t approached zero.

E.4 Unknown implementation safeguards

Whether the numerical implementation rejected invalid parameter draws, projected them onto the positive-semidefinite cone, worked with a Cholesky factor, or took some other precaution, is not described in the published paper and cannot be determined from it. We record this as unknown. It is entirely possible that the optimiser never encountered an invalid matrix, either because the safeguards existed or because, as Section E.2 suggests, the feasible region around the estimates is comfortably interior. Nothing in this appendix should be read as a claim about what the implementation did.

Table 12: Feasibility and validity of the four-dimensional Brownian correlation matrix under the complete variance identity $v_t^2 = Q_t + C_t + \epsilon_t^2$, at the constant-risk-price estimates of Table 3 of Brandt and Santa-Clara (2002). Rates are multiples of their long-run means. A candidate point is algebraically feasible when the implied ϵ_t^2 is nonnegative. The final row of each block uses $v_t = \sqrt{Q_t}$, which requires $C_t = 0$ and is therefore infeasible at these estimates; it is shown to separate what the algebraic restrictions exclude from what they permit.

System	State	Q_t	C_t	ϵ_t^2	v_t	Feas.	$\max\lvert\rho_{ij}\rvert$	$\lambda_{\min}(R_t)$
U.S. vs. U.K.	means, $v_t = \bar{v}$	0.0011	0.0030	0.0065	0.1030	yes	0.313	+0.6834
	means, v_t min	0.0011	0.0030	0.0000	0.0640	yes	0.503	+0.4875
	rates ×2, at min	0.0022	0.0030	0.0000	0.0720	yes	0.633	+0.3540
	rates ×3, at min	0.0033	0.0030	0.0000	0.0792	yes	0.704	+0.2799
	rates ×10, at min	0.0109	0.0030	0.0000	0.1179	yes	0.863	+0.1149
	rates ×100, at min	0.1090	0.0030	0.0000	0.3346	yes	0.962	+0.0128
	rates ×3, $v_t = \sqrt{Q_t}$	0.0033	0.0030	−0.0030	0.0572	**no**	0.975	−0.0009
U.S. vs. Germany	means, $v_t = \bar{v}$	0.0009	0.0030	0.0076	0.1070	yes	0.263	+0.6390
	means, v_t min	0.0009	0.0030	0.0000	0.0622	yes	0.453	+0.4762
	rates ×2, at min	0.0017	0.0030	0.0000	0.0689	yes	0.578	+0.3596
	rates ×3, at min	0.0026	0.0030	0.0000	0.0749	yes	0.651	+0.2907
	rates ×10, at min	0.0087	0.0030	0.0000	0.1083	yes	0.823	+0.1252
	rates ×100, at min	0.0872	0.0030	0.0000	0.3004	yes	0.938	+0.0130
	rates ×3, $v_t = \sqrt{Q_t}$	0.0026	0.0030	−0.0030	0.0512	**no**	0.954	−0.0027

F Minimum incompleteness is an identifying selection rule

The exchange-rate dynamics do not separately identify one risk-premium component and the degree of market incompleteness. The paper resolves this by choosing the unidentified quantity, described as chosen “in a least-squares sense”, to minimise $\sum_t \epsilon_t^2$ subject to $\epsilon_t^2 \geq 0$ at every date. Under constant pure-currency risk prices the free quantity is the product $\sigma_{z^*}\epsilon_{C^*}$; under time-varying prices it is the correlation ρ_{zz^*}.

Although the criterion is expressed as a sum of squared incompleteness shocks, it is affine in the selected reduced-form parameter, because ϵ_t^2 is itself the residual rather than a squared residual in that parameter. Its solution is therefore a boundary point of the feasible set, the minimum or maximum feasible value of the scalar parameter, rather than an interior least-squares projection. Whether the problem is a *linear programme*, and correspondingly whether “vertex” is the right word, depends on which feasible set is meant. Over $\mathcal{C}_{\text{pub}}$ in (86) every constraint is affine, the set is a polyhedron, and both terms apply. Over $\mathcal{C}_{\text{obs}}$ and $\mathcal{C}_{\text{glob}}$ the semidefiniteness requirements are not affine, the feasible set need not be polyhedral or even connected, and the problem is an affine-objective constrained optimisation problem whose optimum is a boundary point. We use “vertex” and “linear programme” only for the first case.

The feasible set matters, and three must be distinguished, not two. Write the free quantity as a scalar c entering linearly, $\epsilon_t^2(c) = A_t - \gamma_t c$ for known loadings γ_t, and let $\mathcal{R}$ collect any range restriction on c itself, such as $c \in [-1, 1]$ when c is a correlation. Then

$$\mathcal{C}_{\text{pub}} = \Big\{ c \in \mathcal{R} : \ \epsilon_t^2(c) \geq 0, \ t = 1, \dots, N \Big\}, \tag{86}$$

$$\mathcal{C}_{\text{obs}} = \mathcal{C}_{\text{pub}} \cap \{ c : \ R_t(c) \succeq 0, \ t = 1, \dots, N \}, \tag{87}$$

$$\mathcal{C}_{\text{glob}} = \mathcal{C}_{\text{pub}} \cap \{ c : \ R(c, x) \succeq 0 \text{ for every admissible state } x \in \mathcal{X} \}. \tag{88}$$

The distinction that matters is not merely one of strength but of *cardinality*. $\mathcal{C}_{\text{pub}}$ is cut out by N affine inequalities together with an interval, so it is a finite intersection and indeed a polyhedron. $\mathcal{C}_{\text{obs}}$ adds N further constraints, one per observed or reconstructed sample state; each is continuous in c, because the smallest eigenvalue of a symmetric matrix depends continuously on its entries, so this too is a finite family of continuous constraints. $\mathcal{C}_{\text{glob}}$ is different in kind: it requires positive semidefiniteness at *every* state of a continuous state domain $\mathcal{X}$, which is an infinite family of constraints indexed by x, and it admits no finite representation of the form used below without further argument.

We do not know which set the implementation enforced. Section E.4 records that as unknown, so no conclusion below assumes that the reported optimiser respected $\mathcal{C}_{\text{obs}}$ or $\mathcal{C}_{\text{glob}}$.

Two questions must therefore be separated, because they need different hypotheses. *Where* the minimiser sits requires only compactness, and so covers all three sets. *Which constraint is active there* is a statement about a representation, and the phrase is empty without one. The proposition is split accordingly.

For the second part we use the representation

$$\mathcal{C} = \{ c \in [c_{\text{L}}, c_{\text{U}}] : \ g_j(c) \leq 0, \ j = 1, \dots, J \}, \tag{89}$$

with $-\infty < c_{\text{L}} \leq c_{\text{U}} < \infty$, each $g_j : \mathbb{R} \to \mathbb{R}$ continuous, J finite and $\mathcal{C} \neq \emptyset$. Both $\mathcal{C}_{\text{pub}}$ and $\mathcal{C}_{\text{obs}}$ are of this form: the nonnegativity requirements give $g_t(c) = \gamma_t c - A_t$, the range restriction $\mathcal{R}$ supplies $[c_{\text{L}}, c_{\text{U}}]$, and each observed-state semidefiniteness requirement gives $g_t(c) = -\lambda_{\min}(R_t(c))$. $\mathcal{C}_{\text{glob}}$ is *not* of this form, since its constraints are indexed by a continuum. Restricting attention to the nonnegativity constraints alone, as an earlier version of this analysis did, can identify the wrong active constraint.

Proposition F.1 (Corner solution of the minimum-incompleteness rule)**.** *Consider the problem*

$$\min_{c \in \mathcal{C}} \sum_{t=1}^{N} \epsilon_t^2(c), \qquad \epsilon_t^2(c) = A_t - \gamma_t c, \tag{90}$$

and suppose $\sum_t \gamma_t \neq 0$. *No convexity of* $\mathcal{C}$ *is assumed.*

Part A (location of the minimiser; compactness only). *Let* $\mathcal{C} \subset \mathbb{R}$ *be any nonempty compact set. Then the minimum is attained, the minimiser is unique, and it is* $\max \mathcal{C}$ *if* $\sum_t \gamma_t > 0$ *and* $\min \mathcal{C}$ *if* $\sum_t \gamma_t < 0$. *In either case it is a boundary point of* $\mathcal{C}$. *This applies to* $\mathcal{C}_{pub}$, $\mathcal{C}_{obs}$ *and* $\mathcal{C}_{glob}$ *alike, whenever the set in question is nonempty and compact.*

Part B (identification of an active constraint; finite representation). *Suppose in addition that* $\mathcal{C}$ *has the representation* (89) *with* J *finite and each* g_j *continuous. Then* $\mathcal{C}$ *is compact, so Part A applies, and at the minimiser* c^* *at least one of the following holds:*

$$c^* = c_{\text{L}}, \qquad c^* = c_{\text{U}}, \qquad g_j(c^*) = 0 \textit{ for at least one } j. \tag{91}$$

If $\mathcal{C}$ *is a finite union of compact intervals, the minimiser is an endpoint of one connected component. Moreover:*

(i) If the active constraint at the optimum is one of the nonnegativity constraints, then $\min_t \epsilon_t^2 = 0$ *exactly.*

(ii) If instead the active constraint is a range restriction or the positive-semidefinite boundary, then $\min_t \epsilon_t^2 > 0$ *is possible and the incompleteness constraints need not bind at all.*

(iii) In the constant-risk-price case $\gamma_t \equiv 1$ *and* $c = \sigma_z \epsilon_c + \sigma_{z^*} \epsilon_{c^*}$*, the objective is* $\sum_t A_t - Nc$*, strictly decreasing in c. If the feasible set is* $\mathcal{C}_{pub}$*, and within it the largest feasible c is determined by the nonnegativity constraints rather than by the range restriction* $\mathcal{R}$*, then*

$$c^* = \min_{1 \le t \le N} A_t, \qquad \epsilon_t^2 = A_t - \min_s A_s. \tag{92}$$

All four conditions are needed: the published feasible set, a decreasing objective, the constraints $c \le A_t$ *as the operative upper bounds, and no stricter range condition active. Under* $\mathcal{C}_{obs}$ *or* $\mathcal{C}_{glob}$ *the semidefiniteness requirement may cut the interval before* $\min_t A_t$ *is reached, and then* (92) *fails.*

Proof. Part A. The objective equals $\sum_t A_t - c \sum_t \gamma_t$, which is affine in c with slope $-\sum_t \gamma_t \neq 0$. An affine function with nonzero slope is strictly monotone, so on a nonempty compact $\mathcal{C} \subset \mathbb{R}$ it attains its minimum at $\max \mathcal{C}$ when the slope is negative and at $\min \mathcal{C}$ when it is positive; both exist by compactness, since a nonempty compact subset of $\mathbb{R}$ contains its supremum and infimum. Strict monotonicity makes the minimiser unique. Neither $\max \mathcal{C}$ nor $\min \mathcal{C}$ is interior to $\mathcal{C}$, so the minimiser is a boundary point. No representation of $\mathcal{C}$ by constraints is used anywhere in this part, which is why it covers $\mathcal{C}_{\mathrm{glob}}$.

Part B. $\mathcal{C}$ is compact: it is bounded, lying in $[c_{\mathrm{L}}, c_{\mathrm{U}}]$, and closed, being the intersection of the closed interval with the preimages $g_j^{-1}((-\infty, 0])$ of a closed set under continuous maps. Part A therefore applies.

It remains to prove (91). Suppose not: then $c^* \in (c_{\mathrm{L}}, c_{\mathrm{U}})$ and $g_j(c^*) < 0$ for every j. By continuity of each g_j and finiteness of J there is $\delta > 0$ with $g_j(c) < 0$ for all j and all $|c - c^*| < \delta$, and shrinking δ keeps c inside $(c_{\mathrm{L}}, c_{\mathrm{U}})$; so $(c^* - \delta, c^* + \delta) \subseteq \mathcal{C}$, contradicting that c^* is an endpoint of $\mathcal{C}$. Finiteness of J is what makes the common δ available: a minimum of finitely many positive numbers is positive, whereas an infimum over infinitely many constraints, as in $\mathcal{C}_{\mathrm{glob}}$, may be zero. That is precisely why Part B is stated for a finite representation and why it does not extend to $\mathcal{C}_{\mathrm{glob}}$ without a further argument, such as compactness of the index set $\mathcal{X}$ together with joint continuity in (c, x).

If $\mathcal{C}$ is a finite union of compact intervals, the minimiser is an endpoint of one component, by the same argument applied to the component containing it. Statements (i) and (ii) follow according to which of the alternatives in (91) occurs. For (iii), $\gamma_t \equiv 1$ gives slope $-N < 0$, so the optimum is $\max \mathcal{C}$; if that equals $\min_t A_t$ then (92) follows and the constraint indexed by $\arg\min_t A_t$ is active. □

The proposition is conditional in a way the earlier version was not. It does not assert that an incompleteness constraint must always bind, because the optimum may instead be stopped by $\mathcal{R}$ or by semidefiniteness. Whether the conditional case applies to the published estimates is an empirical question about interiority, and the honest answer is that we cannot settle it. What is available is the following, and it falls short of identifying the active constraint in $\mathcal{C}_{\mathrm{obs}}$, let alone in $\mathcal{C}_{\mathrm{glob}}$:

- the original paper reports that the nonnegativity constraint binds in the time-varying specifications;
- where Table 4 of Brandt and Santa-Clara (2002) reports ρ_{zz^*} directly, the estimates 0.89 and 0.94 are interior to $[-1, 1]$, so the range restriction is slack *at those points*;

- Section E finds the reconstructed correlation matrices positive definite, with minimum eigenvalue bounded away from zero, at every reference state and candidate grid point examined;
- the observed state series and the implementation are unavailable.

The third item is the weak link. Slackness of a constraint at the reported optimum and at a collection of candidate points is not slackness along the whole feasible set, and the active constraint is a property of the set, not of a sample of its points. Accordingly: for the published optimisation problem, the authors report that the nonnegativity constraint binds in the time-varying specifications, and our affine-objective analysis explains why a boundary solution is expected there. We cannot independently determine the active constraint in $\mathcal{C}_{\text{obs}}$ without the observed state sequence, nor in $\mathcal{C}_{\text{glob}}$ without both that and an argument covering an infinite constraint family.

Table 4 of Brandt and Santa-Clara (2002) reports the minimum of the inferred excess volatility as 0.0000 in all six specifications, that is for models A, B and C in both the U.S.–U.K. and the U.S.–Germany systems. This is consistent with case (i), but it should be read carefully, and the three sources of support must be kept apart. A reported 0.0000 is a rounded value at four decimal places and is not by itself proof of exact binding. Proposition F.1 proves that the optimum is a boundary point of whichever compact feasible set the optimiser used; for the published affine feasible set $\mathcal{C}_{\text{pub}}$ that boundary point is a vertex, and in the constant-risk-price case with the stated conditions it gives the exact corner formula (92). That the optimum sits on a nonnegativity boundary rather than on a range or semidefiniteness boundary is, for the time-varying specifications, what the original authors report rather than something we verify. Table 13 sets out the status of the claim $\min_t \epsilon_t^2 = 0$ in each case, and every statement elsewhere in this paper conforms to it.

Table 13: Status of the claim that the incompleteness boundary is attained exactly, by specification and by source. The distinction is not pedantic: only the first row is a theorem of this paper, and only under its stated conditions. Rows two and three are statements by the original authors that we repeat rather than verify, and the last row is open.

Specification	Source of the boundary claim	Exact status
Constant prices (model A)	Proposition F.1(iii), its conditions	proved conditionally
Time-varying, model B	original authors report binding	reported statement
Time-varying, model C	original authors report binding	reported statement
Table 4 minimum 0.0000	rounded published output	corroborative only
$\mathcal{C}_{\text{obs}}$, $\mathcal{C}_{\text{glob}}$	state path, implementation unavailable	unresolved

The mechanism is acknowledged in part in the original paper, which observes on p. 193 that with constant risk prices the inferred excess volatility is essentially $\epsilon_t = \{v_t^2 - \min_s v_s^2\}^{1/2}$, and draws from this the conclusion that a volatile v_t leaves a large average residual. Proposition F.1 formalises that observation, shows that it is a property of the optimisation rather than of the data, and extends it conditionally to the time-varying specifications, where the same corner structure holds for a different free parameter *whenever the aggregate affine slope $\sum_t \gamma_t$ is nonzero.* That slope depends on the observed state sequence, which we do not reconstruct, so we do not verify the condition; for the published estimates Brandt and Santa-Clara (2002) report that the nonnegativity constraint binds, and we neither reconstruct the aggregate slope nor identify the active constraint independently. The consequence for interpretation is that the dispersion of v_t^2, rather than its level, mechanically drives much of the estimated excess volatility: the residual is the observed variance minus its sample minimum after other adjustments, so its average size is a measure of that dispersion and is bounded below by construction.

The rule is transparent, but it is not identification by the likelihood of the original structural model. It selects a member of an observationally equivalent or weakly identified class by imposing the smallest feasible incompleteness. The paper describes it as a "prior of sorts". Mathematically, the final estimator combines simulated likelihood, auxiliary least squares from bond yields, a minimum-incompleteness corner rule, and inequality constraints. A further consequence, used in Section C, is that under case (i) the reconstructed incompleteness state sits exactly on its own degenerate boundary at the minimising date.

G Binding constraints are not covered by interior asymptotics

The empirical optimisation assigns zero likelihood to parameter values for which any reconstructed ϵ_t^2 is negative, and the original authors report that the constraint binds in the time-varying specifications.

A finite-sample optimum on an active inequality boundary does not by itself determine the asymptotic distribution. Whether the limit is nonstandard depends on the population or pseudo-true problem, not on the realised sample: a constraint can bind in every finite sample drawn and still be slack in the limit, in which case ordinary interior asymptotics apply after all. What the finite-sample binding does show is that the standard unconstrained interior argument is insufficient as stated, and that tangent-cone or mixture limits may arise if the population problem is also on or near a boundary. Boundary maximum-likelihood estimators and likelihood-ratio statistics commonly have such limits rather than the standard interior distributions; see Self and Liang (1987).

Stacking first-order conditions and reporting ordinary GMM standard errors does not automatically account for:

- a sample-dependent feasible set defined by N inequalities;
- changes in the active set;
- nondifferentiability of an inner constrained optimisation;
- structural parameters selected through minimum incompleteness;
- an optimum on or near the boundary.

We do not claim that the estimator has a nonstandard distribution, nor that the reported standard errors are invalid; establishing either would require the population problem, which we do not have. The claim is narrower: the standard interior asymptotic justification is incomplete, and a separate constrained-estimation analysis is required before conventional symmetric standard errors and Wald tests can be justified.

H The argmax question in a location model

This appendix records what can be proved about the simulated-likelihood maximiser in the most favourable possible model, and where the argument stops. The conclusion is negative: we do not obtain a counterexample to argmax consistency, and the manuscript makes no such claim. Two intermediate results are complete and are stated here because they clarify what the endpoint construction does to the estimation problem.

Consider

$$dY_t = \theta\, dt + dW_t, \qquad Y_t \in \mathbb{R}^K, \tag{93}$$

observed at unit intervals $t = 0, 1, \ldots, N$. The Euler scheme is exact for every M, the coefficients satisfy Assumptions 1 and 2 of Brandt and Santa-Clara (2002) exactly, and the exact maximum likelihood estimator is the mean increment $\widehat{\theta}_N = N^{-1} \sum_n \Delta Y_n$. Common random numbers across θ are used throughout, as the published implementation requires for a smooth objective: the innovations $\xi_{n,s} \sim \mathcal{N}(0, I_K)$ are drawn once and held fixed as θ varies.

Proposition H.1 (Common random numbers give an exact kernel estimator)**.** *Set $h = 1/M$, $u_n(\theta) = \Delta Y_n - \theta$ and $a_{n,s} = \sqrt{1-h}\,\xi_{n,s}$. Then for every θ,*

$$\widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta) = \frac{1}{S} \sum_{s=1}^{S} \varphi_K(u_n(\theta) - a_{n,s}; 0, hI_K)\,. \tag{94}$$

That is, the simulated transition density is exactly a Gaussian kernel density estimate with bandwidth $\sqrt{h}$, built from a sample that does not depend on θ and evaluated at the θ-dependent point $u_n(\theta)$.

Proof. The Euler recursion for (93) is $Y^M_{m+1} = Y^M_m + \theta h + \sqrt{h}\,\varepsilon_{m+1}$, so after $M-1$ steps from $Y^M_0 = Y_n$ the preterminal state is $Z_{n,s}(\theta) = Y_n + \theta(1-h) + \sqrt{1-h}\,\xi_{n,s}$, where $\sqrt{1-h}\,\xi_{n,s}$ collects the $M-1$ independent increments. The one-step density (3) evaluated at Y_{n+1} has mean $Z_{n,s}(\theta) + \theta h$ and covariance hI_K, and

$$Y_{n+1} - Z_{n,s}(\theta) - \theta h = \Delta Y_n - \theta - \sqrt{1-h}\,\xi_{n,s} = u_n(\theta) - a_{n,s}.$$

Substituting into (5) gives (94). □

Proposition H.1 is an identity, not an approximation, and it recasts Theorem 3.3 in familiar terms: a kernel density estimate with S atoms and bandwidth $\sqrt{h}$ is uninformative when $Sh^{K/2} \to 0$.

Proposition H.2 (Uniform comparison with a nearest-atom objective)**.** *Write $b_{n,s} = \Delta Y_n - \sqrt{1-h}\,\xi_{n,s}$ and*

$$D_N(\theta) = \sum_{n=0}^{N-1} \min_{s \le S} \|\theta - b_{n,s}\|^2.$$

Then for every θ,

$$-2Nh \log S \le 2h \log \widehat{\mathcal{L}}_{N,M,S}(\theta) + D_N(\theta) + NhK \log(2\pi h) \le 0. \tag{95}$$

Proof. Fix n and let $d_n = \min_s \|\theta - b_{n,s}\|^2$. Factoring the minimising term out of the sum in (94),

$$\frac{1}{S} \sum_s e^{-\|\theta - b_{n,s}\|^2/(2h)} = e^{-d_n/(2h)} \cdot \frac{1}{S} \sum_s e^{-(\|\theta - b_{n,s}\|^2 - d_n)/(2h)}.$$

Every exponent in the second factor is nonpositive and at least one is zero, so that factor lies in $[1/S, 1]$ and its logarithm lies in $[-\log S, 0]$. Taking logarithms, adding the constant $-\frac{K}{2}\log(2\pi h)$ from the Gaussian prefactor, multiplying by $2h$ and summing over n gives (95). □

The bound is uniform in θ with slack $2Nh \log S$. After division by N that slack is $2h \log S$, which along $M = S = n$ equals $2(\log n)/n$ and therefore vanishes. What this does *not* settle is whether the comparison is sharp enough to control the maximiser, because that is a question about the slack relative to the variation of D_N near its optimum, and the order and local geometry of D_N are exactly what remain unresolved. An earlier version of this discussion asserted the slack was negligible relative to D_N; that was not proved and is withdrawn.

So the statement is the one in the title: the scaled simulated log likelihood lies uniformly within $2Nh \log S$ of a transformed nearest-atom criterion. The simulated criterion is not itself a nearest-neighbour functional; it is a smooth log-sum-exp of NS terms, and Proposition H.2

says only that after scaling and the addition of a constant it is uniformly approximated by one. The exact log likelihood in this model is by contrast the quadratic $-\frac{1}{2}\sum_n \|\theta - \Delta Y_n\|^2$ with minimiser $\widehat{\theta}_N$. Whether the approximation controls the maximiser depends on the unresolved local scale of D_N, so any argument that the two maximisers agree must be made rather than assumed.

Five things the uniform bound alone does not establish, and which are not claimed anywhere in this paper: convergence of the two maximisers; equality of their finite-sample values; equivalence of local curvature at the optimum; the order of D_N; or a contradiction to Lemma 5 of Brandt and Santa-Clara (2002).

What is missing is the limit of D_N, and it is worth being precise about how much is missing, because an earlier version of this appendix misidentified the obstacle.

The candidate limit is computable in closed form. By Proposition H.1 the atoms $b_{n,s} = \Delta Y_n - \sqrt{1-h}\,\xi_{n,s}$ do not depend on θ at all; given ΔY_n they are independent across s with the Gaussian density f_n, and they are independent across n. Classical nearest-neighbour asymptotics for an independent sample from a fixed density then give

$$\mathbb{E}\Big[\min_{s\le S}\|\theta - b_{n,s}\|^2 \;\Big|\; \Delta Y_n\Big] \;\sim\; \Gamma(1+\tfrac{2}{K})\,(\omega_K S)^{-2/K} f_n(\theta)^{-2/K}, \tag{96}$$

with ω_K the volume of the unit ball. Since $f_n(\theta)^{-2/K} = 2\pi(1-h)\exp\{\|\theta-\Delta Y_n\|^2/(K(1-h))\}$ and $\Delta Y_n \sim \mathcal{N}(\theta_0, I)$ in this model, averaging over ΔY_n with the Gaussian identity $\mathbb{E}e^{a\|W\|^2} = (1-2a)^{-K/2}e^{a\|m\|^2/(1-2a)}$ at $a = 1/K$ gives the candidate

$$N^{-1}S^{2/K}D_N(\theta) \;\longrightarrow\; 2\pi\,\Gamma(1+\tfrac{2}{K})\,\omega_K^{-2/K}\Big(\frac{K}{K-2}\Big)^{K/2}\exp\Big(\frac{\|\theta-\theta_0\|^2}{K-2}\Big), \tag{97}$$

finite exactly when $a < 1/2$, that is when $K > 2$. The threshold is not a coincidence of two calculations: it is the integrability boundary of the same Gaussian moment that drives Theorem 3.3, appearing here as the condition for the averaged nearest-neighbour distance to be finite.

Two of the three ingredients check out numerically. The conditional statement (96) is accurate: at $K = 4$ and $K = 6$, for centres at distance 0 and 1.5 from θ, simulated values agree with the formula to within 0.5 per cent at $S = 50{,}000$, improving monotonically in S. The Gaussian moment identity is exact by computation.

What is *not* established is the interchange: passing from (96) to (97) requires uniform integrability of $S^{2/K}\min_s\|\theta - b_{n,s}\|^2$ over the distribution of ΔY_n, and that is a genuine obstacle rather than a formality. The limit is dominated by centres far from θ: at $K = 4$ the nearest 90 per cent of centres contribute only 58 per cent of (97), and the outermost 1 per cent contributes 15 per cent. Those are exactly the centres for which (96) needs the largest S, since its accuracy is governed by $Sf_n(\theta)$, the expected number of atoms near θ, and $f_n(\theta)$ is exponentially small there. Direct simulation of the unconditional quantity is correspondingly slow: at $K = 4$ and $S = 5{,}000$ it reaches only about 89 per cent of (97), approaching it from below as S grows.

An earlier version of this appendix gave a different and incorrect reason for stopping, namely that the atoms are “exchangeable, not independent”. That is wrong. By Proposition H.1 the atoms are free of θ and independent across s given the data, which is precisely the setting classical nearest-neighbour asymptotics assume. The real requirements are the uniform integrability just described and a uniform-in-θ upgrade of the pointwise law, both of which are standard in kind if delicate in execution. We have not carried them out, and (97) remains a conjecture; but it is a better-specified conjecture than before, with the constant explicit and the single missing step named.

If it were established, the consequence would be substantial. The limit criterion is minimised at θ_0, so the simulated argmax would be *consistent*; but its curvature at θ_0 is $2/(K-2)$ against the exact maximum-likelihood curvature 2, and the fluctuation of $N^{-1}S^{2/K}D_N$ about its limit is $O_{\mathbb{P}}(N^{-1/2})$ with a covariance not depending on S. Then $\sqrt{N}(\widetilde{\theta}_{N,M,S} - \widehat{\theta}_N)$ would converge to a nondegenerate limit rather than to zero, for every admissible (M_n, S_n) including the regime in which Theorem 2 of Brandt and Santa-Clara (2002) operates. That would refute the *conclusion* of Lemma 5, not merely its proof, and would change the standing of this paper's main claim.

Remark H.3 (Why Proposition H.2 cannot deliver this)**.** The uniform sandwich has slack $2h \log S$ after division by N. Against the *level* $D_N/N \asymp S^{-2/K}$ that slack is negligible whenever $M \gg S^{2/K} \log S$, which along $M = S$ with $K > 2$ holds comfortably. So Proposition H.2 is strong enough to transfer consistency, if the limit above were known. It is not strong enough for the $\sqrt{N}$ statement, because there the comparison is with the *variation* of D_N/N over a ball of radius $N^{-1/2}$ around the minimiser, where the limit criterion is flat and the slack is not negligible. Any route to the $\sqrt{N}$ conclusion must therefore go through the limit (97) rather than through the sandwich.

Remark H.4 (A likely shorter route)**.** Proposition H.1 makes the simulated transition density an exact kernel density estimate, so the simulated score is $\nabla_\theta \log \widehat{f}_n(\theta)$ and the estimating equation is $\sum_n \nabla_\theta \log \widehat{f}_n(\theta) = 0$. The asymptotics of log-density-gradient estimators at bandwidth $\sqrt{h}$ from S draws are standard, and would deliver directly the order of the discrepancy between the simulated and exact scores. That is exactly what hypothesis (iii) of Proposition 6.2 requires, and it is what Lemma 4 of Brandt and Santa-Clara (2002) asserts without proof. This route stays inside Section 6 rather than relying on nearest-neighbour theory, and we expect it to be both easier and more general. We record it as the first thing to try.

The status of the ingredients above should be kept distinct.

Proposition H.1 is an exact identity. Proposition H.2 is an exact uniform bound, and nothing more: it compares two criteria, it does not compare their maximisers. The limit of D_N is open, and with it the argmax question.

The nearest-neighbour limit is a *conjecture*. It is stated as such in the accompanying notes, it is used nowhere in this paper, and we do not assert that $N^{-1}S^{2/K}D_N(\theta)$ converges to the shape given there. A proof would require uniform-in-θ nearest-neighbour asymptotics for a triangular array whose atoms are exchangeable rather than independent, since they share the observation innovation, together with control of the dependence that common random numbers induce across θ and an argmax-continuity argument. None of these is available to us.

The accompanying notes also report *numerical observations*. Across the designs examined the simulated maximiser appears to approach θ_0, while the scaled gap $\sqrt{N}\|\widetilde{\theta}_{N,M,S} - \widehat{\theta}_N\|$ shows no downward trend. Those runs use a single seed per design, cover one dimension, and cannot reach the regime $N^4 \ll S \ll M^2$ in which the published theorem operates. They are not a demonstration of consistency, and they are deliberately confined to the notes rather than reported as a finding here.

Accordingly the manuscript claims only what Section 7 establishes: Lemmas 2 and 3 are directly refuted, and the proofs of Theorems 1 and 2 therefore fail.

I Reproducibility

The accompanying directory contains:

- `main.tex`, the manuscript source;
- `references.bib`, the bibliography;
- `code/generate_results.py`, the fully reproducible simulation and figure code;
- vector PDF and PNG versions of all figures;
- CSV and LaTeX versions of generated numerical tables.

The code uses a fixed seed, vectorised Gaussian simulation, and log-sum-exp evaluation of the endpoint estimator. No proprietary data are used.

Environment and cost. The results in this paper were produced with Python 3.13.5, NumPy 2.3.5, SciPy 1.15.3, pandas 2.3.3 and Matplotlib 3.10.8, from the single seed 20260726 set in `code/generate_results.py`. The full artefact regeneration, which includes the counterexample path of Table 4 and the six-design grid sweep of Table 9, takes a few minutes on one desktop core; no run requires special hardware, and none is long enough to warrant checkpointing.

Verification. Three checks are automated and can be run by a reader. `make verify` regenerates every figure and table into a scratch directory and compares them with the committed copies, field-wise at relative tolerance 10^{-9} for CSV output and by SHA-256 for PDF and PNG; figure output carries no embedded timestamp, so the comparison is exact rather than approximate. `make test` runs the analytical test suite, in which every closed form quoted here is checked against an independent computation. `code/verify_dois.py` checks each bibliography DOI against the Crossref registry by title.

Archive. The code and manuscript are deposited, not merely described. The concept DOI `10.5281/zenodo.21719868` resolves to the most recent version and is the one to cite; each release additionally receives its own version DOI, and superseded versions remain resolvable and are listed with the specific correction that supersedes them.

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